Regularisation in Functional Linear Regression: Theory, Methods, and Applications

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Abstract

Scalar-on-function linear regression is an inverse problem in which the predictor covariance operator determines both identification and the stability of slope recovery. This thesis examines how regularisation, numerical representation, and the purpose of an analysis interact. It compares functional principal component regression, raw and Arnoldi implementations of functional partial least squares, and a conjugate-gradient-type procedure based on moment-residual minimisation.

The theoretical discussion distinguishes ordinary slope error, covariance-weighted prediction error, and the residual approximation needed for inference. Monte Carlo experiments with 5,000 replications in each of three controlled designs show that the implemented early-stopping rule can reduce slope variance while leaving predictive signal unresolved. Raw and orthogonalised Krylov fits agree closely when their coordinates are reliable, but rare ill-conditioned raw fits create severe error tails. A separate inference study shows that an estimation-oriented stopping variant can leave a substantial moment residual on the root-n scale. Late iteration makes this algorithmic discrepancy negligible in the studied designs; calibration remains conditional on the reference distribution, discretisation, and finite-sample approximation.

An application uses instrument-1 near-infrared spectra to predict assay for pharmaceutical tablets, with 195 development observations and a designated 460-tablet benchmark. Raw and Arnoldi FPLS achieve benchmark RMSEs of 4.607 and 4.643 in the supplied assay units. Their predictions are very close, while Arnoldi uses a stable five-dimensional representation and the selected raw representation is severely ill-conditioned. The benchmark had been inspected during dataset selection, and the uncertainty intervals condition on the fitted models. The findings support matching regularisation and stopping to the statistical target, monitoring numerical reliability, and distinguishing controlled inferential benchmarks from feasible data analysis.

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Chapter 1

Introduction

Many modern data sets contain observations that are most naturally represented as functions rather than as finite collections of unrelated measurements. Examples include growth trajectories, spectrometric curves, electricity-demand profiles, biomedical signals, and records collected continuously over time or space. Treating such observations as functional data preserves their ordering and smooth structure and makes it possible to study features of the complete trajectory. Functional data analysis provides the statistical framework for this purpose; see, for example, Ramsay and Silverman (2005). One of its central regression models relates a scalar response to a functional predictor. For a square-integrable random function X on a compact interval $\mathcal{I}$, the scalar-on-function linear model is

$$Y = \alpha + \int_{\mathcal{I}} X(s)\beta(s)\,\mathrm{d}s + \varepsilon, \tag{1.1}$$

where Y is a scalar response, α is an intercept, β is an unknown slope function, and ε is an error term with conditional mean zero. The function β describes how variation in the predictor at different locations of its domain is associated with the response. This model is conceptually similar to multiple linear regression, but estimation is substantially more difficult because the unknown parameter belongs to an infinite-dimensional function space (Cardot et al. 1999; Hall and Horowitz 2007).

After centring X and Y and, for notational simplicity, denoting the centred variables again by X and Y, the population normal equation corresponding to (1.1) can be written as

$$K\beta = r, \tag{1.2}$$

where K is the covariance operator of X and $r = \mathbb{E}(YX)$ is the cross-covariance element between the response and the predictor. If K has a non-trivial null space, the data identify β only up to additions from $\mathrm{Null}(K)$; the conventional target is the minimum-norm representative $\beta^\dagger = K^\dagger r$ in the identified space $\mathcal{H}_{\mathrm{id}} := \mathrm{Null}(K)^\perp$. Unless the distinction is material, the notation β below refers to this identified representative. In typical functional settings, K is compact. When it has infinite rank, its positive eigenvalues decrease to zero and its generalised inverse is unbounded: small sampling errors in directions associated with small eigenvalues may be greatly magnified when (1.2) is solved. Estimating the identified slope is therefore an ill-posed inverse problem even when it is unique on the chosen parameter space. A direct solution of the empirical normal equation is generally unstable, including after the functional observations have been represented on a finite basis. Some form of regularisation is indispensable (Engl et al. 1996; Hall and Horowitz 2007).

Regularisation deliberately restricts or modifies the inverse problem so that the estimator does not attempt to recover directions that are weakly identified by the data. In functional principal component regression (FPCR), the covariance operator is replaced by a truncated spectral expansion and the inverse is computed only on the span of selected empirical eigenfunctions. This produces an interpretable and widely used estimator, and the truncation level acts as the regularisation parameter. Its components, however, are ordered according to variation in the predictor alone; a direction that explains relatively little variation in X may still be important for predicting Y. Functional partial least squares (FPLS) instead constructs response-informed directions and has consequently become an important alternative to purely spectral regularisation (Preda and Saporta 2005; Delaigle and Hall 2012; Febrero-Bande et al. 2017).

An especially useful view of FPLS is obtained through Krylov subspaces. At iteration m, the estimator is sought in

$$\mathcal{K}_m(K, r) = \operatorname{span}\{r, Kr, K^2 r, \dots, K^{m-1} r\}. \tag{1.3}$$

This formulation shows that the method progressively explores directions determined jointly by the covariance structure of X and its relationship with Y. It also connects FPLS with the conjugate-gradient (CG) algorithm for linear systems and inverse problems (Hestenes and Stiefel 1952; Phatak and Hoog 2002; Babii et al. 2026). The connection is statistically important because the number of Krylov iterations is itself a regularisation parameter. Early iterations recover well-identified signal directions, whereas later iterations increasingly expose the solution to sampling noise in small-eigenvalue directions. Early stopping can therefore regularise the functional inverse problem without explicitly truncating an eigendecomposition (Blanchard and Krämer 2016; Babii et al. 2026).

The mathematical equivalence of alternative Krylov formulations does not imply identical numerical behaviour. A direct implementation of (1.3) uses successive powers of the empirical covariance operator. These vectors can become nearly linearly dependent, causing the associated Gram matrix to be severely ill-conditioned. In exact arithmetic, this construction spans the intended space; in finite-precision arithmetic, however, loss of independence can produce unstable coefficients and occasionally very large estimation errors. Orthogonal Krylov algorithms such as Arnoldi replace the raw power basis by an orthonormal basis, while CG generates the relevant iterates through short recurrences. The practical performance of regularisation therefore has both a statistical and a numerical dimension: one must decide how far to iterate and how to represent the space explored by those iterations.

This thesis studies that interaction through four estimators: FPCR, raw FPLS based on unorthogonalised Krylov powers, Arnoldi FPLS with two-pass classical Gram–Schmidt orthogonalisation, and a direct CG–FPLS implementation with a data-dependent stopping rule. These estimators represent two broad regularisation principles. FPCR uses a spectral cut-off, while the three FPLS implementations use response-informed Krylov spaces and iterative regularisation. At a common component count and under the required rank conditions, raw and Arnoldi FPLS target the same restricted response fit, so their comparison isolates the effect of numerical stabilisation. CG–FPLS, in turn, permits a direct examination of early stopping and of the computational geometry underlying FPLS.

The study is organised around four questions. First, how do spectral and iterative regularisation methods compare in slope estimation and out-of-sample prediction when the covariance structure and slope smoothness change? Second, when theoretically equivalent FPLS formulations are implemented in finite precision, how strongly do Krylov-basis conditioning and orthogonalisation affect their reliability? Third, how does the CG stopping index affect the validity of statistical inference, and why may an inference-compatible index differ from one chosen for estimation or prediction? Fourth, how do these regularisation and stability trade-offs appear in a real spectroscopic prediction problem?

The third question reflects a fundamental distinction. For estimation and prediction, stopping CG early controls variance and prevents weakly identified directions from dominating the slope estimate. For inference on the moment equation, the remaining algorithmic residual must instead be negligible relative to sampling uncertainty. Following Babii et al. (2026), this thesis considers, for $b \in \mathcal{H}_{\mathrm{id}}$, the global hypothesis

$$H_0 : \beta = b \qquad \text{against} \qquad H_1 : \beta \neq b, \tag{1.4}$$

using the statistic

$$\mathcal{T}_n(b) = n \left\| \widehat{K}\left(\widehat{\beta}_m - b\right) \right\|_{L^2}^2 . \tag{1.5}$$

For this statistic, iteration must continue until the moment residual $\widehat{r} - \widehat{K}\widehat{\beta}_m$ is negligible at the root-n sampling scale. Under the conditions developed later, this makes $\mathcal{T}_n(b)$ asymptotically equivalent to a direct-moment statistic and supports testing, power analysis, and confidence sets obtained by inversion. The essential contrast is one of purpose: early stopping regularises estimation, whereas inference requires the algorithmic residual to be asymptotically negligible.

The simulation programme has two connected components. The first is a reproducible Monte Carlo comparison of the four regularised estimators across three functional linear models. Performance is evaluated through integrated squared error for the slope, mean squared prediction error, selected complexity, conditioning, and complete error distributions. Particular attention is paid to rare numerical failures, since averages or medians alone can conceal the practical consequences of an unstable Krylov basis. The second component studies CG–FPLS inference through null calibration, empirical size, power, confidence-set coverage, and convergence of the finite-iteration statistic towards its direct-moment limit. These controlled studies are complemented by a concise spectroscopy application, which asks whether the statistical and numerical trade-offs isolated in simulation remain visible with observed functional predictors. Chapter 8 reports the principal empirical evidence; Appendix D records the complete protocol, diagnostics, and sensitivity analysis.

The contribution is therefore methodological, computational, and expository rather than a new asymptotic theory. The thesis brings spectral regularisation, raw and orthogonalised Krylov formulations, direct CG iteration, and CG-based inference into a common, reproducible analysis. Its evidence yields conditional trade-offs rather than a universal method ranking: performance depends jointly on the target geometry, tuning rule, numerical representation, and data structure. The later application provides a first assessment of how those mechanisms transfer beyond the controlled designs.

The argument proceeds in four stages. Chapters 2 and 3 develop the functional-analytic foundation and turn it into four fully specified estimators. Chapters 4 and 5 use controlled changes in the slope and covariance structure to study estimation, prediction, and numerical stability. Chapters 6 and 7 then develop and evaluate the late-stopped inference procedure. Chapter 8 applies the four methods to spectroscopic data and relates the empirical findings to the mechanisms isolated in simulation; Appendix D provides the full supporting analysis. Chapter 9 closes by synthesising the results, limitations, and implications for further work.

Chapter 2

Literature and Functional Regression Background

The scalar-on-function linear model is easy to state but cannot be understood fully through finite-dimensional regression intuition alone. Its regressor is a random element of a function space, its covariance is a linear operator rather than a matrix, and estimation of the slope requires the inversion of that operator. This chapter introduces the mathematical objects needed to formulate the problem precisely and places the regularisation methods studied in this thesis within the functional regression literature. The presentation draws on the functional-analysis material developed in the MIT 18.102 lecture notes (Lin 2021), the operator-theoretic treatment of Hsing and Eubank (2015), and the statistical accounts of Ramsay and Silverman (2005), Horváth and Kokoszka (2012), and Kokoszka and Reimherr (2017).

Definitions and proofs of the foundational results used below are collected in Appendix A. Keeping the arguments there makes the assumptions and derivations available for reference while allowing this chapter to retain a continuous statistical narrative.

2.1 Functional observations from discrete measurements

In an ordinary multivariate data set, the observational unit is a vector with a fixed number of coordinates. In functional data analysis, the observational unit is a curve or another structured function. The available data are still recorded at finitely many points, but those measurements are regarded as observations of an underlying object $X(t)$ defined over a continuum $t \in \mathcal{I}$. This viewpoint is appropriate when the order and proximity of the measurement locations have substantive meaning and when values at neighbouring locations are related. Daily demand profiles, growth trajectories, spectra, and biomedical signals are typical examples (Ramsay and Silverman 2005; Kokoszka and Reimherr 2017).

Suppose that the ith curve is observed at grid points $t_1, \ldots, t_T$. A functional representation may be obtained directly from a dense grid, by interpolation or smoothing, or through a finite basis expansion

$$X_i(t) \approx \sum_{j=1}^{J} x_{ij}\phi_j(t), \tag{2.1}$$

where $\phi_1, \ldots, \phi_J$ may be Fourier, spline, wavelet, or other basis functions. The coefficient vector in (2.1) is a computational representation of the curve, not a change in the conceptual model: the statistical object remains a function. Increasing the grid size or the number of basis functions allows progressively richer variation and therefore exposes the infinite-dimensional character of the problem.

Functional observations may be densely and regularly sampled, sparsely sampled, or contaminated by measurement error. These regimes require different preliminary reconstruction methods. The simulations in this thesis use densely evaluated curves generated from a known basis, so reconstruction uncertainty is deliberately separated from regularisation of the regression slope. This controlled setting allows the comparison to focus on the inverse problem and on the numerical behaviour of the four estimators. The spectroscopy application in Chapter 8 will state the preprocessing choices needed to interpret its principal findings, while Appendix D will document them in full, since smoothing, registration, and scaling can form an additional layer of regularisation with observed curves.

Once a curve has been reconstructed, its statistical analysis requires a geometry in which distances, projections, and linear functionals are well defined. The square-integrable function space introduced next supplies that geometry.

2.2 Hilbert-space foundations

2.2.1 The space of square-integrable functions

Let $\mathcal{I} = [0, 1]$ and consider real-valued functions on this interval. The interval can be replaced by any compact interval through an affine transformation. The natural state space for the present model is

$$\mathcal{H} = L^2(\mathcal{I}) = \left\{ f : \mathcal{I} \to \mathbb{R} : \int_{\mathcal{I}} f(t)^2 \, \mathrm{d}t < \infty \right\}, \tag{2.2}$$

where functions that differ only on a set of Lebesgue measure zero are identified. The space is equipped with the inner product

$$\langle f, g \rangle = \int_{\mathcal{I}} f(t) g(t) \, \mathrm{d}t \tag{2.3}$$

and induced norm $\|f\| = \langle f, f \rangle^{1/2}$. The Cauchy–Schwarz inequality,

$$|\langle f, g \rangle| \leq \|f\| \, \|g\|, \tag{2.4}$$

ensures, among other things, that the regression functional $f \mapsto \langle f, \beta \rangle$ is continuous whenever $\beta \in L^2(\mathcal{I})$.

Definition 2.1 (Hilbert space)**.** An inner-product space is a *Hilbert space* if it is complete under the norm induced by its inner product. Completeness means that every Cauchy sequence has a limit in the space.

The space $L^2(\mathcal{I})$ is a separable Hilbert space. Completeness makes limits of approximating sequences mathematically well defined, while separability guarantees the existence of a finite or countable orthonormal basis. Both properties are essential in functional statistics. They permit functions to be approximated by finite expansions without reducing the population model itself to a fixed finite dimension (Lin 2021; Hsing and Eubank 2015).

The use of L^2 is also matched to the regression model. The quantity $\langle X, \beta \rangle$ is the integrated contribution of the predictor curve to the response, and $\|\widehat{\beta} - \beta\|^2$ is the integrated squared error used later to evaluate slope recovery. Other function spaces may encode smoothness or pointwise behaviour more directly, but L^2 supplies the geometry in which covariance operators, principal components, and the CG–FPLS moment equation have particularly convenient forms.

2.2.2 Orthogonality, bases, and projections

A sequence $\{e_j\}_{j \geq 1} \subset \mathcal{H}$ is orthonormal if $\langle e_j, e_k \rangle = 0$ for $j \neq k$ and $\|e_j\| = 1$. It is an orthonormal basis if its closed linear span equals $\mathcal{H}$. For such a basis, every $f \in \mathcal{H}$ has the expansion

$$f = \sum_{j=1}^{\infty} \langle f, e_j \rangle e_j, \tag{2.5}$$

where convergence is in the L^2 norm. Parseval's identity gives

$$\|f\|^2 = \sum_{j=1}^{\infty} |\langle f, e_j \rangle|^2. \tag{2.6}$$

Thus, the coefficient sequence contains the same norm information as the function itself.

For the finite-dimensional subspace $V_m = \operatorname{span}\{e_1, \ldots, e_m\}$, the orthogonal projection is

$$P_m f = \sum_{j=1}^{m} \langle f, e_j \rangle e_j. \tag{2.7}$$

It satisfies the best-approximation property

$$\|f - P_m f\| = \inf_{v \in V_m} \|f - v\|. \tag{2.8}$$

This basic geometric result appears repeatedly in functional regression. A predetermined basis approximation restricts the slope to a chosen subspace; FPCR projects onto a subspace estimated from the covariance operator; and Arnoldi FPLS constructs an orthonormal basis for a response-informed Krylov subspace. The distinction between these methods therefore lies less in whether projection occurs than in how the target subspace is selected.

The Riesz representation theorem provides a second connection between Hilbert-space geometry and regression. Every continuous linear functional $\ell : \mathcal{H} \to \mathbb{R}$ can be written uniquely as $\ell(f) = \langle f, b \rangle$ for some $b \in \mathcal{H}$. Consequently, assuming a linear and continuous effect of a functional predictor naturally leads to an inner-product model with slope β. This representation is the infinite-dimensional analogue of expressing a linear functional on $\mathbb{R}^p$ as multiplication by a coefficient vector (Lin 2021; Hsing and Eubank 2015).

Figure 2.1 places this Hilbert-space geometry within the broader family of ℓ^p norms. In the two-dimensional coordinate space $\mathbb{R}^2$, write

$$\|x\|_p = (|x_1|^p + |x_2|^p)^{1/p}, \quad 1 \le p < \infty, \qquad \|x\|_\infty = \max\{|x_1|, |x_2|\}.$$

The corresponding metric is $d_p(x, z) = \|x - z\|_p$, so a unit-ball boundary is also a unit-distance contour about the origin; translating and rescaling it gives the contour about any other centre. The diamond, circle, and square in the left panel therefore show both different unit balls and different ways of measuring the same displacement. These are finite-dimensional sections used for visualisation. Although all norms on $\mathbb{R}^2$ are equivalent, the infinite-dimensional sequence spaces ℓ^p are genuinely different.

More formally, if $1 \le p < \infty$ and q satisfies $p^{-1} + q^{-1} = 1$, then $(\ell^p)^*$ is isometrically ℓ^q and

$$\|y\|_q = \sup_{\|x\|_p \le 1} \left| \sum_{j\ge 1} x_j y_j \right|.$$

The case $p = q = 2$ is the self-dual member of this conjugate family. For an orthonormal basis, Parseval's identity identifies L^2 functions with their ℓ^2 coefficient sequences without changing distances, while Riesz identifies continuous linear effects with elements of the same space. This common geometry is particularly well matched to orthogonal projections, covariance operators, and the conjugate-gradient constructions used later in the thesis. It does not assert that L^2 is universally preferable to every other function space: in particular, the prediction criterion later uses a covariance-weighted seminorm. The full sequence-space statement and its finite-dimensional polar interpretation are proved in Theorem A.13; Remark A.14 records the exceptional endpoint behaviour of $(\ell^\infty)^*$ (Lin 2021).

2.3 Random functions and covariance operators

2.3.1 Random elements in L^2

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A functional observation is represented by a random element $X : \Omega \to \mathcal{H}$. Assume throughout that

$$\mathbb{E}\|X\|^2 < \infty. \tag{2.9}$$

Its mean $\mu_X \in \mathcal{H}$ is defined by the relation

$$\langle \mu_X, f\rangle = \mathbb{E}\langle X, f\rangle, \qquad f \in \mathcal{H}. \tag{2.10}$$

This definition does not require pointwise expectations to exist at every t; it characterises the mean as an element of the Hilbert space. Write $X^c = X - \mu_X$ for the centred predictor.

For $u, v \in \mathcal{H}$, define the rank-one operator $u \otimes v$ by

$$(u \otimes v)f = \langle v, f\rangle u. \tag{2.11}$$

The covariance operator of X is

$$K = \mathbb{E}(X^c \otimes X^c), \qquad Kf = \mathbb{E}[\langle X^c, f\rangle X^c]. \tag{2.12}$$

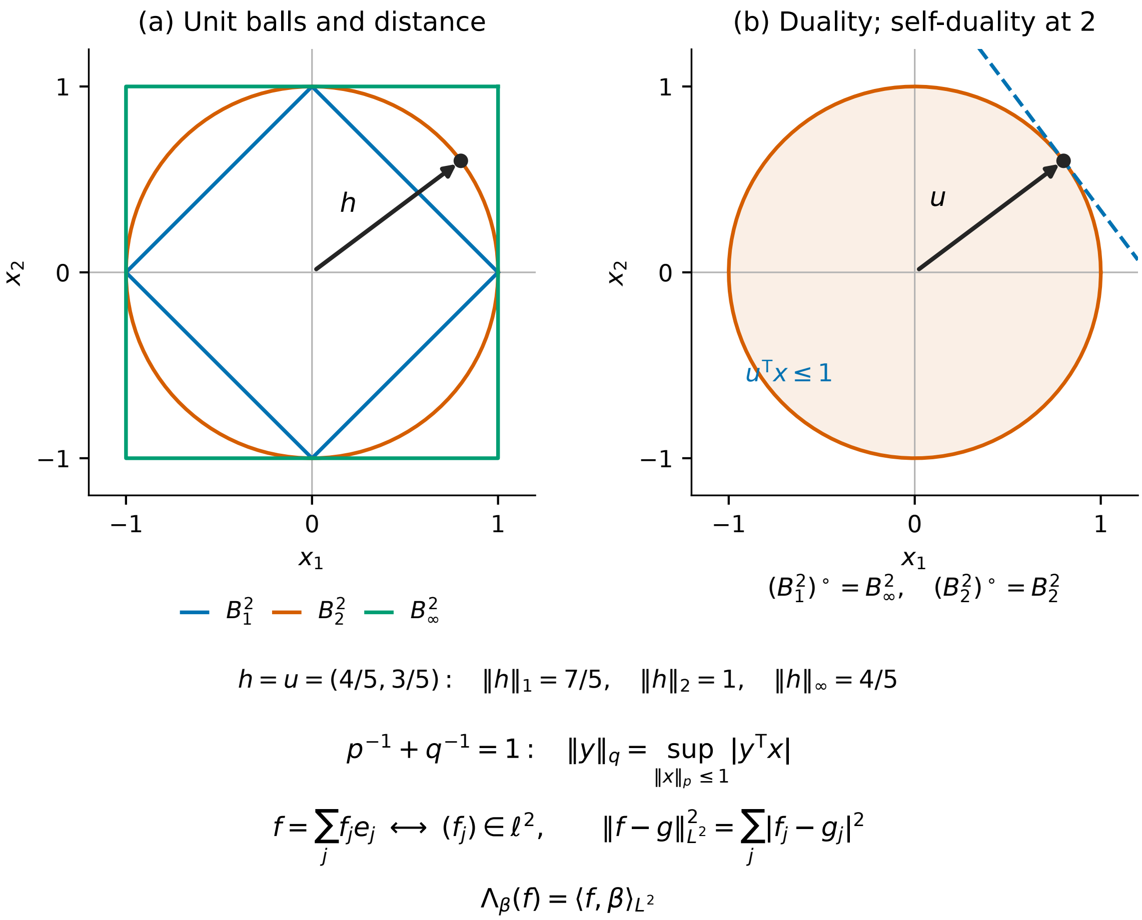


Figure 2.1: Unit-ball geometry, norm-induced distance, and duality. The left panel plots $B_p^2 = \{x \in \mathbb{R}^2 : \|x\|_p \le 1\}$ for $p = 1, 2, \infty$ and assigns three different lengths to the same displacement $h = (4/5, 3/5)$. In the right panel, q is the Hölder-conjugate exponent and $(B_p^2)^\circ = B_q^2$ under the standard finite-dimensional pairing. The dashed line $u^{\mathsf{T}}x = 1$ supports the Euclidean ball at $u = (4/5, 3/5)$; the same u is both the boundary point and its norming vector, illustrating $(B_2^2)^\circ = B_2^2$. The lower strip records how Parseval's identity and Riesz representation transfer this self-dual coefficient geometry to $L^2(\mathcal{I})$. The plotted balls are two-dimensional analogues, not depictions of infinite-dimensional sets.

If a jointly measurable representative of X^c has pointwise second moments, define

$$c(s,t) = \mathrm{Cov}\{X(s), X(t)\}.$$

Provided expectation and integration may be interchanged—for example, if, for the function f under consideration,

$$\int_{\mathcal{I}} \mathbb{E}|X^c(s)X^c(t)|\,|f(t)|\,\mathrm{d}t < \infty \qquad \text{for almost every } s,$$

the same operator can be written as

$$(Kf)(s) = \int_{\mathcal{I}} c(s,t) f(t)\,\mathrm{d}t. \tag{2.13}$$

The qualification is important because an element of L^2 is an equivalence class rather than a pointwise-defined curve. The operator formulation is therefore preferable: it remains meaningful without choosing pointwise values or imposing sample-path smoothness.

2.3.2 Properties and spectral representation

The covariance operator is bounded, self-adjoint, and non-negative:

$$\langle Kf, g\rangle = \langle f, Kg\rangle, \qquad \langle Kf, f\rangle = \mathbb{E}\langle X^c, f\rangle^2 \geq 0. \tag{2.14}$$

Under (2.9), it is also trace class and hence compact, with

$$\mathrm{tr}(K) = \mathbb{E}\|X^c\|^2 < \infty. \tag{2.15}$$

These facts place K within the compact self-adjoint operator theory covered by the spectral theorem (Lin 2021; Hsing and Eubank 2015; Horváth and Kokoszka 2012).

Proposition 2.2 (Spectral representation of the covariance operator)**.** *Let $\mathcal{J}_K$ be either a possibly empty finite index set $\{1,\dots,q\}$, with $q \geq 0$, or $\mathbb{N}$. There exist positive eigenvalues $\{\lambda_j : j \in \mathcal{J}_K\}$ and orthonormal eigenfunctions $\{v_j : j \in \mathcal{J}_K\}$ spanning* $\overline{\mathrm{Range}(K)}$ *such that*

$$Kf = \sum_{j \in \mathcal{J}_K} \lambda_j \langle f, v_j\rangle v_j, \qquad f \in \mathcal{H}, \tag{2.16}$$

where the eigenvalues are ordered non-increasingly. If $\mathcal{J}_K = \mathbb{N}$, then $\lambda_j \to 0$; on $\mathrm{Null}(K)$ *the operator is zero.*

The eigenfunctions define the functional principal component directions. With scores

$\xi_j = \langle X^c, v_j \rangle$, the Karhunen–Loève representation is

$$X = \mu_X + \sum_{j \in \mathcal{J}_K} \xi_j v_j, \tag{2.17}$$

with convergence in mean square. The scores are uncorrelated, $\mathbb{E}\xi_j = 0$, $\mathbb{E}(\xi_j \xi_k) = 0$ for $j \neq k$, and $\mathbb{E}\xi_j^2 = \lambda_j$. When at least m positive eigenvalues exist, the first m eigenfunctions span the m-dimensional subspace that minimises the expected squared reconstruction error of X among all m-dimensional subspaces. This optimality concerns reconstruction of the predictor; it does not by itself guarantee that the same directions are the most informative for a response variable.

Given independent observations $X_1, \ldots, X_n$, the empirical mean and covariance operator are

$$\overline{X} = \frac{1}{n} \sum_{i=1}^{n} X_i, \tag{2.18}$$

$$\widehat{K} f = \frac{1}{n} \sum_{i=1}^{n} \langle X_i - \overline{X}, f \rangle (X_i - \overline{X}). \tag{2.19}$$

Even though $\widehat{K}$ acts on an infinite-dimensional space, its rank is at most $n-1$. It is therefore non-invertible on $\mathcal{H}$. A basis or grid representation replaces it by a finite matrix, but this does not remove the underlying difficulty: the resulting matrix is singular or increasingly ill-conditioned as more directions are admitted.

2.4 The scalar-on-function linear model

Let (Y, X) be a scalar–function pair with $\mathbb{E}(Y^2) < \infty$ and $X \in \mathcal{H}$. The scalar-on-function linear model is

$$Y = \alpha + \langle X, \beta \rangle + \varepsilon, \qquad \mathbb{E}(\varepsilon \mid X) = 0, \tag{2.20}$$

where $\beta \in \mathcal{H}$ is the slope function. If $\mu_Y = \mathbb{E}Y$, then

$$\alpha = \mu_Y - \langle \mu_X, \beta \rangle. \tag{2.21}$$

Consequently, centring gives

$$Y^c = Y - \mu_Y = \langle X^c, \beta \rangle + \varepsilon. \tag{2.22}$$

All subsequent derivations can therefore be carried out for centred variables without an intercept.

Define the cross-covariance element

$$r = \mathbb{E}(Y^c X^c) \in \mathcal{H}. \tag{2.23}$$

For any $f \in \mathcal{H}$, the conditional mean restriction implies

$$\begin{aligned} \langle r, f\rangle &= \mathbb{E}[Y^c \langle X^c, f\rangle] && (2.24)\\ &= \mathbb{E}[\langle X^c, \beta\rangle\langle X^c, f\rangle] = \langle K\beta, f\rangle. && (2.25) \end{aligned}$$

Since this holds for every f, the population normal equation is

$$K\beta = r. \tag{2.26}$$

This is the operator counterpart of the finite-dimensional normal equations: the covariance matrix and cross-covariance vector are replaced by K and r, respectively.

The same equation follows from least squares. For a candidate $b \in \mathcal{H}$, let

$$R(b) = \mathbb{E}\left(Y^c - \langle X^c, b\rangle\right)^2. \tag{2.27}$$

The error orthogonality condition gives the risk identity

$$R(b) - R(\beta) = \mathbb{E}\langle X^c, b-\beta\rangle^2 = \langle K(b-\beta), b-\beta\rangle = \left\|K^{1/2}(b-\beta)\right\|^2. \tag{2.28}$$

Thus, prediction error and slope estimation error use different geometries. Integrated squared error is $\|b-\beta\|^2$, whereas excess prediction risk is the weaker covariance-weighted seminorm $\|K^{1/2}(b-\beta)\|^2$ on $\mathcal{H}$. It becomes a norm after the parameter space is restricted to $\mathrm{Null}(K)^{\perp}$. Directions with small eigenvalues contribute little to prediction but may contribute substantially to integrated slope error. The distinction is central to the theoretical results of Cai and Hall (2006) and motivates reporting both criteria in the simulation study.

The empirical counterparts of r and the normal equation are

$$\begin{aligned} \widehat{r} &= \frac{1}{n}\sum_{i=1}^{n}(Y_i - \overline{Y})(X_i - \overline{X}), && (2.29)\\ \widehat{K}\widehat{\beta} &= \widehat{r}. && (2.30) \end{aligned}$$

Equation (2.30) is the common starting point for all four estimators. Before regularising it, however, one must separate which part of the slope is identified from whether that identified part can be recovered stably.

2.5 Identification and ill-posedness

2.5.1 Identification of the slope

For two slopes β and $\widetilde{\beta}$, put $h = \beta - \widetilde{\beta}$. They produce the same conditional mean if and only if

$$h \in \mathrm{Null}(K). \tag{2.31}$$

Indeed, if $h \in \mathrm{Null}(K)$, then

$$\mathbb{E}\langle X^c, h\rangle^2 = \langle Kh, h\rangle = 0, \tag{2.32}$$

and hence $\langle X^c, h\rangle = 0$ almost surely. Conversely, equality of the conditional means implies this almost-sure equality and therefore $Kh = \mathbb{E}\{\langle X^c, h\rangle X^c\} = 0$. Thus the observationally identified parameter is the equivalence class $[\beta] = \beta + \mathrm{Null}(K)$. Equivalently, one may work on the canonical parameter space

$$\mathcal{H}_{\mathrm{id}} = \mathrm{Null}(K)^{\perp} = \overline{\mathrm{Range}(K)}, \tag{2.33}$$

where the conditional-mean map is injective. The identified representative is the unique minimum-norm element of $[\beta]$; it is denoted by $\beta^{\dagger}$. Full identification on all of $\mathcal{H}$ occurs if and only if $\mathrm{Null}(K) = \{0\}$. When the distinction is not material, later chapters write β for the canonical representative $\beta^{\dagger}$.

Expand $\beta^{\dagger}$ and r in the positive-eigenvalue system of K:

$$\beta^{\dagger} = \sum_{j \in \mathcal{J}_K} \beta_j v_j, \qquad r = \sum_{j \in \mathcal{J}_K} r_j v_j. \tag{2.34}$$

For positive eigenvalues, (2.26) implies

$$r_j = \lambda_j \beta_j, \qquad \beta_j = \frac{r_j}{\lambda_j}. \tag{2.35}$$

Together with $r \perp \mathrm{Null}(K)$, a square-integrable solution exists if and only if the coefficients satisfy the Picard condition

$$\sum_{j \in \mathcal{J}_K} \frac{r_j^2}{\lambda_j^2} < \infty. \tag{2.36}$$

Under the correctly specified model, $r = K\beta$ ensures this population condition for the true square-integrable slope. It does not ensure that replacing r and K by noisy empirical quantities yields a stable estimator.

2.5.2 Why inversion is unstable

Compactness is the source of the inverse problem. If K has infinite rank, its positive eigenvalues converge to zero. The appropriate inverse for the identified parameter is the Moore–Penrose inverse. On its natural domain it is

$$K^\dagger f = \sum_{j \in \mathcal{J}_K} \frac{\langle f, v_j \rangle}{\lambda_j} v_j, \qquad \mathcal{D}(K^\dagger) = \left\{ f \in \mathcal{H} : \sum_{j \in \mathcal{J}_K} \frac{|\langle f, v_j \rangle|^2}{\lambda_j^2} < \infty \right\}, \tag{2.37}$$

with $K^\dagger f = 0$ for $f \in \mathrm{Null}(K)$. The Picard condition ensures that $r \in \mathcal{D}(K^\dagger)$ and the canonical slope is $\beta^\dagger = K^\dagger r$. The factors $1/\lambda_j$ are nevertheless unbounded. To see the consequence directly, perturb the right-hand side of the normal equation by $a_j v_j$. Although the perturbation has norm $|a_j|$, its effect after inversion has norm $|a_j|/\lambda_j$. For high-index directions, an arbitrarily small error in r can therefore generate a large error in the recovered slope.

Sampling error affects both $\widehat{r}$ and $\widehat{K}$. Empirical eigenvalues fluctuate, empirical eigenfunctions are estimated imperfectly, and components of $\widehat{r}$ along low-variance directions can be dominated by noise. Simply increasing the basis dimension does not solve these problems. It permits more small-eigenvalue directions to enter the calculation and may increase variance and numerical conditioning problems. Functional linear regression is consequently ill posed in the sense that the solution does not depend continuously on the observed moments (Engl et al. 1996; Hall and Horowitz 2007).

Ill-posedness should be distinguished from non-identification. An injective compact covariance operator can identify every coefficient of β at the population level while still possessing an unbounded inverse. Identification states that two different slopes imply different population conditional means. Stability asks whether a small change in the moments produces only a small change in the recovered slope. Regularisation is needed for the second property even when the first holds.

There is also a finite-dimensional numerical manifestation. After discretisation in a J-dimensional basis, the covariance operator becomes a positive semidefinite matrix. Its condition number is the ratio between its largest and smallest retained eigenvalues. As J grows, the smallest eigenvalues typically become very small and the condition number becomes large. Rounding error and statistical noise can then be amplified by a direct linear solve. This link explains why the theoretical inverse problem and the numerical conditioning diagnostics in the simulation study describe the same underlying geometry at different levels.

2.6 Regularisation principles in functional regression

Regularisation replaces the unstable inverse by a family of stable approximate inverses. The amount of regularisation controls a bias–variance trade-off. Strong regularisation suppresses poorly identified directions and introduces bias; weak regularisation admits more directions and increases sensitivity to noise. A tuning parameter, truncation level, or stopping index determines the operating point.

For orientation, suppose that $\widehat{K}$ has eigenpairs $(\widehat{\lambda}_j, \widehat{v}_j)$. A broad class of spectral estimators has the form

$$\widehat{\beta}_\eta = \sum_{j:\widehat{\lambda}_j>0} g_\eta(\widehat{\lambda}_j)\langle \widehat{r}, \widehat{v}_j\rangle \widehat{v}_j, \tag{2.38}$$

where g_η is a bounded approximation to $\lambda \mapsto 1/\lambda$ on the positive spectrum and is taken to satisfy $g_\eta(0) = 0$. Different choices of g_η lead to different regularisation schemes. Krylov estimators need not be computed through the empirical eigendecomposition, but they can still be understood as iteration-dependent polynomial approximations to this inverse.

Table 2.1: Regularisation principles relevant to functional linear regression.

Method family	Restricted or modified object	Main tuning quantity	Statistical interpretation
Basis and roughness penalty	Predetermined basis and a penalty on size or derivatives	Basis dimension and penalty weight	Encodes structural smoothness and stabilises correlated coefficients
FPCR	Leading empirical covariance eigenfunctions	Number of components	Removes directions explaining little predictor variation
Tikhonov regularisation	Shifted operator $\widehat{K} + \rho I$	Penalty ρ	Continuously shrinks small-eigenvalue directions
FPLS and CG	Response-informed Krylov subspace	Component count or stopping rule	Uses iteration as regularisation and prioritises directions linked to the response

2.6.1 Basis penalties and Tikhonov regularisation

One approach restricts β to a predetermined finite basis and estimates its coefficients by penalised least squares. When L is an unbounded operator such as differentiation, the optimisation is understood over $b \in \mathrm{Dom}(L)$, or over a finite differentiable basis contained in that domain. A typical criterion is

$$\frac{1}{n}\sum_{i=1}^{n}\left\{Y_i - \overline{Y} - \langle X_i - \overline{X}, b\rangle\right\}^2 + \rho\|Lb\|^2, \tag{2.39}$$

where L is often a derivative operator and $\rho > 0$ controls smoothness. When $L = I$, the population analogue yields the Tikhonov solution $(K + \rho I)^{-1}r$, whose filter on the positive spectrum is $g_\rho(\lambda) = (\lambda + \rho)^{-1}$. We retain the convention $g_\rho(0) = 0$ in (2.38). This gives the same solution because $r \perp \mathrm{Null}(K)$ under the model; likewise, $\widehat{r} \perp \mathrm{Null}(\widehat{K})$ empirically. Every positive eigendirection is retained, with stronger shrinkage where the eigenvalue is small. Penalised basis and spline estimators for functional linear regression were developed, among others, by Cardot et al. (2003) and Crambes et al. (2009). Although these estimators are not part of the four-method Monte Carlo comparison, they clarify that regularisation can encode smoothness directly rather than through covariance components.

2.6.2 Spectral cut-off and FPCR

Functional principal component regression uses the truncated inverse

$$\widehat{\beta}_m^{\mathrm{FPCR}} = \sum_{j=1}^{m}\frac{\langle \widehat{r}, \widehat{v}_j\rangle}{\widehat{\lambda}_j}\widehat{v}_j. \tag{2.40}$$

The cut-off prevents division by empirical eigenvalues beyond the retained rank. FPCR is attractive because it is transparent, computationally convenient, and tied directly to the Karhunen–Loève representation. Its main limitation is that $\widehat{v}_j$ and their order are determined only by the marginal variation of X. A direction with modest predictor variance but a large slope coefficient may be important for the response and nevertheless enter late in the principal component ordering.

Cardot et al. (1999) developed functional principal-component estimation, while their later spline construction combines basis expansion with a roughness penalty (Cardot et al. 2003). Hall and Horowitz (2007) analysed convergence rates for principal-component and ridge-type regularisation under eigenvalue and smoothness conditions. Cai and Hall (2006) showed why estimating the slope and predicting a response can have substantially different difficulty, a distinction already visible in (2.28). FPCR therefore provides both a canonical estimator and a benchmark against which response-informed regularisation can be assessed.

2.6.3 Response-informed Krylov regularisation

Partial least squares selects directions using both X and Y. In functional form, the relevant population Krylov space is

$$\mathcal{K}_m(K, r) = \operatorname{span}\{r, Kr, \ldots, K^{m-1}r\}. \tag{2.41}$$

An m-component FPLS solution has the polynomial form

$$\beta_m = p_{m-1}(K)r \tag{2.42}$$

for a polynomial p_{m-1} of degree at most $m-1$. Its moment residual is

$$r - K\beta_m = \pi_m(K)r, \qquad \pi_m(\lambda) = 1 - \lambda p_{m-1}(\lambda). \tag{2.43}$$

The iteration constructs an approximation to the inverse only along directions represented in the cross-covariance r. This explains the supervised nature of FPLS and why it can require fewer components than FPCR in some prediction problems.

Preda and Saporta (2005) formulated PLS regression for stochastic-process predictors. Delaigle and Hall (2012) developed an alternative functional PLS representation using successive powers of the covariance operator and established its statistical properties. In finite-dimensional PLS, the connection with Lanczos and conjugate-gradient algorithms was studied by Phatak and Hoog (2002). More recently, Babii et al. (2026) formulated functional PLS explicitly as a CG regularisation method, developed a data-driven early-stopping rule, and derived estimation and inference theory.

The common organising object is the Krylov space, but not every algorithm defines the same finite-step estimate. Raw FPLS uses the displayed power sequence directly, while Arnoldi replaces it by an orthonormal basis of the same space. Under the usual full-rank conditions, classical score-and-deflation FPLS, often implemented by a NIPALS-type algorithm, yields the same restricted response-space fit at a common component count.

Babii's CG–FPLS recurrence also explores these nested spaces without forming the complete power basis, but it minimises the empirical moment residual rather than the response sum of squares. It is therefore closely related to, but not generally identical with, the finite-step NIPALS/APLS estimator. Even among formulations with the same exact-arithmetic target, the coordinates matter numerically: repeated covariance powers can approach an active dominant eigenspace and become nearly dependent. Because the objectives and coordinates differ, the rule that selects the iteration or component count is part of the procedure rather than an interchangeable afterthought.

2.6.4 Tuning and early stopping

In every method in Table 2.1, tuning determines how much information from weak directions enters the estimator. For FPCR and component-based FPLS, m is often selected by cross-validation or a generalised cross-validation criterion. For iterative inverse methods, a discrepancy rule can stop when the residual is comparable with an estimate of the sampling noise. The precise rule matters: allowing m to increase without control ultimately moves the estimator towards the unstable empirical inverse.

The statistical role of early stopping parallels spectral shrinkage. Initially, the iterative estimator tends to recover directions associated with larger effective eigenvalues. Later iterations fit increasingly delicate features of the sample moment equation. Stopping before this occurs introduces regularisation bias but limits variance. This is the sense in which CG is not merely a faster linear-system solver in the present thesis; its iteration path is a family of regularised estimators.

2.7 Estimation, prediction, and inference

The literature reviewed above shows that the quality of a regularisation method depends on the target. With β now denoting the canonical representative, slope estimation evaluates $\|\widehat{\beta} - \beta\|$, whereas prediction evaluates the covariance-weighted seminorm $\|K^{1/2}(\widehat{\beta} - \beta)\|$. Inference introduces a third objective: the algorithmic approximation error must be negligible at the scale of the sampling distribution. These objectives do not generally select the same amount of regularisation.

For the hypothesis $H_0 : \beta = b$, with $b \in \mathcal{H}_{\mathrm{id}}$, Babii et al. (2026) base inference on the covariance-weighted difference appearing in $\mathcal{T}_n(b)$ from (1.5):

$$\widehat{K}(\widehat{\beta}_m - b) = (\widehat{r} - \widehat{K}b) - (\widehat{r} - \widehat{K}\widehat{\beta}_m). \tag{2.44}$$

The first term is the sample moment under the null; the second is the residual left by stopping CG after m iterations. Early stopping may minimise an estimation or prediction criterion while leaving the second term too large for valid inference. Under independent and identically distributed sampling, $\mathbb{E}(\varepsilon X^c) = 0$, $\mathbb{E}\|\varepsilon X^c\|^2 < \infty$, and the standard measurability conditions for the Hilbert-space central limit theorem,

$$\sqrt{n}\,(\widehat{r} - \widehat{K}b) \rightsquigarrow G \qquad \text{under } H_0, \tag{2.45}$$

where G is a centred Gaussian element. Sample centring contributes only a lower-order term under the corresponding second-moment conditions. To ensure that the algorithmic residual does not alter this limit, the inferential requirement is

$$\sqrt{n}\,\|\widehat{r} - \widehat{K}\widehat{\beta}_m\| \xrightarrow{\mathbb{P}} 0. \tag{2.46}$$

This generally calls for later stopping than estimation or prediction. Under (2.46), the finite-iteration statistic has the same first-order limit as the direct-moment statistic $n\|\widehat{r} - \widehat{K}b\|^2$, and the spectral representation of G leads to a weighted chi-square law. Chapter 6 states the full testing and calibration argument; Chapter 7 then evaluates its finite-sample behaviour.

This separation also clarifies why late stopping does not claim to improve the slope estimate. The inferential statistic depends on $\widehat{K}\widehat{\beta}_m$, for which the weakly identified slope directions are attenuated by $\widehat{K}$. Late iteration is used to solve the sample moment equation accurately, not to recover β accurately in the unweighted L^2 norm. These target-dependent requirements have so far been expressed at the operator level. Their implementation introduces approximation and floating-point effects in addition to sampling error, motivating the distinction considered next.

2.8 Discretisation and the population–computation distinction

Theoretical objects in this chapter live in $L^2(\mathcal{I})$, whereas computation uses vectors and matrices. Let $t_1, \ldots, t_T$ be a numerical grid with quadrature weights $w_1, \ldots, w_T$. Then

$$\langle f, g\rangle \approx \sum_{\ell=1}^{T} w_\ell f(t_\ell) g(t_\ell). \tag{2.47}$$

Equivalently, a J-term basis representation yields a finite coefficient vector and a basis Gram matrix. Both routes approximate the Hilbert-space geometry; neither changes the population estimand.

Three errors should therefore be kept conceptually separate. Approximation error arises from representing a function on a finite grid or basis. Statistical error arises because K and r are estimated from a finite sample. Numerical error arises from floating-point calculations, particularly when a matrix or basis is ill-conditioned. The simulation design uses a sufficiently rich common basis and grid for all methods, so comparisons are not driven by method-specific discretisations. The raw-versus-Arnoldi comparison then isolates numerical stability within the same theoretical Krylov space, while the comparison across component counts describes statistical regularisation.

This distinction is also important for interpreting a successful matrix computation. A finite matrix may be invertible even though it approximates an operator with an unbounded inverse. Conversely, a stable population regularisation principle may be implemented poorly if the chosen coordinates are nearly dependent. A defensible functional regression analysis must therefore align the population theory, statistical tuning, and numerical linear algebra.

2.9 Position of the thesis in the literature

The functional linear model and its principal-component estimators are well established (Cardot et al. 1999; Hall and Horowitz 2007). Penalised and spline approaches provide alternative ways to encode smoothness (Crambes et al. 2009), while the different statistical difficulty of slope recovery and prediction has been studied explicitly (Cai and Hall 2006). The FPLS literature introduced response-guided dimension reduction and later supplied a Krylov representation and convergence theory (Preda and Saporta 2005; Delaigle and Hall 2012). Comparative evidence indicates that neither FPCR nor FPLS can dominate independently of the covariance spectrum, slope alignment, and tuning design (Febrero-Bande et al. 2017). The question pursued here lies where those statistical principles meet finite-precision computation: a Krylov subspace can remain mathematically well defined even when its coordinates have become numerically unusable.

The study isolates that interaction through controlled contrasts. FPCR changes how the search space is selected; raw and Arnoldi FPLS hold the Krylov space and response objective fixed while changing its coordinates and linear solver; and CG–FPLS retains the nested spaces but changes the finite-step objective and stopping rule. The separate inference study then asks how far the moment-residual Krylov path must be continued when the target changes from regularised estimation to testing. With the operator, identification, and target geometries now in place, Chapter 3 turns these contrasts into four fully specified estimators.

Chapter 3

FPLS, Krylov Methods, and Regularisation

Chapter 2 showed that functional linear regression leads to a compact-operator equation whose direct inverse is unstable. This chapter develops the four regularised estimators used in the Monte Carlo study: FPCR, raw FPLS, Arnoldi FPLS, and the CG–FPLS procedure of Babii et al. (2026). Particular attention is paid to a terminological point that otherwise causes confusion. Classical functional PLS can be computed by a sequential score-and-deflation algorithm of NIPALS type or represented as an alternative PLS (APLS) problem on a Krylov subspace. Raw and Arnoldi FPLS are two numerical implementations of this same response-space APLS estimator. Babii's CG–FPLS explores the same nested Krylov spaces but uses a different finite-iteration objective. The methods are consequently closely related without being interchangeable in every respect.

To align the theoretical comparison with the computation, the definitions below use the same centring, discretisation, rank safeguards, and tuning conventions. Proofs of the Krylov-space identities and variational characterisations are collected in Appendix A.

3.1 The empirical operator problem

Let $X_i^c = X_i - \mu_X$ and $y_i = Y_i - \mu_Y$. In an empirical application the unknown means would normally be replaced by their sample counterparts. The controlled simulation designs impose $\mu_X = \mu_Y = 0$, so the generated curves and responses enter directly and no intercept is estimated. Define the empirical sampling operator $T_n : \mathcal{H} \to \mathbb{R}^n$ and its adjoint by

$$(T_n b)_i = \langle X_i^c, b\rangle, \tag{3.1}$$

$$T_n^* a = \frac{1}{n}\sum_{i=1}^{n} a_i X_i^c, \qquad a \in \mathbb{R}^n, \tag{3.2}$$

where $\langle a, c\rangle_n = n^{-1} a^\top c$ is used in response space. The empirical covariance operator and cross-covariance element are then

$$\widehat{K} = T_n^* T_n, \qquad \widehat{r} = T_n^* y, \tag{3.3}$$

so unregularised least squares would attempt to solve $\widehat{K}b = \widehat{r}$. With population-centred curves, $\widehat{K}$ is generated by n rank-one operators and hence has rank at most n; when the curves are centred by their sample mean, the sharper bound is $n-1$. The simulations use the known zero means, so the former bound applies. In either case, $\widehat{K}$ is finite rank and approximates a compact population operator, so this equation cannot be used as a stable unrestricted inverse. Unless K is injective, all slope estimands below are understood as the minimum-norm representatives in $\mathcal{H}_{\mathrm{id}} = \mathrm{Null}(K)^\perp$; components in $\mathrm{Null}(K)$ are not identified by the data.

The simulation evaluates functions on an equally spaced grid $t_1, \ldots, t_T$ and uses the discrete inner product

$$\langle f, g\rangle_T = \frac{1}{T}\sum_{\ell=1}^{T} f(t_\ell) g(t_\ell) = \frac{1}{T} f^\top g. \tag{3.4}$$

If X denotes the centred $n \times T$ matrix of curves, the operators in (3.3) are represented exactly as

$$\widehat{K} = \frac{X^\top X}{nT}, \qquad \widehat{r} = \frac{X^\top y}{n}, \qquad T_n b = \frac{Xb}{T}. \tag{3.5}$$

These scaling factors matter: omitting T^{-1} would change the numerical operator and the stopping threshold. All four estimators receive the same sample, grid, zero-mean convention, and inner product.

Two residuals will be used repeatedly. The response residual at b is $y - T_n b$, whereas the moment residual is

$$e(b) = \widehat{r} - \widehat{K}b = T_n^*(y - T_n b). \tag{3.6}$$

The latter is the response residual after it has been mapped back to function space. Although the two contain related information, minimising their norms is not the same finite-dimensional optimisation problem.

3.2 Functional principal component regression

FPCR is the spectral benchmark. Let $(\widehat{\lambda}_j, \widehat{v}_j)$ denote the empirical eigenpairs of $\widehat{K}$, ordered by decreasing eigenvalue. For a component count m, the estimator is

$$\widehat{\beta}_m^{\mathrm{FPCR}} = \sum_{j=1}^{m} \frac{\widehat{v}_j^\top \widehat{r}}{\widehat{\lambda}_j} \widehat{v}_j, \tag{3.7}$$

where the coordinate eigenvectors are Euclidean-normalised. Formula (3.7) is the least-squares solution restricted to the first m empirical principal directions. The component count is therefore a hard spectral cut-off: components beyond m are assigned an inverse weight of zero instead of the potentially large value $1/\widehat{\lambda}_j$. In the implementation, an empirical eigenvalue is retained only when it exceeds $\varepsilon_{\mathrm{mach}} T \max_j |\widehat{\lambda}_j|$; this is a numerical rank safeguard, not an additional statistical tuning parameter. The reported path is further capped at $\min\{m_{\max}, n-1, T-1\}$ before GCV, and at the number of numerically positive eigenvalues.

The reported FPCR estimator chooses m by response-space generalised cross-validation (GCV). If $\widehat{y}_m = T_n \widehat{\beta}_m^{\mathrm{FPCR}}$, the criterion is

$$\mathrm{GCV}_Y(m) = \frac{n^{-1}\|y - \widehat{y}_m\|_2^2}{(1 - m/n)^2}. \tag{3.8}$$

The numerator rewards in-sample fit and the denominator penalises the effective dimension. This is response GCV, not a criterion based on the moment residual. Section 3.7 introduces a separate moment-space FPCR pilot used only to estimate the noise scale for CG stopping.

FPCR orders directions by predictor variance. It is therefore expected to be effective when the slope signal is concentrated in leading covariance eigenfunctions. It may be less parsimonious when a response-relevant direction has modest predictor variance, which motivates supervised PLS directions.

3.3 Classical functional PLS and the NIPALS algorithm

Partial least squares originated as a component method for multivariate regression. NIPALS, conventionally expanded as *non-linear iterative partial least squares*, is one of its classical score-and-deflation algorithms (Wold et al. 1984; Geladi and Kowalski 1986). The present model has one scalar response and therefore corresponds to PLS1. Its functional version replaces predictor vectors by functions and ordinary dot products by the L^2 inner product.

To see the construction, start with residual curves $X_i^{(0)} = X_i^c$ and residual responses $y_i^{(0)} = y_i$. At component $h = 1, \ldots, m$, a NIPALS-type FPLS algorithm performs the following steps.

1. Form and normalise the residual cross-covariance direction,

$$c_h = \frac{1}{n} \sum_{i=1}^{n} y_i^{(h-1)} X_i^{(h-1)}, \qquad w_h = \frac{c_h}{\|c_h\|}. \tag{3.9}$$

2. Project the residual predictor curves onto this direction to obtain the score

$$t_{ih} = \langle X_i^{(h-1)}, w_h \rangle. \tag{3.10}$$

3. Regress both residual objects on the score. The functional predictor loading and scalar response loading are

$$p_h = \frac{\sum_i t_{ih} X_i^{(h-1)}}{\sum_i t_{ih}^2}, \qquad \ell_h = \frac{\sum_i t_{ih} y_i^{(h-1)}}{\sum_i t_{ih}^2}. \tag{3.11}$$

4. Deflate the predictor and response,

$$X_i^{(h)} = X_i^{(h-1)} - t_{ih} p_h, \qquad y_i^{(h)} = y_i^{(h-1)} - t_{ih} \ell_h. \tag{3.12}$$

At the first step, $c_1 = \widehat{r}$, so the first PLS weight is the normalised sample cross-covariance. Subsequent weights are computed after deflation and are therefore not simply displayed as raw powers of $\widehat{K}$. If $W_m a = \sum_{h=1}^{m} a_h w_h$ and $P_m a = \sum_{h=1}^{m} a_h p_h$, the corresponding slope can be reconstructed as

$$\widehat{\beta}_m^{\text{NIPALS}} = W_m (P_m^* W_m)^{-1} \ell, \qquad \ell = (\ell_1, \ldots, \ell_m)^\top, \tag{3.13}$$

provided the small loading matrix is nonsingular.

The deflation formulas provide the classical algorithmic view: each score explains the part of the response not captured by earlier scores. Under the usual full-rank conditions, NIPALS and SIMPLS yield the same fixed-m response fit despite using different intermediate coordinates (Jong 1993). The next section identifies that common fit directly as an optimisation over a Krylov subspace. NIPALS is therefore a reference description, not an additional estimator in the simulation.

3.4 APLS and the Krylov representation

Delaigle and Hall (2012) established an alternative functional PLS representation that exposes the operator geometry hidden by sequential deflation. Define the empirical Krylov subspace

$$\widehat{\mathcal{K}}_m = \mathcal{K}_m(\widehat{K}, \widehat{r}) = \operatorname{span}\{\widehat{r}, \widehat{K}\widehat{r}, \dots, \widehat{K}^{m-1}\widehat{r}\}. \tag{3.14}$$

Standard response-space FPLS is characterised by

$$\widehat{\beta}_m^{\mathrm{APLS}} \in \arg\min_{b \in \widehat{\mathcal{K}}_m} \|y - T_n b\|_n^2. \tag{3.15}$$

Its first-order condition is

$$\widehat{r} - \widehat{K}\widehat{\beta}_m^{\mathrm{APLS}} \perp \widehat{\mathcal{K}}_m. \tag{3.16}$$

The NIPALS scores span the image $T_n\widehat{\mathcal{K}}_m$, so the sequential algorithm and (3.15) produce the same fitted responses at a common component count. When the response risk is strictly convex on the restricted space, their slope functions are equal as well. This is the precise similarity the present thesis uses: classical NIPALS-type FPLS and APLS are algorithmic and operator-theoretic descriptions of the same standard PLS1 fit. A proof is given in Proposition A.37.

Every element of (3.14) is a polynomial in the empirical covariance operator applied to the cross-covariance:

$$b = p_{m-1}(\widehat{K})\widehat{r}. \tag{3.17}$$

Unlike FPCR, this space is response-informed because $\widehat{r} = T_n^* y$ determines its starting direction. The iteration does not retain the m largest-variance directions automatically. It constructs a polynomial approximation to the inverse in directions that are visible in the empirical relationship between X and Y.

Let $g_j = \widehat{K}^{j-1}\widehat{r}$ for $j = 1, \dots, m$ and write $b = \sum_{j=1}^{m} a_j g_j$. The normal equations for (3.15) are

$$H_m^Y a = h_m^Y, \tag{3.18}$$

with entries

$$(H_m^Y)_{jk} = \langle \widehat{r}, \widehat{K}^{j+k-1}\widehat{r}\rangle, \qquad (h_m^Y)_j = \langle \widehat{r}, \widehat{K}^{j-1}\widehat{r}\rangle. \tag{3.19}$$

The matrix is Hankel because each entry depends only on $j + k$. The same system is obtained by regressing y on the design columns $T_n g_1, \dots, T_n g_m$. This representation reveals both the close relationship with PLS and the numerical weakness of raw covariance powers.

3.5 Raw FPLS

Raw FPLS implements (3.15) literally. It forms the matrix

$$G_m = [\widehat{r}, \widehat{K}\widehat{r}, \ldots, \widehat{K}^{m-1}\widehat{r}] \tag{3.20}$$

without rescaling or orthogonalising its columns, constructs $Z_m = T_n G_m$, and solves

$$(Z_m^\top Z_m)a = Z_m^\top y, \qquad \widehat{\beta}_m^{\mathrm{raw}} = G_m a. \tag{3.21}$$

In exact arithmetic, this is simply APLS in the power basis. It is included because it is the most direct translation of the theoretical Krylov span and because it exposes a numerical issue that can remain hidden when only final prediction errors are reported.

Repeated multiplication by $\widehat{K}$ progressively emphasises the leading empirical eigendirections. To see this, expand $\widehat{r} = \sum_j \widehat{r}_j \widehat{v}_j$. Then

$$\widehat{K}^k \widehat{r} = \sum_j \widehat{\lambda}_j^k \widehat{r}_j \widehat{v}_j. \tag{3.22}$$

As k increases, ratios $(\widehat{\lambda}_j/\widehat{\lambda}_1)^k$ contract whenever $\widehat{\lambda}_j < \widehat{\lambda}_1$. More precisely, let $\widehat{\lambda}_\star$ be the largest empirical eigenvalue whose eigenspace has a non-zero projection of $\widehat{r}$, and let $P_\star$ denote that projection. Then, in the finite empirical spectrum,

$$\frac{\widehat{K}^k \widehat{r}}{\|\widehat{K}^k \widehat{r}\|} \longrightarrow \frac{P_\star \widehat{r}}{\|P_\star \widehat{r}\|}$$

whenever $\widehat{\lambda}_\star > 0$. Thus the normalised raw columns eventually become nearly parallel. The effect can be slow at modest k when several active eigenvalues are close, but the condition number of G_m can nevertheless grow rapidly. Moreover, whenever Z_m has full column rank, $\kappa_2(Z_m^\top Z_m) = \kappa_2(Z_m)^2$ exactly; if it is rank deficient, both sides are infinite under the usual extended convention. Small floating-point perturbations can then create large changes in a, despite the fact that the intended exact subspace remains well defined.

The raw path is deliberately not repaired by a pseudoinverse or hidden orthogonalisation in the simulation. A component count is marked invalid if its normal-equation solve fails or produces non-finite values. Five-fold cross-validation is evaluated only on component counts valid in the full sample and every fold. This makes numerical failure part of the empirical comparison rather than silently changing the estimator.

3.6 Arnoldi FPLS and reorthogonalisation

Arnoldi FPLS retains the response-space objective (3.15) but changes the coordinates used for the Krylov space. Starting from $q_1 = \widehat{r}/\|\widehat{r}\|$, Arnoldi forms $z = \widehat{K}q_j$ and removes its projections onto the existing basis. The implementation uses two passes of classical Gram–Schmidt (CGS2):

$$z \leftarrow z - Q_j(Q_j^\top z), \tag{3.23}$$

$$z \leftarrow z - Q_j(Q_j^\top z), \tag{3.24}$$

$$q_{j+1} = z/\|z\|, \tag{3.25}$$

where $Q_j = [q_1, \ldots, q_j]$. A second pass is inexpensive at the component counts considered here and reduces the loss of orthogonality caused by rounding. The implementation terminates basis expansion when

$$\|z\|_2 \le \varepsilon_{\text{rank}} \max\{\|\widehat{K}\|_\infty, \eta_{\text{tiny}}\}, \qquad \varepsilon_{\text{rank}} = 10^{-10}, \tag{3.26}$$

where η_{tiny} is the smallest positive normal floating-point number. Expansion also terminates when the candidate norm is non-finite. The resulting columns span the same $\widehat{\mathcal{K}}_m$ as the raw powers in exact arithmetic, as proved in Proposition A.35.

The displayed Arnoldi operations use the Euclidean norm of the grid coordinates, exactly as in the code. On an equally weighted grid this norm differs from the discrete L^2 norm in (3.4) only by the common factor $\sqrt{T}$. The factor changes neither the generated span nor the orthogonality and normalised-condition diagnostics.

For each available m, Arnoldi FPLS solves

$$\widehat{a}_m \in \arg\min_{a\in\mathbb{R}^m} \|y - T_n Q_m a\|_n^2, \qquad \widehat{\beta}_m^{\text{Arnoldi}} = Q_m \widehat{a}_m. \tag{3.27}$$

The response-space least-squares problem is solved by QR decomposition, not by forming another Gram matrix. If $Z_m^A = T_n Q_m = Q_{Z,m} R_{Z,m}$ denotes the reduced QR factorisation, the response solve is accepted through component j only while

$$|(R_{Z,m})_{jj}| > \varepsilon_{\text{rank}} \max\{|(R_{Z,m})_{11}|, \eta_{\text{tiny}}\}. \tag{3.28}$$

On failure of this response-rank test, or after a non-finite solve, the path is frozen at its last valid fitted estimate for larger requested component counts. Thus, orthogonalisation protects the functional basis, QR protects the subsequent response solve, and the two stated tests handle numerical rank loss. These are fixed computational safeguards rather than data-driven regularisation choices.

Figure 3.1 illustrates the distinction for the population geometry of Model 1. At $m = 12$, every pair among the normalised raw powers has absolute inner product above 0.997, and their condition number is of order 10^{16}. The CGS2 Arnoldi basis has no off-diagonal inner product exceeding the displayed threshold and an orthogonality defect below 10^{-15}. The precise floating-point values can vary slightly across linear-algebra platforms. The figure does not show different statistical information: both panels represent the same Krylov space. It shows why the choice of basis can determine whether that space is numerically usable.

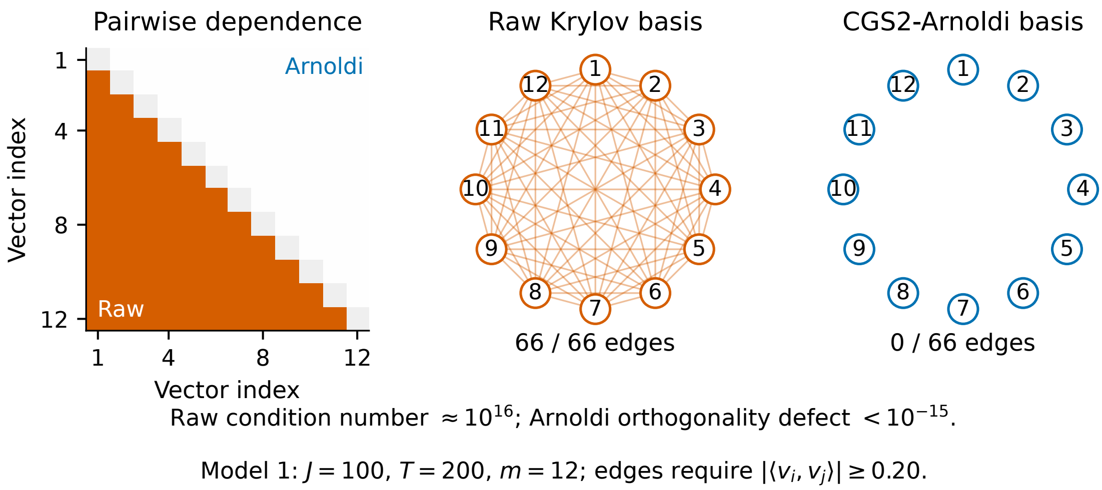


Figure 3.1: Raw and CGS2-Arnoldi representations of the same 12-dimensional population Krylov space in Model 1. An edge marks an absolute pairwise inner product above 0.20. The raw power vectors are almost collinear, whereas reorthogonalisation yields an orthonormal coordinate system. The illustration uses $J = 100$ basis coefficients and a grid of $T = 200$ points.

At a fixed common m, a successful raw solve and a full-rank Arnoldi solve must give the same exact-arithmetic response fit because both minimise the same criterion over the same subspace. Their selected estimates need not be identical in the Monte Carlo study, however. Each method is tuned by its own five-fold cross-validation curve; finite-precision instability can alter the raw curve, invalidate candidate dimensions, and lead to a different selected component count. The comparison therefore separates exact subspace equivalence from the behaviour of complete, tuned numerical procedures.

3.7 Conjugate-Gradient FPLS

The fourth estimator follows the construction of Babii et al. (2026). It searches over the same spaces $\widehat{\mathcal{K}}_m$ but defines the finite-m iterate by

$$\widehat{\beta}_m^{\mathrm{CG}} \in \arg\min_{b \in \widehat{\mathcal{K}}_m} \|\widehat{r} - \widehat{K} b\|^2. \tag{3.29}$$

This is PLS measured in the norm of the empirical normal-equation residual. Its first-order condition is

$$\widehat{r} - \widehat{K}\widehat{\beta}_m^{\mathrm{CG}} \perp \widehat{K}\widehat{\mathcal{K}}_m, \tag{3.30}$$

or, equivalently, $\widehat{K} e_m \perp \widehat{\mathcal{K}}_m$. Compare this with $e_m \perp \widehat{\mathcal{K}}_m$ in (3.16). The subspace is the same, but the projection condition is not.

The difference is also visible in raw coordinates. With $b = \sum_{j=1}^m a_j g_j$, the moment objective produces

$$H_m^M a = h_m^M, \tag{3.31}$$

where

$$(H_m^M)_{jk} = \langle \widehat{r}, \widehat{K}^{j+k} \widehat{r} \rangle, \qquad (h_m^M)_j = \langle \widehat{r}, \widehat{K}^j \widehat{r} \rangle. \tag{3.32}$$

Relative to (3.19), every operator power has shifted by one. Hence the NIPALS/APLS and CG estimates generally differ before convergence. Proposition A.38 proves these two characterisations.

The estimator is computed without explicitly constructing either Hankel matrix. Initialise

$$\widehat{\beta}_0 = 0, \qquad e_0 = \widehat{r}, \qquad d_0 = \widehat{r}. \tag{3.33}$$

For $m = 0, 1, \ldots$, the released recurrence is

$$\alpha_m = \frac{\langle e_m, \widehat{K} e_m \rangle}{\|\widehat{K} d_m\|^2}, \tag{3.34}$$

$$\widehat{\beta}_{m+1} = \widehat{\beta}_m + \alpha_m d_m, \tag{3.35}$$

$$e_{m+1} = e_m - \alpha_m \widehat{K} d_m, \tag{3.36}$$

$$\gamma_m = \frac{\langle e_{m+1}, \widehat{K} e_{m+1} \rangle}{\langle e_m, \widehat{K} e_m \rangle}, \tag{3.37}$$

$$d_{m+1} = e_{m+1} + \gamma_m d_m. \tag{3.38}$$

Its short recurrence avoids the explicit, ill-conditioned power-basis normal equations. It should be noted that in standard numerical-linear-algebra terminology, the residual-minimising updates in (3.34)–(3.38) are the conjugate-residual recurrence, rather than the textbook energy-norm CG recurrence used in Section 3.8. The displayed updates are well defined in exact arithmetic while their quadratic-form numerator and squared-norm denominator are positive. The implementation symmetrises $\widehat{K}$ and freezes the remaining path at the last valid iterate if either scalar is non-finite or non-positive, or if

$$\|\widehat{K}d_m\|_2 \leq 10^{-12} \max\{\|\widehat{K}\|_\infty, \eta_{\text{tiny}}\} \max\{\|d_m\|_2, \eta_{\text{tiny}}\}, \tag{3.39}$$

using the same η_{tiny} as above. A non-finite updated iterate or recurrence coefficient raises a floating-point error. These are numerical-breakdown guards, not the statistical stopping rule.

The theoretical discrepancy rule in Babii et al. (2026) uses the population noise scale

$$\nu_n^\circ = \tau \left\{ \frac{2\sigma^2 \mathbb{E}\|X^c\|^2}{\delta n} \right\}^{1/2}, \qquad m^\circ = \min\{m \geq 0 : \|e_m\| \leq \nu_n^\circ\}. \tag{3.40}$$

Here $\sigma^2 = \mathbb{E}\varepsilon^2$ in the homoskedastic simulation design. This ideal rule states the theoretical benchmark. Because σ^2 and $\mathbb{E}\|X^c\|^2$ are not supplied to an estimator, the implemented workflow instead uses the plug-in scale

$$\widehat{\nu}_n = \tau \left\{ \frac{2\widehat{\sigma}^2 \widehat{\mathbb{E}}\|X^c\|^2}{\delta n} \right\}^{1/2}, \qquad \widehat{\mathbb{E}}\|X^c\|^2 = \frac{1}{n}\sum_{i=1}^n \|X_i^c\|^2. \tag{3.41}$$

and selects

$$\widehat{m} = \min\left\{ m \geq 1 : \|\widehat{r} - \widehat{K}\widehat{\beta}_m^{\text{CG}}\| \leq \widehat{\nu}_n \right\}, \tag{3.42}$$

with the maximum permitted component count used if the threshold is not reached. Here $\tau > 1$ is a safety multiplier, $\delta \in (0, 1)$ controls the probability bound underlying the rule, and $\widehat{\sigma}^2$ is the mean squared response residual from an internal FPCR pilot. The pilot dimension is chosen by moment-space GCV solely to estimate the noise level; it does not enter the method comparison as a separate estimator.

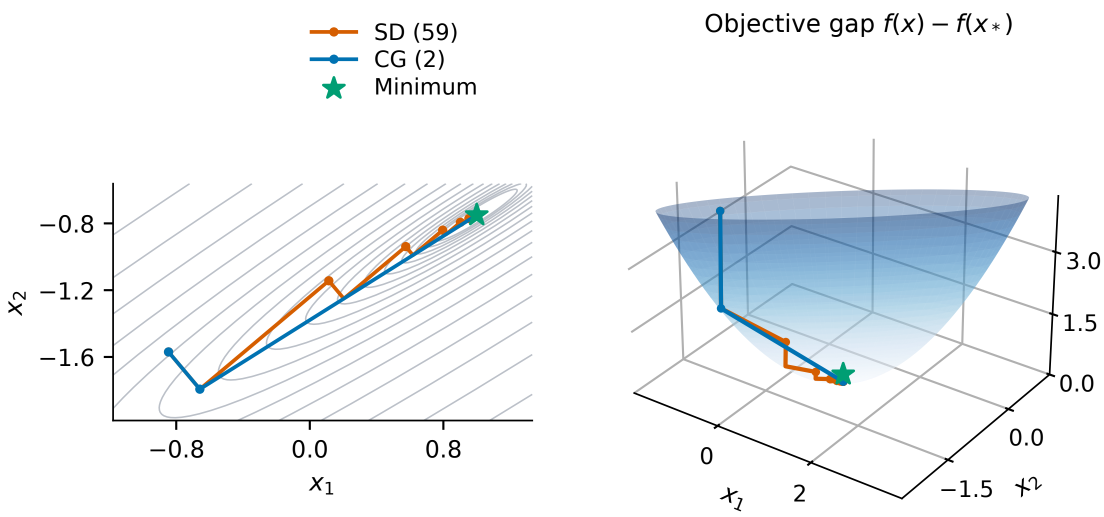
3.8 Why conjugate directions improve on steepest descent

The following comparison supplies geometric intuition for conjugate directions; it does not assert equivalence with the released CG–FPLS recurrence, which uses the conjugate-residual updates specified above. Consider an ordinary symmetric positive-definite quadratic,

$$f(x) = \frac{1}{2}x^\top Ax - b^\top x, \qquad A = A^\top \succ 0. \tag{3.43}$$

Steepest descent moves in the negative-gradient direction $r_k = b - Ax_k$ and, even with exact line search, often zig-zags across a narrow elliptical valley. Its progress deteriorates as the condition number $\kappa_2(A)$ increases. Standard CG instead constructs directions satisfying $p_j^\top A p_k = 0$ for $j \neq k$. Minimisation along a new direction does not undo minimisation along earlier conjugate directions, and exact-arithmetic CG terminates in at most the dimension of the system (Hestenes and Stiefel 1952).

Figure 3.2 uses the same initial point, quadratic, and convergence tolerance for both methods. The matrix has condition number 50. Steepest descent is given its optimal exact line-search step at every iteration, yet takes 59 iterations; CG reaches the two-dimensional minimiser in two. The contour and surface views depict the identical trajectories from different perspectives.



(a) Contour view.

(b) Quadratic-surface view.

Figure 3.2: Steepest descent and conjugate gradient on the same two-dimensional positive-definite quadratic with $\kappa_2(A) = 50$. Steepest descent uses exact line search. The objective gap is measured relative to the common minimiser.

In this limited role, the illustration motivates conjugate directions without predicting the statistical ranking of the estimators. That ranking also depends on sampling error, ill-posedness, the restricted objective, and tuning—the distinctions assembled after the stopping rule is considered.

3.9 Early stopping as regularisation

In exact finite-dimensional arithmetic, continued Krylov iteration can eventually solve the estimable part of the empirical equation. That is not the statistical objective. The right-hand side $\widehat{r}$ and operator $\widehat{K}$ contain sampling noise, and late iterates increasingly fit directions associated with small empirical eigenvalues. The initial iterates are typically smoother and more stable; the sequence may then enter a semi-convergent regime in which the algebraic residual continues to fall while slope error begins to rise. Iteration number is therefore a regularisation parameter rather than only a measure of computational effort (Hanke 1995; Engl et al. 1996).

The discrepancy principle formalises this idea by comparing the moment residual with a sampling-noise scale. If the residual is larger than that scale, the iterate has not yet solved the reliable part of the empirical equation. Once it is comparable with sampling uncertainty, further reduction is more likely to reproduce noise than stable signal. Under the source and moment conditions imposed by Babii et al. (2026), the ideal rule in (3.40) adapts to unknown slope regularity and yields convergence rates for both the unweighted slope norm and covariance-weighted norms. The source condition has the generic form $\beta = K^{\mu}g$: a larger μ means that the slope is more concentrated in well-identified covariance directions. The practical rule (3.42) replaces the unknown scale by a pilot estimate. Its behaviour is evaluated as part of the complete simulation procedure; the theorem for the ideal threshold should not be read as an automatic guarantee for this particular plug-in pilot without additional consistency and probability-control arguments.

The discrepancy rule above is calibrated for estimation. Inference instead uses the stricter residual requirement derived in Section 2.7, and therefore follows the same path to a later stopping index. For a literal hypothesis $H_0 : \beta = b$, either K must be injective or both β and b must be restricted to $\mathcal{H}_{\mathrm{id}}$; otherwise the data identify only the moment hypothesis $H_0 : K\beta = Kb$.

3.10 Equivalence boundaries and tuning design

The comparison is organised by three decisions: which subspace is searched, which norm is minimised within that space, and how its dimension is chosen. Table 3.1 makes these dependencies explicit; a common component count implies a common estimate only when both the subspace and restricted objective coincide.

Tuning is part of each complete estimator and is therefore retained rather than artificially equalised. Separate tuning means that differences in selected complexity are outcomes to interpret alongside prediction error, slope error, and conditioning. In particular, the raw–Arnoldi contrast asks whether coordinate instability changes an otherwise common APLS target, whereas the Arnoldi–CG contrast changes the finite-step objective itself.

Table 3.1: Roles and exact equivalence boundaries of the methods discussed in this chapter.

Procedure	Search space	Finite-m criterion	Relationship and tuning
FPCR	Leading eigenspace of $\widehat{K}$	Response least squares	Spectral rather than response-informed; response GCV selects m
NIPALS-type FPLS	Sequential scores spanning $T_n\widehat{\mathcal{K}}_m$	Response least squares	Classical PLS1 reference; not an extra reported estimator
Raw FPLS	Raw powers spanning $\widehat{\mathcal{K}}_m$	Response least squares	Same fixed-m APLS fit as NIPALS under full rank; five-fold CV
Arnoldi FPLS	CGS2 basis spanning $\widehat{\mathcal{K}}_m$	Response least squares solved by QR	Same exact fixed-m fit as raw FPLS; five-fold CV applied separately
CG–FPLS	Recursive directions spanning $\widehat{\mathcal{K}}_m$	Moment-residual least squares	Same nested spaces but generally a different finite-m iterate; discrepancy stopping

These contrasts determine the Monte Carlo design in Chapter 4, which holds the observations and candidate path common while preserving each procedure's objective, tuning rule, and numerical representation.

Chapter 4

Monte Carlo Design

This chapter specifies the experiment used to compare the four estimators developed in Chapter 3. Its purpose is to separate the scientific design from the numerical findings. The data-generating process, component-selection rules, diagnostic thresholds, loss functions, and random number streams recorded here define the simulation setting. The design retains the three functional linear models and principal numerical settings used in the replication study of Babii et al. (2026), while restricting the comparison to the four estimators relevant to this thesis: CG–FPLS, raw FPLS, Arnoldi FPLS, and FPCR.

The three models form controlled comparisons. Model 2 changes only the slope relative to Model 1, whereas Model 3 changes only the score-variance sequence that determines the predictor covariance. Within every Monte Carlo replication, all four estimators receive exactly the same training and test observations. Differences between methods within a model therefore reflect estimation, tuning, and numerical implementation rather than different simulated samples.

4.1 Discretised functional data-generating process

4.1.1 Grid, basis, and inner product

The functional domain is $\mathcal{I} = [0, 1]$. Each curve is observed without additional measurement error on the endpoint-inclusive grid

$$s_t = \frac{t-1}{T-1}, \qquad t = 1, \dots, T, \qquad T = 200. \tag{4.1}$$

All integrals used by the estimators and evaluation criteria are represented by the equal-weight discrete inner product

$$\langle f, g\rangle_T = \frac{1}{T}\sum_{t=1}^{T} f(s_t)g(s_t). \tag{4.2}$$

This is the endpoint average used in the simulation. Consequently, every displayed finite-sample formula is interpreted under the same numerical geometry as the estimators.

Generated dataset contains $J = 100$ functions. Following the indexing in the our simulation, it is

$$\phi_1(s) = 1, \qquad \phi_j(s) = \sqrt{2}\cos(j\pi s), \quad j = 2, \ldots, J. \tag{4.3}$$

The indexing in (4.3) is intentional. The simulation first constructs cosine frequencies $1, \ldots, J$ and then replaces its first column by the constant function; hence $\sqrt{2}\cos(\pi s)$ is not included. Stating this convention explicitly prevents an apparently harmless re-indexing from creating a different experiment. In continuous $L^2[0, 1]$ the retained functions are mutually orthonormal. On the finite endpoint grid their inner products are evaluated exactly by (4.2) and are not assumed to equal their continuum values.

4.1.2 Predictors and responses

For observation i, draw independent standard normal scores

$$Z_{ij} \overset{\text{iid}}{\sim} \mathcal{N}(0, 1), \qquad j = 1, \ldots, J, \tag{4.4}$$

and construct the functional predictor

$$X_i(s_t) = \sum_{j=1}^{J} \sqrt{\lambda_j}\, Z_{ij}\phi_j(s_t). \tag{4.5}$$

Here λ_j is exactly the variance multiplier of the independent score Z_{ij}. In the underlying continuous orthonormal system these quantities are covariance eigenvalues. Because the endpoint-grid inner product does not make the retained dictionary exactly orthonormal, however, they should be read as score variances—not as the exact eigenvalues of the finite $T \times T$ discrete covariance matrix. This distinction does not affect any formula below, which is evaluated directly under the discrete geometry. The scalar response is generated from

$$Y_i = \langle X_i, \beta \rangle_T + \varepsilon_i = \frac{1}{T} \sum_{t=1}^{T} X_i(s_t)\beta(s_t) + \varepsilon_i, \qquad \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2), \tag{4.6}$$

where $\sigma = 1$ and the errors are independent of all scores. The population means of X_i and Y_i are therefore zero and the intercept is known to be zero. The simulation passes the generated observations directly to all four estimators; it does not subtract realised sample means. This common convention isolates regularisation and numerical effects from an additional centring step.

For a coefficient vector $b = (b_1, \ldots, b_J)^\top$, the slope evaluated on the grid is

$$\beta(s_t) = \sum_{j=1}^{J} b_j \phi_j(s_t). \tag{4.7}$$

Equations (4.5)–(4.7) define the data directly at the T grid points. There is thus no preliminary smoothing, basis estimation, or interpolation stage in this controlled study.

4.2 The three controlled models

Define baseline slope coefficients and score variances by

$$b_j^{(0)} = \frac{4}{j^{2.7}}, \qquad \lambda_j^{(0)} = \frac{2}{j^{1.1}}, \qquad j = 1, \ldots, J. \tag{4.8}$$

Two targeted modifications are then introduced:

$$b_j^{(1)} = \begin{cases} 4, & j \leq 5, \\ 4j^{-2.7}, & j > 5, \end{cases} \tag{4.9}$$

$$\lambda_j^{(1)} = \begin{cases} 2, & j \leq 5, \\ 2j^{-1.1}, & j > 5. \end{cases} \tag{4.10}$$

The resulting combinations are summarised in Table 4.1.

Figure 4.1 shows both the true slopes and the first twenty coefficient magnitudes. Models 1 and 3 have the same target function. Model 2 distributes substantial slope energy across the first five directions and is therefore more oscillatory. This distinction tests whether a supervised method can exploit response-relevant directions differently from a method ordered only by predictor variance.

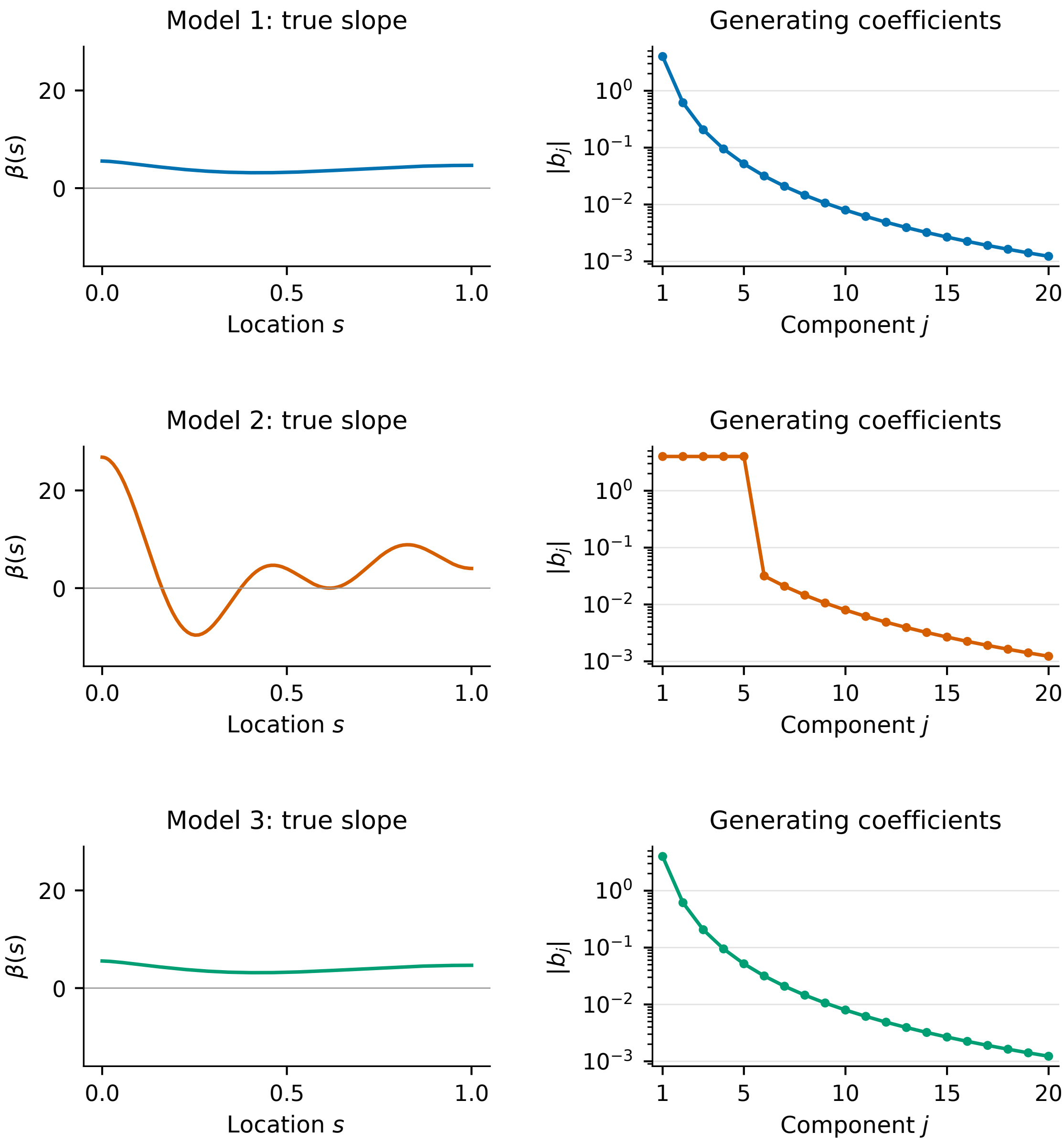


Figure 4.1: True slope functions and coefficient magnitudes in the three Monte Carlo models. The curves use $J = 100$ functions and are evaluated on the $T = 200$ grid. Only the first twenty coefficients are shown in the right-hand panels; all one hundred enter the data-generating process. Rows correspond to Models 1–3, with the slope functions on the left.

Table 4.1: Definition and controlled role of the three simulation models.

Model	Slope coefficients	Score variances	Controlled role
Model 1	$b^{(0)}$	$\lambda^{(0)}$	Baseline: rapidly decaying slope coefficients and a power-law score-variance sequence.
Model 2	$b^{(1)}$	$\lambda^{(0)}$	Slope-only change: the first five coefficients are set to 4, so coefficients 2–5 increase while the first remains unchanged; the predictor distribution is unchanged.
Model 3	$b^{(0)}$	$\lambda^{(1)}$	Predictor-only change: the first five score variances are set to 2, so variances 2–5 increase while the first remains unchanged; the true slope is unchanged.

Figure 4.2 isolates the predictor-side change. Models 1 and 2 share the power-law score-variance sequence $2j^{-1.1}$. Model 3 has a flat leading block of five score variances followed by the same tail. The second panel reports cumulative predictor variation, not predictive signal: these two quantities need not rank directions in the same way.

4.3 Signal strength and effective directionality

The scale of a slope alone does not determine prediction difficulty. Let

$$c_j = \langle \beta, \phi_j \rangle_T, \qquad q_j = \lambda_j c_j^2. \tag{4.11}$$

Because the scores are independent and standard normal, q_j is the exact discrete contribution of component j to the variance of the noiseless response. In particular,

$$\mathbb{E}\|X\|_T^2 = \sum_{j=1}^{J} \lambda_j \|\phi_j\|_T^2, \tag{4.12}$$

$$\operatorname{Var}\{\langle X, \beta \rangle_T\} = \sum_{j=1}^{J} q_j, \tag{4.13}$$

$$\operatorname{Var}(Y) = \sum_{j=1}^{J} q_j + \sigma^2. \tag{4.14}$$

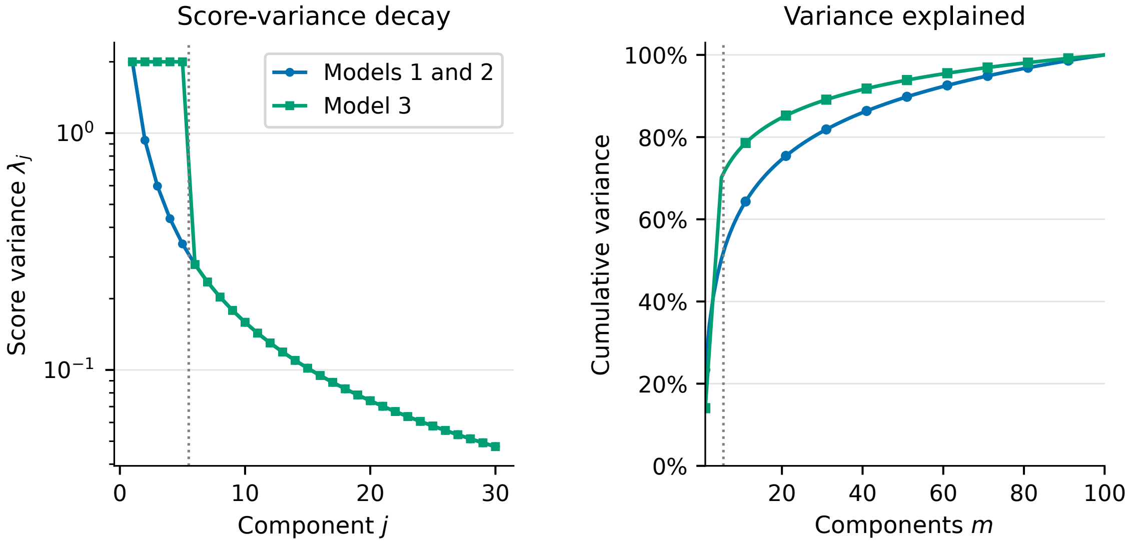


Figure 4.2: Designed covariance structure. The left panel displays the first thirty score variances—the covariance eigenvalues in the continuous design—on a logarithmic scale; the right panel shows their cumulative share over all $J = 100$ components. The dotted vertical line separates the leading block from the common tail.

Detailed proof of these identities is represented in Appendix Proposition A.28. The signal-to-noise ratio is defined as

$$\mathrm{SNR} = \frac{\sum_{j=1}^{J} q_j}{\sigma^2}. \tag{4.15}$$

Table 4.2 reports values computed from the finite $J = 100$, $T = 200$ design itself; they are not Monte Carlo estimates. Since $\sigma^2 = 1$, signal variance and SNR are numerically identical. The substantially larger signal variance in Model 2 also means that raw prediction errors should primarily be compared across methods within a model, not used as an unnormalised measure of which model is intrinsically easiest.

Table 4.2: Exact derived properties under the discrete data-generating process. The final column gives the first component counts reaching 90%, 95%, and 99% of predictive signal.

Model	$\lVert\beta\rVert_T^2$	$\mathbb{E}\lVert X\rVert_T^2$	$\mathrm{Var}\{\langle X,\beta\rangle_T\}$	SNR	First-five share	$m_{90}/m_{95}/m_{99}$
Model 1	16.484	8.589	32.520	32.520	99.986%	1/1/2
Model 2	81.439	8.589	71.306	71.306	99.937%	4/5/5
Model 3	16.484	14.311	33.061	33.061	99.986%	1/1/2

The allocation is made explicit in Figure 4.3. Model 2 needs five components to reach 99% of the exact predictive signal, whereas Models 1 and 3 reach that level after two. This statement describes the population design; it does not predetermine how many empirical components a noisy, data-driven estimator will select.

Representative predictor curves are shown in Figure 4.4. Models 1 and 2 have the same

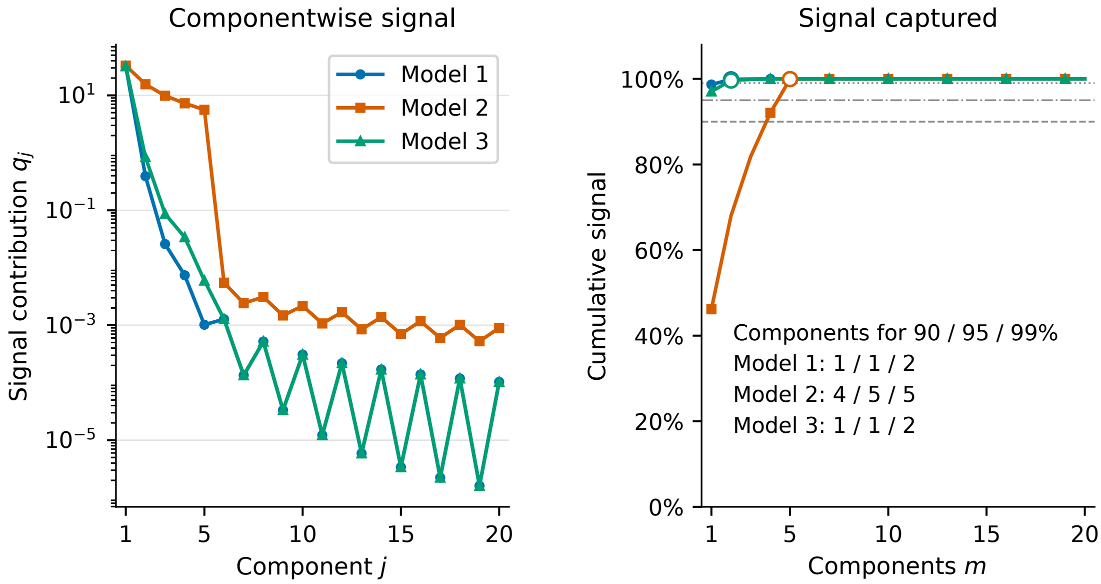


Figure 4.3: Componentwise and cumulative predictive-signal allocation, calculated as $q_j = \lambda_j \langle \beta, \phi_j \rangle_T^2$ under the exact discrete design. Open markers denote the first component count exceeding 99% of total signal.

predictor distribution and therefore share a panel. For this setup illustration only, the same Gaussian score draws are used in both panels so that the visibly larger variation for Model 3 can be attributed to its modified spectrum. The shaded regions are exact pointwise 90% Gaussian bands under the corresponding finite data-generating process. They are not estimated confidence bands and are not used by any estimator.

These population contrasts define the signal each estimator must recover. The Monte Carlo protocol now determines how sampling variation and data-driven tuning act on those contrasts.

4.4 Monte Carlo protocol

The full experiment uses $R = 5{,}000$ independent replications for each of the three models. Within replication r of a given model, the following protocol is applied.

1. Generate a training sample of $n = 100$ curve–response pairs from (4.5)–(4.6).
2. Construct the empirical moments

$$\widehat{K} = \frac{X^\top X}{nT}, \qquad \widehat{r} = \frac{X^\top y}{n}, \tag{4.16}$$

   and fit all four methods to the same training sample.

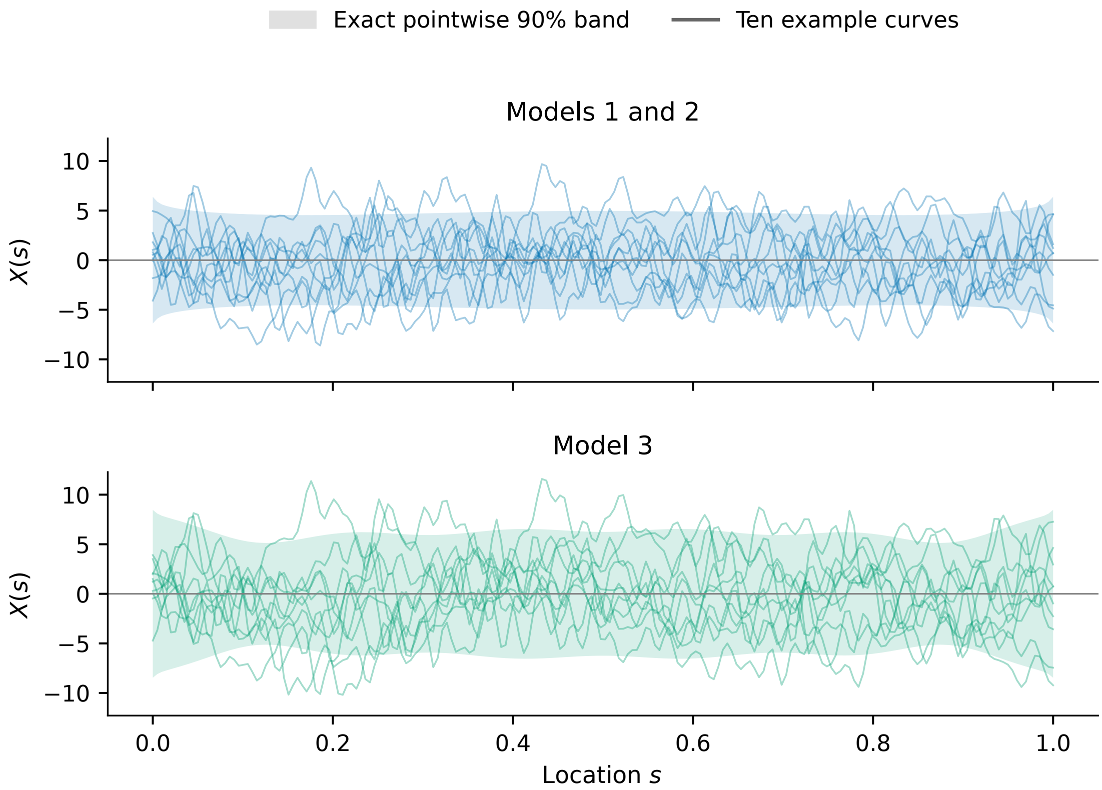


Figure 4.4: Representative functional predictors and their exact pointwise 90% bands: Models 1 and 2 above, Model 3 below. The shared score draws make the spectrum change visually comparable; the actual Monte Carlo models use independent random-number streams.

3. Generate an independent test sample of $n_{\text{test}} = 100$ from the same model. No test observation is used for tuning.

4. Store the selected dimension, slope and prediction errors, and method-specific numerical diagnostics.

The candidate set is $m \in \{1, \ldots, 70\}$ for every method. Thus $m = 0$ is never reported as an estimator. For CG–FPLS, absence of a threshold crossing returns the $m = 70$ estimate. The common upper bound keeps the candidate complexities comparable while remaining below the empirical rank limit; the point selected along that path is nevertheless method-specific. Methods are paired within a replication, but the three model streams are independent.

4.5 Estimator-specific tuning

The method labels refer to complete procedures, including their own tuning rules. Table 4.3 provides an overview; the details below fix the criteria and constants.

Table 4.3: The four reported estimators and their tuning mechanisms.

Method	Restricted problem	Selection of m	Numerical construction
CG–FPLS	Moment-residual least squares	First discrepancy-threshold crossing; $\tau = 1.01$, $\delta = 0.1$	Released short recurrence; internal FPCR noise pilot
Raw FPLS	Response least squares on $\widehat{\mathcal{K}}_m$	Five-fold prediction cross-validation	Unscaled covariance powers and normal equations
Arnoldi FPLS	Same response least-squares problem as raw FPLS	Five-fold prediction cross-validation	CGS2 Arnoldi basis and response-space QR; rank tolerance 10^{-10}
FPCR	Response least squares on leading empirical eigenvectors	Response-space GCV	Symmetric eigendecomposition and spectral cut-off

4.5.1 Five-fold tuning of raw and Arnoldi FPLS

One random partition $I_1, \ldots, I_5$ is generated in each replication and is used by both FPLS implementations. For method $a \in \{\text{Raw}, \text{Arnoldi}\}$, let $\widehat{\beta}_{m,a}^{(-k)}$ be the estimate fitted without fold I_k. Its criterion is

$$\mathrm{CV}_a(m) = \frac{1}{n}\sum_{k=1}^{5}\sum_{i\in I_k}\left\{Y_i - \langle X_i, \widehat{\beta}_{m,a}^{(-k)}\rangle_T\right\}^2. \tag{4.17}$$

Each method minimises its own curve. The common folds remove avoidable Monte Carlo variation, but separate minimisation allows finite-precision instability in the raw basis to affect the complete procedure honestly. A raw component count is eligible only when its fit is finite and valid in the full sample and in every training fold. Arnoldi uses two-pass classical Gram–Schmidt (CGS2) with rank tolerance 10^{-10}. If several eligible values attain the same minimum numerically, the smallest is retained.

4.5.2 Response-GCV tuning of FPCR

FPCR uses the response-space criterion already introduced in (3.8):

$$\mathrm{GCV}_Y(m) = \frac{n^{-1}\|y - X\widehat{\beta}_m^{\mathrm{FPCR}}/T\|_2^2}{(1-m/n)^2}. \tag{4.18}$$

Only numerically positive empirical eigenvalues are eligible. Under the present $m_{\max} = 70 < n-1$ design, the shared upper bound is the active path limit. The smallest minimiser of (4.18) is selected.

4.5.3 Discrepancy tuning of CG–FPLS

For CG–FPLS, define the discrete moment norm and predictor-energy estimate by

$$\|e_m\|_T = \left(\frac{1}{T}e_m^\top e_m\right)^{1/2}, \qquad \widehat{E}_X = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{T}X_i^\top X_i, \tag{4.19}$$

where $e_m = \widehat{r} - \widehat{K}\widehat{\beta}_m^{\mathrm{CG}}$. The noise pilot is constructed from the same FPCR path but is selected internally by moment-space GCV,

$$\mathrm{GCV}_M(q) = \frac{T^{-1}\|\widehat{r} - \widehat{K}\widehat{\beta}_q^{\mathrm{FPCR}}\|_2^2}{(1-q/T)^2}. \tag{4.20}$$

If $\widehat{q}$ minimises this criterion, then

$$\widehat{\sigma}^2 = \frac{1}{n}\|y - X\widehat{\beta}_{\widehat{q}}^{\mathrm{FPCR}}/T\|_2^2. \tag{4.21}$$

This moment-GCV calculation is used only to estimate the threshold scale; it is neither the reported FPCR fit nor a fifth comparator.

For interpretation, an ideal version of the discrepancy scale would replace the unknown quantities below by the population error variance and predictor energy. The implemented rule is feasible instead: it uses the internally estimated $\widehat{\sigma}^2$ in (4.21) and the sample energy $\widehat{E}_X$. Thus $\widehat{\nu}_n$ is a plug-in threshold, not an oracle threshold supplied with the true σ^2 or the population covariance. The selected component count is the first $m \geq 1$ satisfying

$$\|e_m\|_T \leq \widehat{\nu}_n = 1.01 \left(\frac{2\widehat{\sigma}^2 \widehat{E}_X}{0.1n} \right)^{1/2}. \tag{4.22}$$

If no iterate crosses the threshold, $m = 70$ is used and the absence of a crossing is recorded. Together with the cross-validation and GCV rules above, this defines the four complete procedures whose consequences are evaluated through slope and prediction loss. The rule in (4.22) is the single CG selection procedure in the final workflow; no oracle or supplementary CG variant is included.

4.6 Performance measures and summaries

4.6.1 Slope estimation and independent prediction

For method a, the integrated squared error in one replication is the discrete loss

$$\mathrm{ISE}_a = \|\widehat{\beta}_a - \beta\|_T^2 = \frac{1}{T} \sum_{t=1}^{T} \{\widehat{\beta}_a(s_t) - \beta(s_t)\}^2. \tag{4.23}$$

The independent test mean squared prediction error is

$$\mathrm{MSPE}_a = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} \{Y_i^{\text{test}} - \langle X_i^{\text{test}}, \widehat{\beta}_a \rangle_T\}^2. \tag{4.24}$$

MSPE includes irreducible response noise. Conditional on a training-sample estimate and averaging over a new test observation,

$$\mathbb{E}\{\mathrm{MSPE}_a \mid \widehat{\beta}_a\} = \sigma^2 + \sum_{j=1}^{J} \lambda_j \langle \phi_j, \widehat{\beta}_a - \beta \rangle_T^2. \tag{4.25}$$

Thus the oracle noise floor is $\sigma^2 = 1$, while the second term is excess prediction risk. Equation (4.25) is the discrete counterpart of Appendix Proposition A.27.

For every model–method pair, simulation result illustrates the mean and its Monte Carlo standard error, the median, the 90th and 99th percentiles, and the maximum of both ISE and MSPE. If $L_1, \ldots, L_{R_f}$ are the finite values of a loss, the reported standard error is

$$\mathrm{MCSE}(\overline{L}) = \frac{s_L}{\sqrt{R_f}}, \tag{4.26}$$

where s_L is their sample standard deviation. Medians and upper quantiles are reported alongside means because rare numerical failures can make the error distributions extremely right-skewed.

4.6.2 Bias–variance decomposition

Unstable raw fits are omitted only from function-valued diagnostics—the bias–variance decomposition, basis-coefficient distributions, pointwise distributions, and normal quantile–quantile plots. They remain in every primary ISE and MSPE summary. With $\overline{\beta}_a = R^{-1} \sum_{r=1}^{R} \widehat{\beta}_{a,r}$, the empirical slope decomposition is

$$\frac{1}{R}\sum_{r=1}^{R} \|\widehat{\beta}_{a,r} - \beta\|_T^2 = \|\overline{\beta}_a - \beta\|_T^2 + \frac{1}{R}\sum_{r=1}^{R} \|\widehat{\beta}_{a,r} - \overline{\beta}_a\|_T^2. \tag{4.27}$$

The two right-hand terms are labelled squared bias and variance. Prediction error is decomposed analogously after weighting the coordinatewise bias and variance by λ_j, with σ^2 added as irreducible noise. Appendix Proposition A.29 proves the finite-R identity. The factor R^{-1}, rather than $(R-1)^{-1}$, is used so the two terms add exactly to the Monte Carlo mean squared error.

The main comparison uses the full set of results to assess slope estimation and prediction accuracy. An additional bias–variance analysis considers only cases where raw FPLS produces a numerically stable fit. To keep this comparison fair, all four methods are assessed using those same cases. This smaller set is clearly identified and provides an additional perspective; it does not replace the main comparison.

4.6.3 Complexity, instability, and tail events

Selected component count is recorded for every fit and summarised by its mean and median. Complexity is not itself an accuracy measure, but it shows how aggressively each tuning rule regularises the inverse problem.

The simulation also distinguishes statistical error from numerical failure. For raw FPLS, let G_m be the raw Krylov power matrix and $Z_m = XG_m/T$ its response-space design.

Let $\widetilde{G}_m$ and $\widetilde{Z}_m$ denote the same matrices after normalising every column to unit Euclidean length. The diagnostic is

$$\Delta_m^{\mathrm{Raw}} = \max\{\kappa_2(\widetilde{G}_m), \kappa_2(\widetilde{Z}_m)\}. \tag{4.28}$$

The column normalisation removes trivial scale growth and focuses the measure on near-collinearity. A selected raw fit is flagged as numerically unstable when its loss is non-finite or

$$\Delta_m^{\mathrm{Raw}} \geq \varepsilon_{\mathrm{mach}}^{-1/2} \approx 6.71 \times 10^7. \tag{4.29}$$

For Arnoldi FPLS the corresponding diagnostic is the spectral-norm orthogonality defect

$$\Delta_m^{\mathrm{Arnoldi}} = \|Q_m^\top Q_m - I_m\|_2. \tag{4.30}$$

A fit is flagged when its loss is non-finite or the defect is at least $\varepsilon_{\mathrm{mach}}^{1/2} \approx 1.49 \times 10^{-8}$. CG–FPLS and FPCR are flagged only if ISE or MSPE is non-finite because their relevant path diagnostics are already governed by the recurrence and positive-eigenvalue checks.

An estimation or prediction error is considered extreme if it exceeds 100 times the positive median error for the same method and model. These unusually large errors remain in the analysis, including the averages, maximum values, and full-distribution figures. Reporting them alongside failure and instability rates shows whether a method that usually performs well also suffers from occasional serious problems. The simulation settings and checking procedures are kept fixed so that the analysis can be reproduced.

4.7 Randomness and reproducibility

The top-level seed is 2026. A NumPy `SeedSequence` deterministically spawns one independent child stream for each model. Within a model, the training sample is generated first, followed by the random five-fold partition and then the independent test sample. The order matters for exact replay. The four methods share the training and test data, and raw and Arnoldi FPLS additionally share the folds. Models do not share their realised samples.

The models vary the slope and covariance structure in a controlled way, while all methods use the same data. This helps distinguish the effects of regularisation from those of numerical computation. Each method follows fixed tuning and evaluation rules, and unusually large errors remain part of the comparison. Chapter 5 examines how they appear in slope recovery and prediction.

Chapter 5

Estimation and Prediction Results

This chapter analyses the experiment described in Chapter 4, using the original tuning rules and diagnostic criteria. It first compares how well the methods estimate the slope and predict new responses. It then examines the number of selected components, unusually large errors, and differences between raw and Arnoldi FPLS on the same data to explain the roles of regularisation and numerical stability. The aim is to understand whether slope estimation and prediction benefit from the same type and strength of regularisation, and how numerical stability affects that choice.

5.1 How the results are assessed

The main comparison includes all 60,000 fits and their slope errors (ISE), prediction errors (MSPE), selected component counts, and unusually large errors. The 445 raw-FPLS fits identified as numerically unstable remain in this comparison. They are excluded only from additional analyses requiring reliable reconstruction of the slope curves, according to the stability rule in Chapter 4. This exclusion is not based on the size of the observed error.

The mean, median, and upper percentiles describe different aspects of performance. The mean includes the influence of rare but very large errors. The median describes a typical result, while the 90th and 99th percentiles show how large errors become among the less successful cases. These percentiles are less sensitive than the mean to a single extreme value. For the numerically stable methods, the Monte Carlo standard errors of mean ISE range from 0.0018 to 0.0475, and those of mean MSPE range from 0.0024 to 0.0088. These standard errors measure uncertainty due to the finite number of simulation runs. For raw FPLS, rare extreme errors can greatly increase both the mean and its standard error.

5.2 Overall estimation and prediction accuracy

Table 5.1 compares the methods using all 5,000 simulation runs for each model, including every raw-FPLS fit identified as numerically unstable. Figure 5.1 shows the same results on logarithmic scales, so that both typical errors and unusually large errors can be seen.

Table 5.1: Slope-estimation and prediction errors across all simulation runs. Means, medians, and 90th percentiles are calculated over all 5,000 runs for each model–method pair.

		Slope ISE			Test MSPE		
Model	Method	Mean	Median	p_{90}	Mean	Median	p_{90}
Model 1	CG–FPLS	0.633	0.597	0.910	1.270	1.230	1.617
	Raw FPLS	334.298	0.539	0.990	21.224	1.114	1.357
	Arnoldi FPLS	0.760	0.539	0.987	1.130	1.113	1.356
	FPCR	0.670	0.532	0.952	1.126	1.112	1.357
Model 2	CG–FPLS	5.506	5.272	7.354	1.999	1.861	2.753
	Raw FPLS	7.594	5.384	9.949	1.518	1.428	1.793
	Arnoldi FPLS	6.065	5.345	8.736	1.442	1.419	1.763
	FPCR	5.662	5.365	7.392	1.495	1.472	1.836
Model 3	CG–FPLS	0.447	0.426	0.608	1.221	1.176	1.534
	Raw FPLS	1.219	0.446	0.861	1.154	1.114	1.365
	Arnoldi FPLS	0.670	0.446	0.856	1.131	1.114	1.363
	FPCR	0.615	0.463	0.930	1.127	1.113	1.349

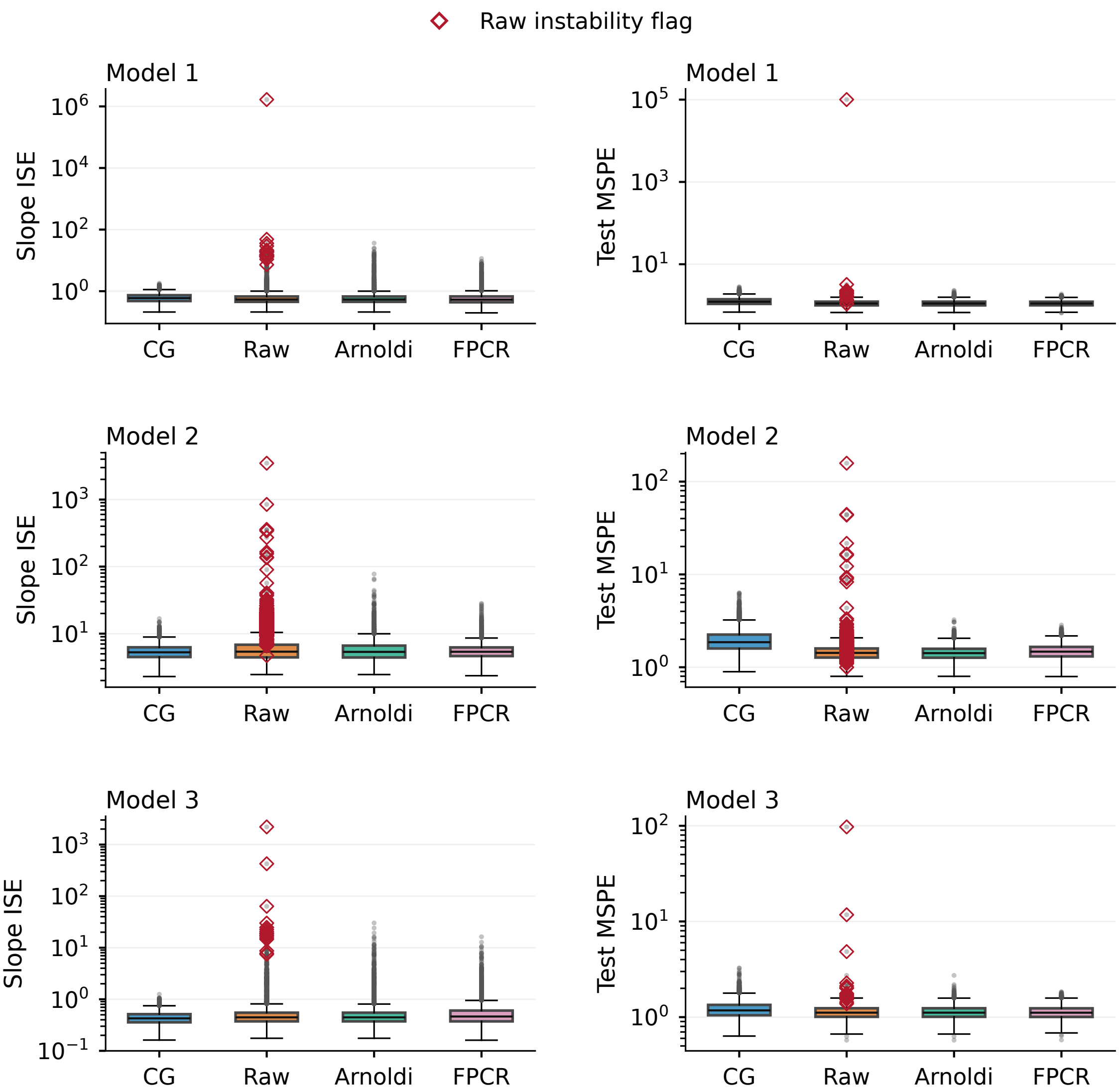


Figure 5.1: Distributions of slope ISE (left column) and independent test MSPE (right column), with one row per model. The boxplots use the conventional 1.5 interquartile-range rule for their whiskers. All axes are logarithmic, and red open diamonds mark raw-FPLS fits identified as numerically unstable. The labels “CG”, “Raw”, and “Arnoldi” refer to the CG–FPLS, raw FPLS, and Arnoldi FPLS procedures defined in Chapter 3. All extreme errors are retained.

5.2.1 Model 1: predictive signal concentrated in the leading directions

In Model 1, the methods have similar typical slope errors. FPCR has the smallest median ISE at 0.532, followed by raw and Arnoldi FPLS at approximately 0.539, and CG–FPLS at 0.597. The comparison changes when we consider the mean. CG–FPLS has mean ISE 0.633, compared with 0.670 for FPCR and 0.760 for Arnoldi FPLS. Its early stopping keeps the slope errors more concentrated and limits unusually poor estimates. Raw FPLS, by contrast, has a mean of 334.298, despite its competitive median. A very small number of severe numerical errors account for this large difference.

FPCR and Arnoldi FPLS perform best in prediction. Their median MSPEs are 1.112 and 1.113, respectively, only slightly above the unavoidable error variance of 1. Raw FPLS has similar typical prediction errors, while CG–FPLS has median MSPE 1.230. A smaller average slope error therefore does not necessarily lead to better predictions. Prediction gives more weight to errors in directions with greater predictor variation, whereas ISE measures the overall error in the slope. Model 2 examines this difference when substantial slope signal is spread across five leading directions.

5.2.2 Model 2: a slope spread across more directions

Model 2 sets the first five slope coefficients to 4 and leaves the predictor score variances unchanged. The first coefficient already equals 4 in Model 1, so only coefficients 2–5 increase. All methods tend to select more components, and their typical slope errors are substantially larger. CG–FPLS has the smallest median and mean ISE, 5.272 and 5.506. FPCR has mean ISE 5.662, followed by Arnoldi FPLS at 6.065. Their medians are closer: 5.272 for CG–FPLS, 5.365 for FPCR, and 5.345 for Arnoldi FPLS. The bias–variance analysis below explains this result: early CG stopping allows more bias in exchange for less variation in the estimated slope, measured in the ordinary L^2 norm.

Arnoldi FPLS performs best in prediction, with mean MSPE 1.442 and median MSPE 1.419. Typical raw-FPLS results are close to these values. FPCR has mean MSPE 1.495, while CG–FPLS reaches 1.999. Some of the directions left out by early CG stopping are therefore useful for predicting the response, even though including them makes the estimated slope more variable. Model 2 shows particularly clearly why a method can perform well in slope estimation and less well in prediction. Model 3 considers a different change: it restores the original slope and increases the leading predictor variances.

5.2.3 Model 3: stronger leading predictor directions

Model 3 keeps the slope of Model 1 and sets the first five score variances to 2. Since the first already equals 2, only score variances 2–5 increase. With more information in these directions, CG–FPLS mean ISE falls from 0.633 in Model 1 to 0.447, the lowest value in Model 3. FPCR and Arnoldi FPLS have mean ISEs of 0.615 and 0.670. The medians are closer: 0.426 for CG–FPLS, approximately 0.446 for both raw and Arnoldi

FPLS, and 0.463 for FPCR.

FPCR and Arnoldi FPLS again predict more accurately than CG–FPLS. Their mean MSPEs are 1.127 and 1.131, compared with 1.221 for CG–FPLS; the corresponding medians are 1.113, 1.114, and 1.176. Greater predictor variation can make a slope direction easier to estimate, but it also makes prediction more sensitive to errors in that direction. Models 1 and 3 thus show that changing the predictor variances can affect estimation and prediction differently even when the true slope stays the same. The number of selected components helps explain how each method responds to these changes.

5.3 Selected regularisation levels

Table 5.2 shows how many components each method selects. The Krylov methods generally use far fewer components than FPCR because their directions use information about both the predictors and the response. FPCR orders its components by predictor variation alone. Component counts therefore need to be interpreted within each method: one Krylov component is not equivalent to one principal component.

Table 5.2: Distribution of the selected component count over 5,000 simulation runs. The maximum records rare selections and should be read together with the numerical diagnostics.

Model	Method	Mean	Median	p_{90}	Maximum
Model 1	CG–FPLS	1.098	1	1	2
	Raw FPLS	2.134	2	2	57
	Arnoldi FPLS	2.010	2	2	14
	FPCR	5.840	5	11	42
Model 2	CG–FPLS	2.886	3	3	3
	Raw FPLS	7.820	5	6	70
	Arnoldi FPLS	4.808	5	6	16
	FPCR	18.350	17	28	60
Model 3	CG–FPLS	1.833	2	2	3
	Raw FPLS	2.956	3	3	68
	Arnoldi FPLS	2.845	3	3	11
	FPCR	7.057	6	11	46

The methods select more components in Model 2, where increasing coefficients 2–5 spreads the predictive signal across more directions. The number of important basis coefficients does not directly determine the number of Krylov components, since a Krylov direction can combine several basis directions. CG–FPLS selects three components in 88.6% of Model 2 runs, compared with one component in 90.2% of Model 1 runs and two components in 83.2% of Model 3 runs. FPCR selects many more components, with a median of 17 in Model 2. Its ordering by predictor variance can require more components to represent the directions that matter for the response.

All 15,000 CG–FPLS fits meet the discrepancy stopping rule before the maximum component count is reached. None therefore needs the fallback value $m = 70$. The noise estimate used in this rule, however, deserves attention. It comes from a preliminary fit chosen by moment-GCV, which selects a median of 70 components in all three models. The resulting mean estimate of the error variance is approximately 0.300, well below the true value of 1.

This preliminary fit usually uses 70 components and divides its residual sum of squares by n, without adjusting for the degrees of freedom used in fitting. An estimate near $1 - 70/100 = 0.30$ is therefore consistent with an overfitting or degrees-of-freedom effect. The preliminary fit does not provide an accurate estimate of the noise variance in this experiment. Since the stopping threshold is proportional to $\widehat{\sigma}$, underestimating the noise makes the threshold harder to reach and can only delay stopping along the same sequence of residuals. The observed stopping after one to three components therefore reflects a rapid reduction in the residual. These results assess the procedure with its estimated noise level; they do not validate the theoretical rule that assumes the noise level is known. The original noise-estimation and stopping rules are retained throughout.

Raw and Arnoldi FPLS usually select similar numbers of components, but their largest selections differ considerably. Raw FPLS sometimes selects 57, 70, and 68 components in Models 1–3, while the Arnoldi maxima are 14, 16, and 11. A large component count alone does not show whether a method is capturing additional signal or responding to numerical problems. The stability checks in the next section help distinguish these possibilities.

5.4 Unusually large errors and numerical stability

Similar median errors in Table 5.1 do not mean that the methods are equally reliable. Table 5.3 shows how often raw FPLS is flagged as numerically unstable. All reported losses are finite, including those from flagged fits. A flag means that the condition number of the normalised raw Krylov basis or its response design exceeds the fixed threshold. In these cases, the normal-equation calculation is too sensitive to small numerical errors to be trusted. The reconstructed slope is therefore excluded from diagnostics that require a reliable estimate of the whole function, while its finite ISE and MSPE remain in the main comparison.

Table 5.3: Numerical instability and the largest errors of raw FPLS. The stable subset contains the simulation runs for which the raw-FPLS slope can be reconstructed reliably. The additional comparison uses these same runs for every method.

Model	Flagged fits	Rate	Maximum ISE	Maximum MSPE	Stable subset
Model 1	20	0.40%	1.668×10^6	1.005×10^5	4,980
Model 2	409	8.18%	3.482×10^3	1.578×10^2	4,591
Model 3	16	0.32%	2.203×10^3	9.737×10^1	4,984

Instability is most common in Model 2, where FPLS typically selects about five Krylov directions rather than two or three. Its raw-FPLS instability rate is 8.18%, compared with 0.40% and 0.32% in Models 1 and 3. Even a much lower rate can have a large effect on average performance. In Model 1, only one raw-FPLS ISE exceeds one hundred times the median for that model and method, but this error reaches approximately 1.67 million. It explains why the median ISE is 0.539 while the mean is 334.298. The median describes the method's usually competitive performance; the mean also reflects the damage caused by a rare, very large error. Both are needed to give a fair account of its reliability.

The combined Arnoldi–QR implementation avoids such extreme numerical errors in this experiment, although its errors still vary across samples. Arnoldi FPLS has no numerical-instability flags, but its maximum ISEs are 36.35, 77.41, and 30.23 across the three models. These larger errors are consistent with ordinary variation in the samples and selected tuning parameters, and they are not accompanied by a loss of orthogonality in the Arnoldi basis. The corresponding FPCR maxima are 11.53, 28.23, and 16.30, while the CG–FPLS maxima are 1.82, 16.79, and 1.25. The lower maximum errors for CG–FPLS are consistent with its strong early stopping. The next section examines how this stopping rule reduces slope variance and how much additional bias it introduces.

5.5 Understanding bias and variance

Figure 5.2 separates the errors of CG–FPLS, Arnoldi FPLS, and FPCR into squared bias and variance using all simulation runs. For raw FPLS, the figure retains the total risk from the full set of saved results. Splitting this total into bias and variance would require reliable reconstruction of the 445 numerically sensitive slopes, so these components are not shown. The same limitation applies to the raw-FPLS coefficient distributions, pointwise distributions, and normal quantile–quantile plots in Appendix B.

Table 5.4 gives the bias and variance components for CG–FPLS, Arnoldi FPLS, and

FPCR. For prediction, the contribution of slope error in each direction is weighted by the corresponding predictor-score variance λ_j, as in (4.25). A slope error therefore matters more when the predictor varies more in that direction. These weights give the exact excess prediction risk for the discrete model used to generate the data. Adding the irreducible noise variance $\sigma^2 = 1$ to the sum of squared bias and variance gives the total prediction risk.

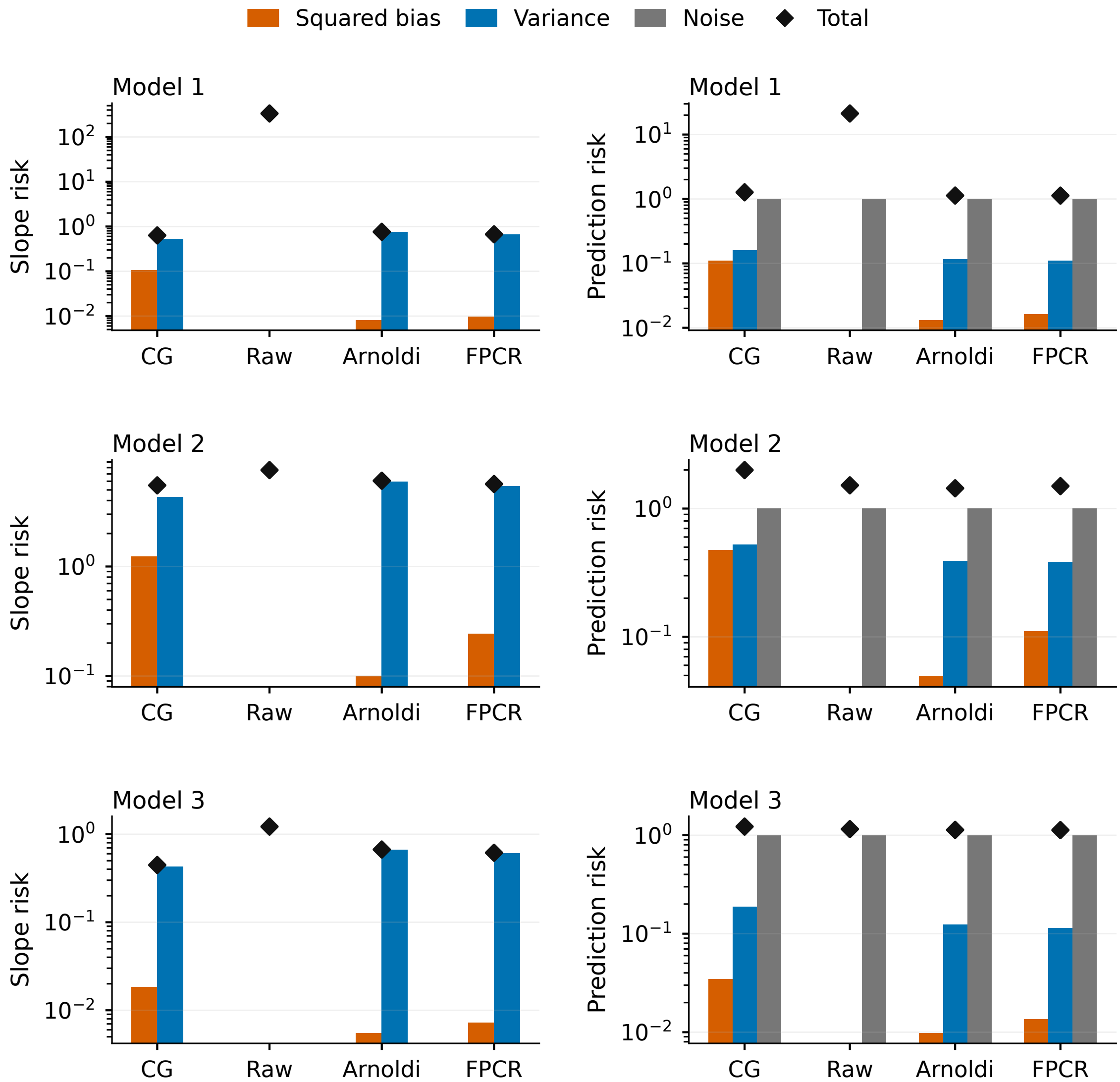


Figure 5.2: Monte Carlo bias–variance decomposition over all 5,000 simulation runs. Each row is one model; slope risk is on the left and prediction risk on the right. Bars show squared bias, variance, and irreducible noise on logarithmic scales; black diamonds show total risk. The full raw-FPLS totals are included, but their components are not shown because the numerically unstable slopes are not considered reliable enough for this reconstruction.

The same pattern appears in all three models. CG–FPLS reduces slope variance but introduces more bias. In Model 1, for example, its slope variance is 0.526, compared with 0.752 for Arnoldi FPLS and 0.660 for FPCR, while its squared bias is 0.107 rather

Table 5.4: Bias–variance components for the three methods whose slopes can be reconstructed reliably in every simulation run. The prediction columns exclude the irreducible noise variance of 1.

		Slope risk		Excess prediction risk	
Model	Method	Squared bias	Variance	Squared bias	Variance
Model 1	CG–FPLS	0.1069	0.5259	0.1102	0.1604
	Arnoldi FPLS	0.0082	0.7520	0.0132	0.1163
	FPCR	0.0098	0.6602	0.0164	0.1102
Model 2	CG–FPLS	1.2263	4.2796	0.4733	0.5224
	Arnoldi FPLS	0.0990	5.9656	0.0491	0.3903
	FPCR	0.2419	5.4197	0.1106	0.3857
Model 3	CG–FPLS	0.0183	0.4282	0.0346	0.1883
	Arnoldi FPLS	0.0055	0.6642	0.0098	0.1230
	FPCR	0.0072	0.6078	0.0135	0.1141

than approximately 0.01. The reduction in variance outweighs the added bias, giving CG–FPLS a lower mean ISE than the other two methods. For prediction, however, both its squared bias and variance are larger: 0.110 and 0.160, compared with 0.013 and 0.116 for Arnoldi FPLS. Adding the common noise variance gives expected MSPEs of approximately 1.271 and 1.130.

Model 2 makes this difference especially clear. CG–FPLS reduces slope variance by about 1.69 relative to Arnoldi FPLS, which more than offsets its larger squared bias when measuring the slope error. Once the errors are weighted for prediction, CG–FPLS has squared bias 0.473 and variance 0.522, both above the Arnoldi values of 0.049 and 0.390. A method can therefore estimate the slope more accurately overall while predicting responses less accurately. Model 3 shows the same pattern on a smaller scale. Early stopping does not always mean more bias and less variance under every measure of error: its effect depends on whether the goal is to recover the slope or to predict the response.

5.6 Raw and Arnoldi FPLS when the raw fit is stable

The preceding results reflect both the tuning rules and the numerical calculations used by each method. To examine the numerical differences more closely, Figure 5.3 and Table 5.5 consider only cases in which the raw Krylov basis satisfies the stability rule. All four methods are evaluated on those same cases. This additional comparison helps explain their behaviour; it does not replace the main comparison using the full set of results.

In these stable cases, raw and Arnoldi FPLS give almost identical results. Their Model 3 mean ISEs differ by less than 3×10^{-7} and their mean MSPEs by less than 10^{-8}. The small differences in Models 1 and 2 arise because each implementation selects

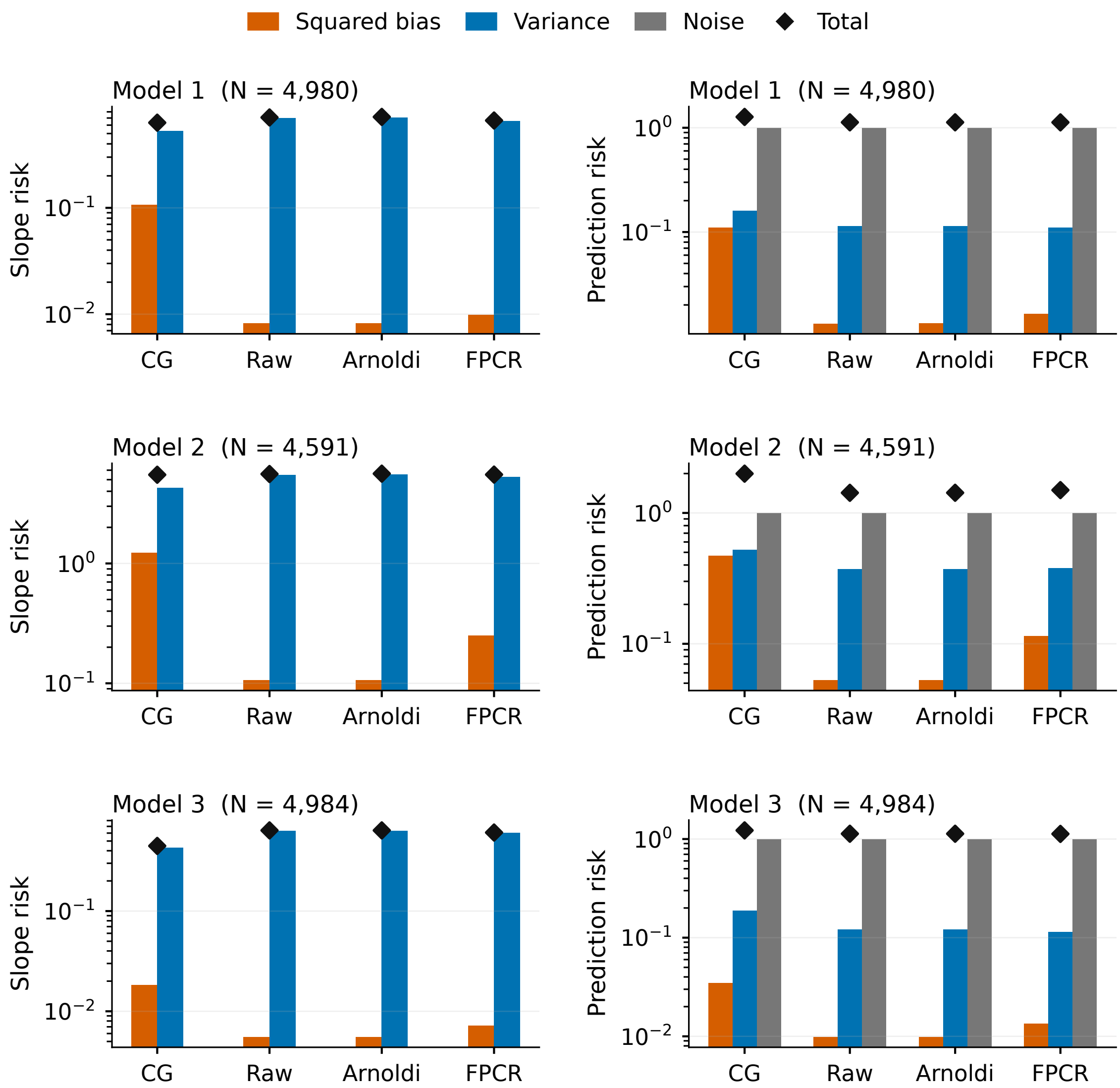


Figure 5.3: Bias–variance decomposition for cases in which raw FPLS satisfies its numerical-stability rule. All four methods are evaluated on exactly the same simulation runs within each model. Each row is one model, with the number of included runs shown; slope risk is on the left and prediction risk on the right. Bar colours and black total-risk diamonds have the same meaning as in Figure 5.2.

Table 5.5: Mean errors of raw and Arnoldi FPLS, using the same simulation runs for both methods and including only cases in which raw FPLS is numerically stable.

		Mean slope ISE		Mean test MSPE	
Model	R_{stable}	Raw	Arnoldi	Raw	Arnoldi
Model 1	4,980	0.7085	0.7166	1.1279	1.1282
Model 2	4,591	5.5739	5.6121	1.4271	1.4282
Model 3	4,984	0.6384	0.6384	1.1302	1.1302

its dimension from its own cross-validation errors. Small numerical differences in those errors can lead to different selected dimensions even before the instability threshold is reached.

This finding agrees with the result in Chapter 3: at the same dimension, when the relevant systems have full rank, raw and Arnoldi FPLS target the same response-least-squares solution in exact arithmetic. Their computed solutions remain close while the raw basis is usable. The advantage of Arnoldi is that its orthogonal basis, combined with a QR solve in the response space, allows this solution to be computed reliably at larger dimensions.

Numerical stabilisation does not change the underlying statistical target or guarantee a smaller error for every sample. Two-pass orthogonalisation (CGS2) gives a reliable basis for the same sequence of Krylov spaces, while QR avoids solving normal equations in the response space. The next section examines how the tuning rules affect individual coefficients and different parts of the estimated slope.

5.7 Distribution of the recovered slope

The coefficient and pointwise plots in Appendix B help explain the overall error measures. Figures B.1– B.3 show the first three basis coefficients, and Figures B.4– B.6 show the estimated slope at five locations. The raw-FPLS panels include only fits that satisfy its stability rule; the other methods use all 5,000 simulation runs.

The plotted coefficients are recovered from the estimated slope values on the finite grid. If $\boldsymbol{\Phi} \in \mathbb{R}^{J\times T}$ has entries $\Phi_{jt} = \phi_j(s_t)$, the coefficient vector is

$$\widehat{\boldsymbol{b}} = (\boldsymbol{\Phi}\boldsymbol{\Phi}^{\mathsf{T}})^{-1}\boldsymbol{\Phi}\widehat{\boldsymbol{\beta}}, \tag{5.1}$$

calculated using a Moore–Penrose pseudoinverse. This is a least-squares recovery on the grid, rather than a calculation using continuous inner products. Because the grid includes both endpoints, it also differs from simply taking discrete inner products with the nominal basis functions.

The coefficient distributions show how early stopping affects the estimated slope. In Model 1, CG–FPLS estimates the second coefficient with mean 0.366 when its true value is 0.616, and the third with mean 0.071 rather than 0.206. Its first coefficient is also shrunk, but much less: 3.867 rather than 4. Since 90.2% of CG fits stop at one component, the method captures the main direction associated with the response while leaving much of the weaker structure unresolved. Raw FPLS, Arnoldi FPLS, and FPCR have less bias in these later coefficients, but their estimated slopes vary more across samples.

The pattern changes with the model. In Model 2, where all first five slope coefficients equal 4, the CG–FPLS means for the first three are 3.930, 3.848, and 3.887. In Model 3, the larger leading score variances are associated with CG–FPLS means of 0.599 and 0.198 for the second and third coefficients, close to the true values 0.616 and 0.206. This agrees with the substantial improvement in CG slope ISE from Model 1 to Model 3.

The pointwise distributions show how these coefficient differences affect the slope itself. CG–FPLS estimates generally vary less, but their distributions can be visibly shifted away from the true slope, particularly in Model 1. Model 2 has the widest distributions for every method. Its oscillatory slope is harder to represent with just a few directions, so several directions must be estimated.

The normal quantile–quantile plots in Figures B.7– B.9 are approximately straight through their centres for most combinations of method and coefficient. Some bend near the extremes, and some CG coefficients show more pronounced curvature. The stopping rule can select different dimensions for different samples, so the resulting distribution combines estimates obtained at different stopping points. These plots should therefore not be taken as evidence that the whole estimated function follows a simple Gaussian distribution.

Together, the results show that performance depends on both the goal of the analysis and numerical reliability. CG–FPLS has the lowest mean slope error in these models, but early stopping can leave useful predictive information unresolved. Arnoldi preserves the full-rank APLS target while keeping the calculation stable as raw power-basis directions become nearly collinear.

These findings are limited to the three chosen models, densely observed curves measured without error, Gaussian scores, $n = 100$, and the stated tuning rules. They do not establish a universal ranking. The spectroscopy study in Chapter 8 considers observed data, with supporting diagnostics and sensitivities in Appendix D.

Choosing a small m can benefit slope estimation. For hypothesis testing, Chapter 6 instead requires the CG residual to be negligible relative to sampling uncertainty of order $n^{-1/2}$. The same stopping rule therefore cannot be used unchanged.

Chapter 6

Inference Methodology

Chapter 5 showed that stopping CG–FPLS early can improve estimation of the slope function. Hypothesis testing places a different demand on the method. If the iterations stop too soon, the error remaining in the fitted equation can affect the test result. The procedure of Babii et al. (2026) therefore continues until the fitted empirical moment $\widehat{K}\widehat{\beta}_m$ is sufficiently close to $\widehat{r}$: their difference must become negligible relative to the root-n sampling fluctuation. This chapter explains why that requirement matters and how it is assessed.

The analysis focuses on the moment-residual CG–FPLS method defined in Section 3.7. Its conclusions apply to this method and do not automatically extend to raw FPLS, Arnoldi response-space FPLS, or FPCR from Chapter 5. The chapter introduces the hypothesis and test statistic, explains the stopping requirement and choice of critical value, and then considers power and confidence sets. The final section describes the simulation design used to assess these ideas in Chapter 7.

6.1 Hypothesis and test statistic

The starting point is the centred scalar-on-function model

$$Y_i^c = \langle X_i^c, \beta \rangle + \varepsilon_i, \qquad \mathbb{E}(\varepsilon_i \mid X_i) = 0, \tag{6.1}$$

and the normal equation $K\beta = r$. The empirical moments $\widehat{K}$ and $\widehat{r}$ are defined in (3.3), with their discrete versions given in (3.5). For a specified candidate slope b, the question is whether the true slope equals this function:

$$H_0 : \beta = b \qquad \text{against} \qquad H_1 : \beta \neq b. \tag{6.2}$$

The hypothesis concerns the whole slope function. It does not ask only whether $\beta(s)$ takes a particular value at one point. Choosing $b = 0$ tests whether the functional predictor has any effect, although other candidate slopes can also be tested. To assess how often the test rejects a true null hypothesis, the simulation uses the known generating slope as b.

There is an identification limit, as explained in Section 2.5. If K is not injective, the observations determine only $K\beta$. They cannot distinguish two slopes whose difference lies in $\mathrm{Null}(K)$. Without further restrictions, the hypothesis that can be tested is therefore

$$H_0 : K\beta = Kb. \tag{6.3}$$

Restricting β and b to $\mathcal{H}_{\mathrm{id}} = \mathrm{Null}(K)^{\perp}$ makes this hypothesis equivalent to (6.2). References to testing the slope below use this restriction. A change lying entirely in $\mathrm{Null}(K)$ cannot be detected by a test based on the functional linear model.

Let $\widehat{\beta}_m^{\mathrm{CG}}$ be the estimate that minimises the empirical moment residual in the mth Krylov space, as defined in (3.29). The test statistic after m iterations is

$$\mathcal{T}_{n,m}(b) = n \left\| \widehat{K}\big(\widehat{\beta}_m^{\mathrm{CG}} - b\big) \right\|^2. \tag{6.4}$$

When the stopping index depends on sample size, write $m = m_n$ and use $\mathcal{T}_n(b) = \mathcal{T}_{n,m_n}(b)$, as in (1.5). The calligraphic symbol $\mathcal{T}_n$ denotes this scalar statistic; T_n continues to denote the sampling operator introduced in Chapter 3.

The multiplication by $\widehat{K}$ has a practical purpose. A test based directly on $\|\widehat{\beta}_m - b\|$ would be affected by the instability of estimating slope components associated with small covariance eigenvalues. For $\widehat{K}(\widehat{\beta}_m - b)$, the sampling fluctuation has a root-n limit under the moment conditions stated below. This allows inference about the identifiable part of the slope without requiring the unweighted slope estimate itself to have such a limit.

6.2 Why inference requires later stopping

Two differences explain the role of stopping. The first measures how well the candidate slope b satisfies the empirical equation; the second is the error remaining after m CG iterations:

$$u_n(b) = \widehat{r} - \widehat{K}b, \qquad e_m = \widehat{r} - \widehat{K}\widehat{\beta}_m^{\mathrm{CG}}. \tag{6.5}$$

They give the exact identity introduced in (2.44):

$$\widehat{K}(\widehat{\beta}_m^{\mathrm{CG}} - b) = u_n(b) - e_m. \tag{6.6}$$

Under the null, $u_n(b)$ reflects sampling variation, while e_m reflects the error left by stopping the iterations. A statistic based directly on the empirical equation is

$$\mathcal{S}_n(b) = n\|u_n(b)\|^2 = n\|\widehat{r} - \widehat{K}b\|^2. \tag{6.7}$$

Expanding the squared norm in (6.6) gives

$$\mathcal{T}_{n,m}(b) - \mathcal{S}_n(b) = -2n\langle u_n(b), e_m\rangle + n\|e_m\|^2, \tag{6.8}$$

and hence

$$|\mathcal{T}_{n,m}(b) - \mathcal{S}_n(b)| \leq 2\sqrt{n}\|u_n(b)\|\sqrt{n}\|e_m\| + n\|e_m\|^2. \tag{6.9}$$

Under the null, $\sqrt{n}\|u_n(b)\| = O_{\mathbb{P}}(1)$. The bound in (6.9) therefore shows that the two statistics become equivalent if the fitting residual decreases faster than $n^{-1/2}$. This gives the sufficient late-stopping condition

$$\sqrt{n}\,\|\widehat{r} - \widehat{K}\widehat{\beta}_{m_n}^{\mathrm{CG}}\| = o_{\mathbb{P}}(1). \tag{6.10}$$

When this condition holds, $\mathcal{T}_{n,m_n}(b) = \mathcal{S}_n(b) + o_{\mathbb{P}}(1)$: the error caused by stopping no longer affects the limiting null distribution. The argument is given in Proposition A.40 and Theorem A.41.

The condition in (6.10) is stronger than the estimation rule in (3.42). That rule stops when $\|e_m\|$ falls below a data-dependent constant multiple of $n^{-1/2}$. Such a threshold can provide useful regularisation, but it does not ensure that $\sqrt{n}\|e_m\|$ tends to zero. Additional iterations can therefore make the unweighted slope estimate less accurate while making the test statistic more accurate. The two tasks place different demands on the fitted moment.

The number of iterations needed depends on the covariance spectrum, the sample and resulting Krylov spaces, and numerical precision. Following the paper's null-distribution experiment, the present null-law study and the fixed-index stopping comparison use $m_{\text{late}} = 70$ at $n = 100$ as a conservative choice to be checked. Power is studied using the direct-moment statistic, while the confidence-set illustration uses $m_C = 10$. Neither fixed iteration count replaces the requirement in (6.10).

At the full empirical Krylov dimension, the residual is zero in exact arithmetic whenever $\widehat{r}$ lies in Range$(\widehat{K})$. Reaching that dimension is unnecessary for first-order inference if the residual is already small enough, and can be undesirable for slope estimation. The stopping study therefore examines how the residual and test statistic change as the iterations continue.

6.3 The null distribution and choice of critical value

Suppose the observations are independent and identically distributed, $\mathbb{E}\|X^c\|^4 < \infty$, and $\mathbb{E}(\varepsilon^2 \mid X) \leq \sigma^2_{\max} < \infty$. Under $H_0 : K\beta = Kb$, and with the known zero means used in the simulations, the empirical null moment is exactly

$$u_n(b) = \frac{1}{n}\sum_{i=1}^{n} \varepsilon_i X_i^c. \tag{6.11}$$

If sample means are used instead, the additional centring term is asymptotically negligible under these moment conditions. The covariance operator of the random score εX^c is

$$V = \mathbb{E}\big\{\varepsilon^2(X^c \otimes X^c)\big\}, \qquad Vh = \mathbb{E}\big[\varepsilon^2\langle X^c, h\rangle X^c\big]. \tag{6.12}$$

The Hilbert-space central limit theorem yields

$$\sqrt{n}\, u_n(b) \rightsquigarrow G, \qquad G \sim \mathcal{N}_{\mathcal{H}}(0, V). \tag{6.13}$$

When the late-stopping condition holds, the test statistic has the same limit as $\mathcal{S}_n(b)$. Letting $\{\omega_j\}$ denote the positive eigenvalues of V, the spectral representation of G and Parseval's identity give

$$\mathcal{T}_n(b) \rightsquigarrow \|G\|^2 = \sum_{j\geq 1} \omega_j Z_j^2, \qquad Z_j \overset{\text{iid}}{\sim} \mathcal{N}(0,1). \tag{6.14}$$

The series is well defined because V is trace class under the stated moment conditions. Its weights depend on the joint distribution of X and the regression error, so the limiting law contains unknown population quantities. This is why a critical value cannot simply be taken from an ordinary chi-square distribution with known degrees of freedom.

When the errors are conditionally homoskedastic, $\mathbb{E}(\varepsilon^2 \mid X) = \sigma^2$, then

$$V = \sigma^2 K, \qquad \omega_j = \sigma^2 \lambda_j, \tag{6.15}$$

where $\{\lambda_j\}$ are the positive eigenvalues of K. Under heteroskedasticity, V and K need not share eigenfunctions. The homoskedastic simplification is proved in Proposition A.42.

Let $q_{1-\alpha}$ denote the $(1-\alpha)$ quantile of the law in (6.14). The level-α rule is

$$\varphi_n(b) = \mathbf{1}\{\mathcal{T}_n(b) > q_{1-\alpha}\}. \tag{6.16}$$

If the limiting law is non-degenerate, it is continuous at this quantile. The probability of rejecting a true null hypothesis then approaches α, as stated in Corollary A.43. This is an asymptotic result; it does not guarantee an exact rejection rate in a finite sample.

The study distinguishes three ways of choosing critical values.

Using the known simulation model: oracle calibration. The simulation specifies the score variances of the continuous model and the error variance. These known values allow independent Z_j to be drawn and combined as

$$Q^{\text{oracle}} = \sum_{j=1}^{J} \sigma^2 \lambda_j Z_j^2, \tag{6.17}$$

and the critical value is estimated from 50,000 draws of this quantity. The term *oracle* indicates that the calculation uses population information supplied by the simulation design. Even with this information, the test need not have exact size at finite n.

There is also a distinction between the continuous model and its numerical grid. The equally weighted grid includes both endpoints, so the basis functions are only approximately orthonormal under the grid inner product. To describe the resulting difference, let $\Phi = (\phi_j(s_t))_{j,t} \in \mathbb{R}^{J\times T}$, $G_T = T^{-1}\Phi\Phi^\top$, and $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_J)$. Under the grid inner product, the non-zero covariance eigenvalues are those of $\Lambda^{1/2} G_T \Lambda^{1/2}$. Writing these eigenvalues as $\nu_{T,j}$, with zeros included if necessary, gives the fixed-grid limit

$$Q_T = \sum_{j=1}^{J} \sigma^2 \nu_{T,j} Z_j^2. \tag{6.18}$$

With $T = 200$ held fixed, a larger sample does not remove the difference between $\nu_{T,j}$ and λ_j. For fixed J, a finer grid gives $G_T \to I_J$ and recovers the continuous-model reference. The reported experiment uses (6.17). Any disagreement with that reference can therefore reflect finite-sample effects, grid approximation, and Monte Carlo error in the simulated distribution and critical value. It cannot be attributed solely to finite-sample error relative to the exact fixed-grid limit.

The oracle draws are generated independently for each model. Models with the same theoretical weights may consequently receive slightly different estimated critical values. These small differences arise from simulation uncertainty and do not indicate different theoretical null distributions.

Using simulated null samples: empirical-null calibration. A second approach repeatedly generates samples under a known null hypothesis and takes the required quantile of their test statistics. This avoids the continuous-model weighted chi-square approximation, although the estimated critical value still has Monte Carlo uncertainty. The power study uses this approach separately for each model and sample size. It is available because the null model is known in the simulation; an observed dataset alone does not provide independent samples from a known null model.

Estimating critical values from observed data. In an application, the unknown features of the null distribution must be estimated. For a specified null, define the residual $\widehat{\varepsilon}_i(b) = Y_i^c - \langle X_i^c, b\rangle$. One route replaces V by the covariance estimator

$$\widehat{V} = \frac{1}{n}\sum_{i=1}^{n} \widehat{\varepsilon}_i(b)^2 (X_i^c \otimes X_i^c) \tag{6.19}$$

and uses its estimated eigenvalues to simulate a weighted chi-square distribution. A non-parametric bootstrap offers another way to approximate the null law. Both approaches are discussed by Babii et al. (2026). A centred-multiplier method is also available as an exploratory option in the accompanying implementation. The reported null-size and confidence-set results, however, use only oracle calibration. Chapter 7 therefore provides no empirical assessment of these methods for estimating critical values from observed data.

6.4 Alternatives and power

Power is the probability that a test detects a departure from the null hypothesis. For a fixed alternative with $K(\beta - b) \neq 0$, and with the required moment convergence and sufficiently late stopping,

$$\frac{1}{n}\mathcal{T}_n(b) \xrightarrow{\mathbb{P}} \|K(\beta - b)\|^2 > 0. \tag{6.20}$$

The statistic therefore grows without bound and the probability of rejection approaches one. The condition $K(\beta - b) \neq 0$ excludes changes in the slope that lie in the null space and are invisible in the data.

A more demanding question is whether the test can detect departures that become smaller as the sample grows. Consider

$$H_{1,n}: \quad \beta_n = b + n^{-1/2}\Delta, \qquad \Delta \in \mathcal{H}. \tag{6.21}$$

Here the departure is in a fixed direction Δ, with magnitude decreasing at rate $n^{-1/2}$. Under the same late-stopping condition,

$$\mathcal{T}_n(b) \rightsquigarrow \|G + K\Delta\|^2. \tag{6.22}$$

Writing $V\phi_j = \omega_j\phi_j$ makes the shift explicit:

$$\|G + K\Delta\|^2 = \sum_{j\geq 1} \omega_j Z_j^2 + 2\sum_{j\geq 1} \sqrt{\omega_j} Z_j \langle \phi_j, K\Delta\rangle + \|K\Delta\|^2. \tag{6.23}$$

The change that enters the limiting statistic is $K\Delta$. Thus the size of a slope difference alone does not determine how easily it can be detected. Under homoskedasticity, writing $\Delta = \sum_j d_j v_j$ in the eigenbasis of K gives

$$\|K\Delta\|^2 = \sum_{j\geq 1} \lambda_j^2 d_j^2. \tag{6.24}$$

For slope changes with comparable coefficient norms, components associated with larger covariance eigenvalues contribute more to this shift. Departures in these directions are typically easier to detect than those concentrated in weak directions. More generally, power also depends on how $K\Delta$ is oriented relative to the covariance operator V, so $\|K\Delta\|$ alone does not determine it. These local results concern each fixed direction separately; they do not guarantee uniform power over all unit-norm directions in an ill-posed problem. The fixed and local alternative results are proved in Proposition A.44.

The finite-sample power experiment uses the perturbation family

$$\beta_\delta(s) = \beta(s) + \delta\Delta(s), \qquad \Delta(s) = s, \qquad \delta \in \{-1, -0.95, \ldots, 0.95, 1\}. \tag{6.25}$$

The candidate being tested remains the baseline $b = \beta$. At $\delta = 0$ the null is true; each fixed non-zero δ gives a fixed departure from it at the chosen sample size. To represent the shrinking alternatives in (6.21), δ would instead need to vary with sample size as $\delta = c/\sqrt{n}$ for a fixed constant c.

The power study uses the direct statistic $\mathcal{S}_n(b)$ in (6.7), evaluated exactly in the generating score coordinates. This keeps the large experiment manageable and measures the finite-sample power of the statistic that sufficiently late-stopped CG–FPLS approximates to first order. The curves do not measure power after a fixed $m = 70$ iterations. A separate stopping study checks the quality of that approximation under the null.

For each model and $n \in \{100, 200\}$, the power study uses an empirical null quantile at the 5% level. Monte Carlo standard errors for the rejection frequencies follow the binomial formula in (4.26). They measure uncertainty conditional on the simulated critical values and do not include the additional uncertainty from estimating an oracle or empirical-null quantile.

6.5 Confidence sets by test inversion

A confidence set can be formed by collecting all candidate slopes that the test does not reject. For a level-α test on an identified parameter space $\Theta \subseteq \mathcal{H}_{\mathrm{id}}$, this gives

$$\mathcal{C}_{1-\alpha} = \{b \in \Theta : \mathcal{T}_n(b) \leq q_{1-\alpha}\} . \tag{6.26}$$

If the test has asymptotic size α, then $\mathbb{P}_\beta\{\beta \in \mathcal{C}_{1-\alpha}\} \to 1 - \alpha$; see Corollary A.45. The identification restriction matters here too. Without it, adding a function in $\mathrm{Null}(K)$ does not change the test, so the set describes slopes that the data cannot distinguish. In a finite sample, $\widehat{K}$ also has finite rank. An unrestricted inverted set therefore extends without bound in directions in $\mathrm{Null}(\widehat{K})$.

To make the set easier to display, candidate functions are restricted to a p-dimensional space:

$$\mathcal{H}_p = \mathrm{span}\{h_1, \ldots, h_p\}, \qquad b_c(s) = \sum_{j=1}^{p} c_j h_j(s), \tag{6.27}$$

where the illustration uses the first $p = 5$ cosine functions from the simulation basis. Let H be the $T \times p$ matrix whose jth column contains h_j on the numerical grid, and define

$$A = \widehat{K} H, \qquad f = \widehat{K}\widehat{\beta}^{\mathrm{CG}}_{m_C}, \tag{6.28}$$

and recall that the discrete squared L^2 norm is $T^{-1}\|\cdot\|_2^2$. The accepted coefficient set is exactly

$$C^{(p)}_{1-\alpha} = \left\{ c \in \mathbb{R}^p : \frac{n}{T}\|f - Ac\|_2^2 \leq q_{1-\alpha} \right\}. \tag{6.29}$$

When A has full column rank, let

$$\widehat{c} = (A^\top A)^{-1} A^\top f, \qquad \rho^2 = \frac{T q_{1-\alpha}}{n} - \|f - A\widehat{c}\|_2^2. \tag{6.30}$$

If $\rho^2 < 0$, no candidate in the restricted space is accepted and the set is empty. Otherwise, the accepted coefficients form the ellipsoid

$$(c - \widehat{c})^\top A^\top A (c - \widehat{c}) \leq \rho^2. \tag{6.31}$$

For the row vector $h(s)^\top = (h_1(s), \ldots, h_p(s))$, the smallest and largest accepted function values at each point s are

$$h(s)^\top \widehat{c} \pm \left\{\rho^2 h(s)^\top (A^\top A)^{-1} h(s)\right\}^{1/2}. \tag{6.32}$$

Proposition A.46 derives (6.31)–(6.32) and describes the cases where A is rank deficient. The analytic calculation gives the exact envelope of the full ellipsoid in $\mathbb{R}^5$. The reference illustration instead searches over 20^5 coefficient vectors in $[0, 4.5]^5$. Both approaches invert the test, but the earlier search also restricts the coefficients to a box and checks only finitely many points. The displayed sets therefore need not coincide.

The confidence illustration uses $m_C = 10$, following the separate choice in the reference study; the null-law experiment uses $m_{\text{late}} = 70$. Ten components do not by themselves establish (6.10). Their adequacy must be assessed through finite-sample coverage and residual diagnostics. This choice is therefore specific to the illustration, and its interpretation as a confidence set remains conditional on a sufficiently small moment residual.

There is also a distinction between the full slope and its five-term approximation. The simulated true slopes have non-zero coefficients beyond the fifth basis function, so they do not belong to the space used for the display. The study accordingly reports three different summaries:

1. how often the test does not reject the full true simulated slope;
2. how often the true slope's five-term projection is accepted within $\mathcal{H}_5$; and
3. the lower and upper pointwise envelope of the accepted five-dimensional coefficient set.

The shaded region in Chapter 7 combines the pointwise medians of 5,000 lower and upper envelopes. It shows a typical range across many simulated samples. It is not a 95% confidence band from one observed sample, and whether it contains the true curve does not measure coverage.

The full-slope acceptance calculation uses the same samples as the null-size experiment, but applies $m_C = 10$ in place of $m_{\text{late}} = 70$. Its average can therefore be compared with the reported late-stopped size, although the two need not add to one. Acceptance and rejection are exact complements only when the stopping index and critical value are the same. Because the samples are shared, this comparison is not an independent check of the size result.

6.6 Implementation and diagnostic design

The inference study uses the three models in Sections 4.1 and 4.2: the same 100-term Gaussian score expansion, 200-point grid, Babii cosine convention, and homoskedastic error standard deviation one. Keeping these features allows the inference results to be related to the estimation results. A separate random seed keeps the two simulation studies separate.

Table 6.1 gives the choices specific to inference. The number of simulated samples applies to every model and, for power, to every combination of sample size and perturbation value. The size study examines nominal levels of 1%, 5%, and 10%; the other parts use 5%.

Table 6.1: Design of the CG–FPLS inference study.

Study branch	Statistic and stopping choice	Sample and replications	Calibration and purpose
Null law and size	$\mathcal{T}_{n,70}$	$n = 100$; $R = 5{,}000$	50,000-draw oracle weighted chi-square law at $\alpha \in \{0.01, 0.05, 0.10\}$; compare the finite-sample null law with the continuous-design reference.
Stopping diagnostics	$\mathcal{T}_{n,m}$ and $\mathcal{S}_n$	$n = 100$; $R = 5{,}000$	Oracle 5% critical value; compare $m \in \{1, 2, 3, 5, 10, 20, 40, 70\}$, the estimation-oriented adaptive variant, and the maximum-rank fit.
Power	Direct $\mathcal{S}_n$	$n \in \{100, 200\}$; $R = 5{,}000$ for every δ	Empirical 5% null quantile for each model and n; evaluate (6.25).
Inverted-set illustration	$\mathcal{T}_{n,10}$; analytic $p = 5$ inversion	$n = 100$; $R = 5{,}000$	Oracle 5% critical value; record full-slope non-rejection, five-term projected acceptance, rank, emptiness, and envelope width.

The stopping comparison includes an adaptive rule intended for estimation. It uses the discrepancy formula and constants from Chapter 3, with $\tau = 1.01$ and $\delta_{\text{stop}} = 0.1$. Its FPCR moment-GCV variance pilot follows (4.20)–(4.21), but uses at most 20 components, compared with 70 in the estimation experiment. The comparison also uses the stabilised moment-residual path described below in place of the estimation study's short recurrence. These differences matter: the smaller pilot cap can change the estimated variance, the discrepancy threshold, and the selected stopping index. This is therefore a variant of the procedure in Chapter 5.

The notation δ_{stop} distinguishes the tuning constant from the power perturbation δ in (6.25). A maximum-rank fit provides another benchmark by projecting $\widehat{r}$ onto the numerical range of $\widehat{K}$. For every stopping rule, the study records

$$\sqrt{n}\|e_m\|, \qquad |\mathcal{T}_{n,m}(b) - \mathcal{S}_n(b)|, \qquad \mathbf{1}\{\mathcal{T}_{n,m}(b) > q_{0.95}\}, \tag{6.33}$$

together with the selected or fixed index. These quantities show how much moment residual remains, how closely the fitted statistic matches the direct statistic, and whether the test rejects. They allow the effect of stopping on null rejection rates to be assessed.

The moment-residual path is computed in a numerically stabilised form because the inference study takes many more iterations than the estimation rule. At each dimension, two-pass reorthogonalisation forms an orthonormal basis Q_m for $\widehat{\mathcal{K}}_m(\widehat{K}, \widehat{r})$, and a QR solve computes

$$\widehat{a}_m \in \arg\min_{a \in \mathbb{R}^m} \|\widehat{r} - \widehat{K} Q_m a\|_2, \qquad \widehat{\beta}_m^{\text{CG}} = Q_m \widehat{a}_m. \tag{6.34}$$

In exact arithmetic, this solves the same minimum-moment-residual problem at each dimension as the original short recurrence. The orthogonal basis improves numerical stability and can change the path in finite precision. It does not change the objective to that of Arnoldi FPLS in Chapter 3. Both methods use an orthogonal Krylov basis, but Arnoldi FPLS minimises response error, while (6.34) minimises the CG–FPLS moment residual. Propositions A.35 and A.38 establish the relevant distinctions between their spaces and objectives.

All random quantities are generated from the master seed 2025. Separate random streams are used for the stopping experiment and the alternative power samples. Some quantities are deliberately shared: the confidence experiment uses the null-size samples, the $n = 100$ empirical power calibration uses the direct null statistics, and the same oracle draws serve all relevant parts of the study. These choices make the calculations repeatable and clarify which comparisons share simulated information.

Interpreting the results first requires checking whether the remaining moment residual is small enough for the fitted statistic to behave like the direct statistic. Chapter 7 therefore begins with the stopping diagnostics, followed by null size, power, and confidence-set behaviour.

Chapter 7

Inference Results

This chapter examines how well the CG–FPLS test performs in the three simulation models. Chapter 6 identified two reasons why its rejection rate may differ from the intended level: stopping the iterations too early, and using an approximate null distribution to choose the critical value. The results first check whether $\mathcal{T}_{n,m}$ is sufficiently close to the direct statistic $\mathcal{S}_n$. This check is needed before differences from the weighted chi-square reference can be attributed mainly to calibration. The chapter then examines rejection under the null, power against departures from it, and the candidate slopes accepted by test inversion.

The focus is the CG–FPLS procedure for inference. The theory studied here depends on continuing the CG iterations until the moment residual is small enough. Raw FPLS, Arnoldi FPLS, and FPCR retain their roles as estimation and prediction methods in Chapter 5; the present results do not provide a further comparison of all four methods.

7.1 Results and reporting conventions

All simulated samples are included in the analysis. None of the recorded numerical results is missing or infinite. Checks based on the saved individual results reproduce the corresponding reported summaries to floating-point precision. The supporting material contains the study settings, individual statistics and diagnostics, five numerical summary tables, and five vector figures.

The four parts of the study follow the design in Table 6.1. The null-distribution study compares $\mathcal{T}_{n,70}$ at $n = 100$ with 50,000 draws from the continuous-model oracle weighted chi-square reference. The stopping study uses separately generated samples to compare fixed iteration counts, an adaptive variant intended for estimation, and the maximum-rank fit. Power is measured using the direct statistic $\mathcal{S}_n$, with an empirical null critical value for each model and sample size. The confidence-set illustration uses $\mathcal{T}_{n,10}$, oracle calibration, and an analytic five-dimensional coefficient ellipsoid. These differences in the statistic and critical value matter when interpreting each set of results.

Unless stated otherwise, reported probabilities are based on 5,000 simulated samples for each model and design point. Monte Carlo standard errors are calculated using (4.26), treating the simulated critical value as fixed. They therefore exclude the uncertainty in estimating either a 50,000-draw oracle quantile or a 5,000-draw empirical null quantile. That additional uncertainty has not been quantified. The error bars show only the Monte Carlo variation conditional on the chosen critical values.

7.2 How stopping affects the test

The stopping study asks whether the discrepancy rule used for estimation also reduces the fitting residual enough for inference. The adaptive rule uses the earlier threshold formula and constants, with the modifications described in Section 6.6: its variance pilot is limited to 20 components instead of 70, and the iterations use a stabilised moment-residual calculation in place of the short recurrence. The results therefore concern a variant of the procedure studied in Chapter 5.

Figure 7.1 shows how three quantities change as the iterations continue: the scaled residual $\sqrt{n}\|e_m\|$, the difference $|\mathcal{T}_{n,m} - \mathcal{S}_n|$, and the frequency with which the test rejects a true null hypothesis.

Table 7.1 reports the adaptive rule and two fixed iteration counts. The adaptive rule chooses a median of one, two, and one components in Models 1–3. It selects $m = 1$ in 99.96% of Model 1 samples, $m = 2$ in 92.94% of Model 2 samples, and $m = 1$ in 63.88% of Model 3 samples. These early choices are consistent with its estimation objective. For testing, however, they lead to null rejection frequencies of 52.70%, 96.02%, and 71.94%. The remaining residual is large enough to cause severe distortion in these designs. As the theory explains, this error can affect the statistic at first order, rather than merely cause a small error in calibration.

The adaptive rule does meet its own stopping criterion: all 15,000 fits satisfy their sample-specific threshold. On the $\sqrt{n}$ scale, the median thresholds are 11.82, 12.57, and 15.25, each above the corresponding median residual in Table 7.1. The rule therefore behaves as intended. Its threshold can be suitable for regularised estimation while still allowing an order-one scaled residual, which fails the vanishing requirement in (6.10).

The rejection rate does not move steadily towards 5% as m increases. For example, $m = 2$ gives 2.64% rejection in Model 1 and 99.82% in Model 2, while $m = 3$ gives

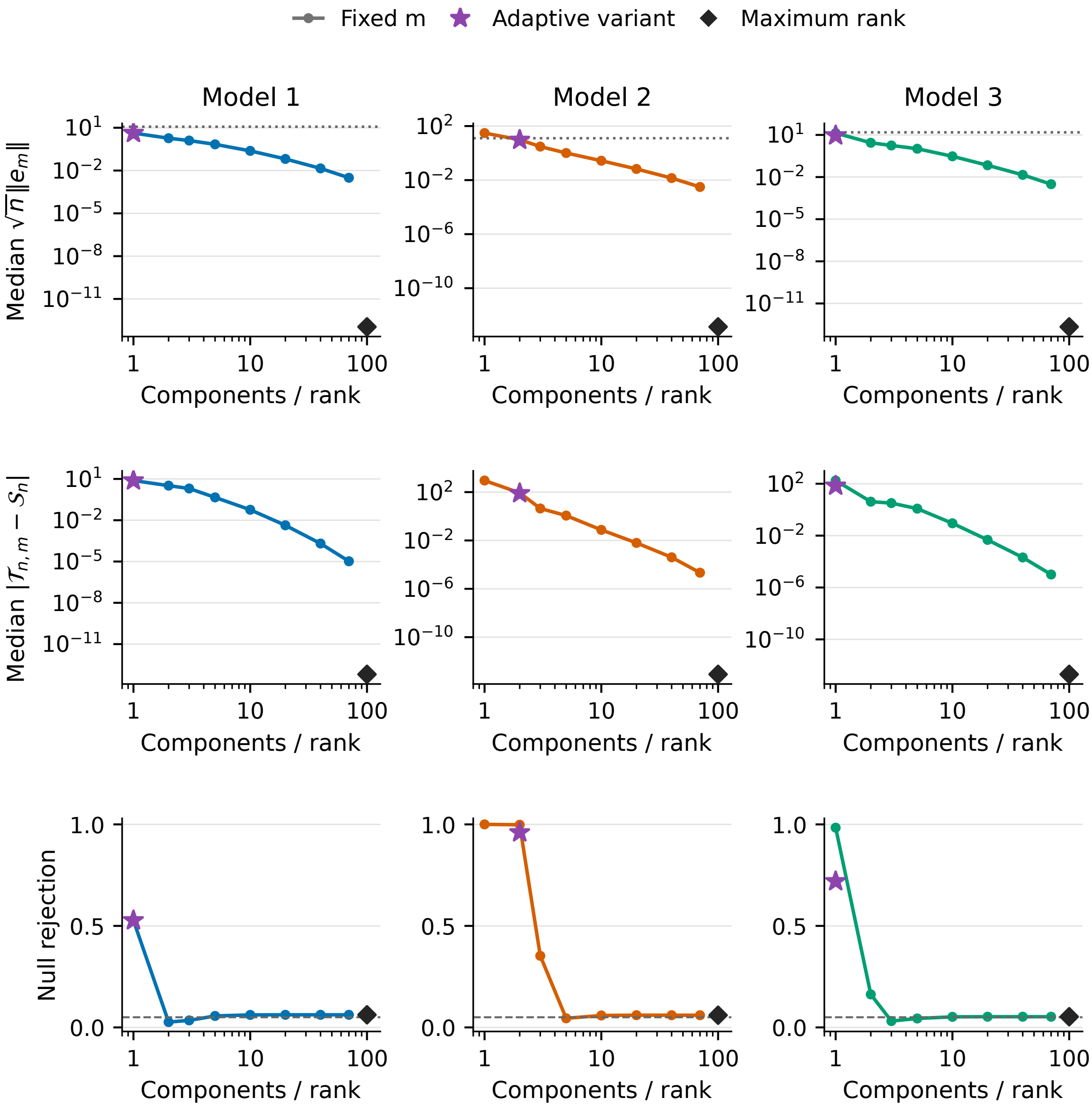


Figure 7.1: Effect of the stopping index on CG–FPLS inference at $n = 100$. The top row shows the median scaled moment residual $\sqrt{n}\|e_m\|$ from (6.5). The dotted line is the median estimation-oriented stopping threshold for $\sqrt{n}\|e_m\|$; it is not the vanishing residual required for inference. The middle row shows the median absolute difference between the finite-m and direct statistics, and the bottom row shows rejection under the null using the oracle 5% critical value. Circles denote fixed m, stars the adaptive variant with a 20-component pilot cap, and diamonds the maximum-rank fit. Each summary uses the 5,000 simulated samples generated for the stopping study.

Table 7.1: Selected stopping diagnostics from the separate study of 5,000 simulated samples per model. The residual and statistic-difference columns report medians; the final column reports the empirical rejection percentage.

Model	Rule	Median m	Median $\sqrt{n}\|e_m\|$	Median $\lvert\mathcal{T}_{n,m} - \mathcal{S}_n\rvert$	Rejection (%)
Model 1	Adaptive	1	4.2452	7.6719	52.70
	$m = 10$	10	0.2392	0.05770	6.16
	$m = 70$	70	0.003113	1.05×10^{-5}	6.20
Model 2	Adaptive	2	9.4103	79.9799	96.02
	$m = 10$	10	0.2711	0.07506	5.88
	$m = 70$	70	0.003109	2.16×10^{-5}	6.00
Model 3	Adaptive	1	9.0848	69.7139	71.94
	$m = 10$	10	0.3005	0.08947	5.20
	$m = 70$	70	0.003113	1.01×10^{-5}	5.26

3.12% in Model 3. At $m = 5$, all three rates lie between 4.40% and 5.64%, even though the median difference between the fitted and direct statistics still ranges from 0.458 to 1.203. A rejection rate close to the intended level at one intermediate step can therefore be misleading. Evidence that the iterations have gone far enough comes from the sustained decrease in the residual and statistic difference.

By $m = 10$, the median statistic differences are below 0.09, and the rejection rates are already close to those obtained with many more iterations. This supports the separate $m_C = 10$ confidence-set illustration for these models, but does not establish ten components as a general rule for inference. At $m = 70$, the median scaled residuals are approximately 0.0031 in all three models, and the median statistic differences are of order 10^{-5}. Continuing to maximum rank reduces both quantities to numerical roundoff, but changes no rejection decision relative to $m = 70$ in the stopping study.

The principal size study, based on separately generated samples, gives a further check. Across its 5,000 samples per model, the largest absolute differences between $\mathcal{T}_{n,70}$ and $\mathcal{S}_n$ are 9.37×10^{-5}, 4.02×10^{-4}, and 1.18×10^{-4}. The two statistics give identical decisions at all of the 1%, 5%, and 10% critical values used. Iteration error is therefore negligible for these decisions. The remaining differences from the nominal rejection rates reflect finite-sample calibration, the fixed-grid approximation in (6.18), and Monte Carlo error in the reference distribution and critical value. More CG iterations cannot remove these other sources of error.

7.3 The null distribution and rejection rates

Figure 7.2 compares the finite-sample distribution of $\mathcal{T}_{n,70}$ with the continuous-model oracle reference in (6.17). Their centres agree closely in all three models. The main differences appear in the upper quantiles that determine rejection.

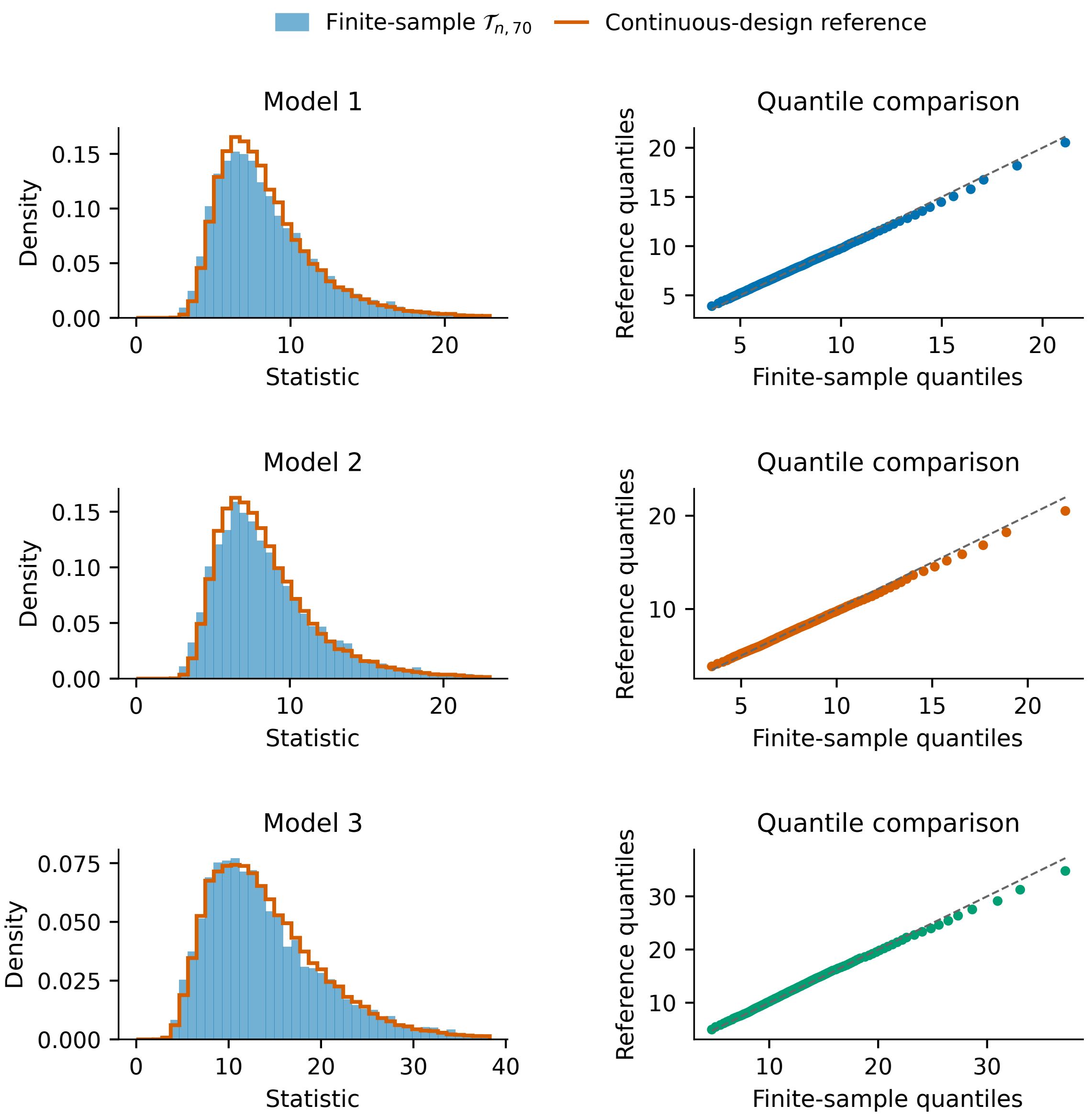


Figure 7.2: Finite-sample null calibration of the late-stopped CG–FPLS test. Rows correspond to Models 1–3. The left column overlays the empirical density of $\mathcal{T}_{n,70}$ from 5,000 samples at $n = 100$ with 50,000 draws from the continuous-design oracle weighted chi-square reference. The right column plots finite-sample quantiles horizontally and reference quantiles vertically against the identity line. Different density scales across models reflect the different null dispersions. Agreement is strongest in the centre; points below the line in the upper tail indicate larger finite-sample quantiles.

In Table 7.2, the empirical and reference means differ by only 0.05, 0.12, and 0.02 in Models 1–3. The finite-sample distributions are more dispersed: their variances exceed the simulated reference values by 11.6%, 20.9%, and 15.6%, respectively. Their 95th percentiles are also higher, by 3.47%, 3.87%, and 3.71%. The Q–Q plots show the same pattern. Points near the centre follow the diagonal, while the upper tail extends beyond what the continuous-model reference predicts. Part of this comparison reflects the numerical grid, on which the basis is only approximately orthonormal. That difference persists as n increases if T remains fixed.

Table 7.2: Null-distribution and empirical-size summary at $n = 100$. Quantities labelled "reference" use the 50,000-draw continuous-design oracle weighted chi-square sample. Rejection entries are percentages based on 5,000 simulated samples.

	Mean		$q_{0.95}$		Rejection probability		
Model	Empirical	Reference	Empirical	Reference	1% level	5% level	10% level
Model 1	8.590	8.543	15.582	15.059	1.20	5.90	11.18
Model 2	8.666	8.542	15.762	15.176	1.48	5.84	11.72
Model 3	14.273	14.253	27.333	26.355	1.46	6.10	10.72

Models 1 and 2 have the same score variances and error variance. Their direct-moment null distributions and weighted chi-square limits are therefore identical despite their different slopes: under the null, the slope cancels from the score average in (6.11). The small differences in the table come from independently simulated samples and oracle draws, with only a negligible numerical contribution from the $m = 70$ residual. Model 3 has a different covariance design and a wider reference distribution. Its 95th percentile is near 26.36, compared with about 15.1 for the other models.

At the principal 5% level, rejection occurs in 5.90%, 5.84%, and 6.10% of samples. The conditional Monte Carlo standard errors are about 0.33–0.34 percentage points. These rates suggest that the oracle-calibrated test rejects a true null slightly too often at $n = 100$. A fuller assessment would also need to include uncertainty in the simulated critical values. The 1% and 10% results show the same broad tendency, although the size of the difference varies across models and levels. Figure C.1 in Appendix C shows the 5% results with conditional Monte Carlo error bars.

This pattern agrees with the quantile comparison: the oracle reference understates the upper tail and sets critical values that are slightly too low for these finite samples. The $m = 70$ and direct statistics already give the same decisions, so further iterations would not correct the difference. The power study uses empirical null critical values to separate the ability to detect slope changes from this modest error in oracle calibration.

7.4 Power against fixed perturbations

The power experiment tests $b = \beta$ using responses generated with slope $\beta_\delta(s) = \beta(s) + \delta s$. Non-zero δ specifies a fixed departure at each sample size, rather than the root-n local sequence in (6.21). The statistic is $\mathcal{S}_n$, rather than the CG statistic stopped at $m = 70$. For each model and n, the critical value is the empirical 95th percentile of a separate null distribution (Section 6.4).

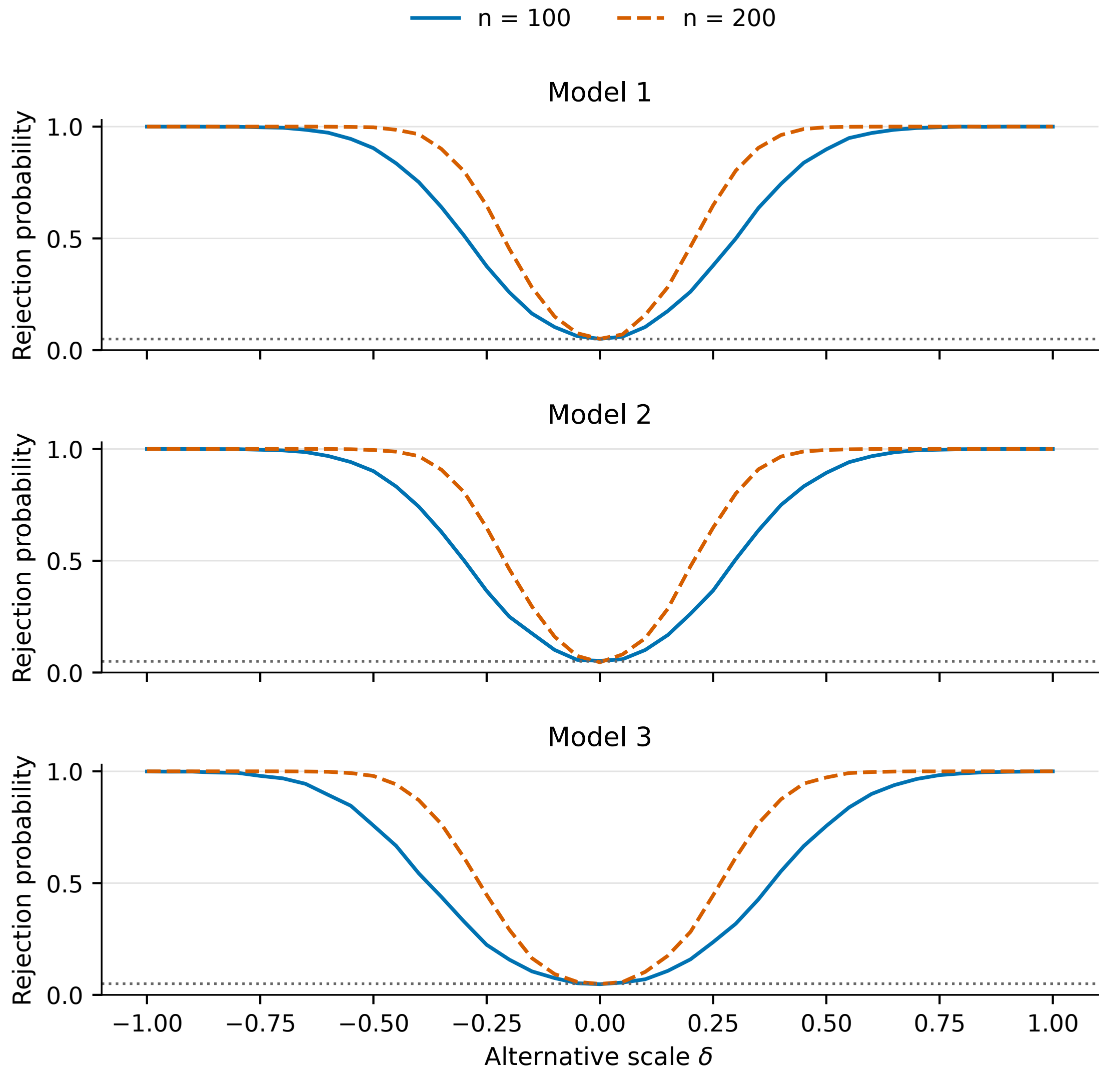


Figure 7.3: Empirical power against $\beta_\delta(s) = \beta(s) + \delta s$. Solid and dashed curves correspond to $n = 100$ and $n = 200$, respectively, and the dotted line marks 5%. Every point uses 5,000 alternative samples and the direct-moment statistic $\mathcal{S}_n$. Its empirical null critical value is calculated separately for each model and sample size. The alternatives use fixed δ; these are not finite-$m = 70$ power curves or a direct simulation of root-n local alternatives.

At $\delta = 0$, rejection stays close to 5% in all six model–sample-size combinations (Table 7.3). It need not equal 5%: the critical value and the plotted $\delta = 0$ rate use separate sets of simulated samples. The approximately symmetric curves agree with the two-sided quadratic test and symmetric Gaussian design. Across the grid, rejection at δ and $-\delta$ differs by at most 1.38 percentage points. These rates use separate samples, so their difference can have a standard error of about one percentage point. The asymmetry is modest relative to simulation variation.

Table 7.3: Selected rejection probabilities from the power curves (percent). The non-zero values shown here are from the positive branch of the approximately symmetric alternative grid.

	$n = 100$			$n = 200$		
Model	$\delta = 0$	$\delta = 0.25$	$\delta = 0.50$	$\delta = 0$	$\delta = 0.25$	$\delta = 0.50$
Model 1	5.08	37.88	89.84	5.08	64.82	99.74
Model 2	5.24	36.70	89.34	4.58	64.80	99.54
Model 3	4.78	23.60	75.58	4.90	44.56	97.26

Doubling sample size increases power in every model. At $\delta = 0.25$, power rises from approximately 37% to 65% in Models 1 and 2, and from 23.6% to 44.6% in Model 3. At $\delta = 0.50$, Models 1 and 2 rise from about 90% to more than 99.5%, and Model 3 from 75.6% to 97.3%. Larger samples make a fixed departure easier to distinguish from random variation.

Models 1 and 2 have almost identical power because they share the same predictor covariance and perturbation direction. Subtracting $\langle X, b\rangle$ leaves $\delta\langle X, s\rangle + \varepsilon$ in both; small differences between their curves reflect simulation variation. Model 3 has lower power for moderate $|\delta|$ along this direction, reflecting both $K\Delta$ and its more dispersed null distribution. This does not imply that Model 3 has lower power against every slope change.

These results agree with the consistency result and show gains from larger samples along $\Delta(s) = s$. They do not verify the finite-m CG approximation under every alternative. The simulated null distributions used for calibration would also be unavailable from a single observed dataset.

7.5 Confidence sets in five dimensions

The final experiment forms confidence sets by inverting $\mathcal{T}_{n,10}$ over the first five simulation basis functions. Figure 7.4 shows the resulting functional envelopes. Table 7.4 separates three summaries needed to interpret them: non-rejection of the complete true slope, acceptance of its five-term truncation, and the pointwise width of the five-dimensional set.

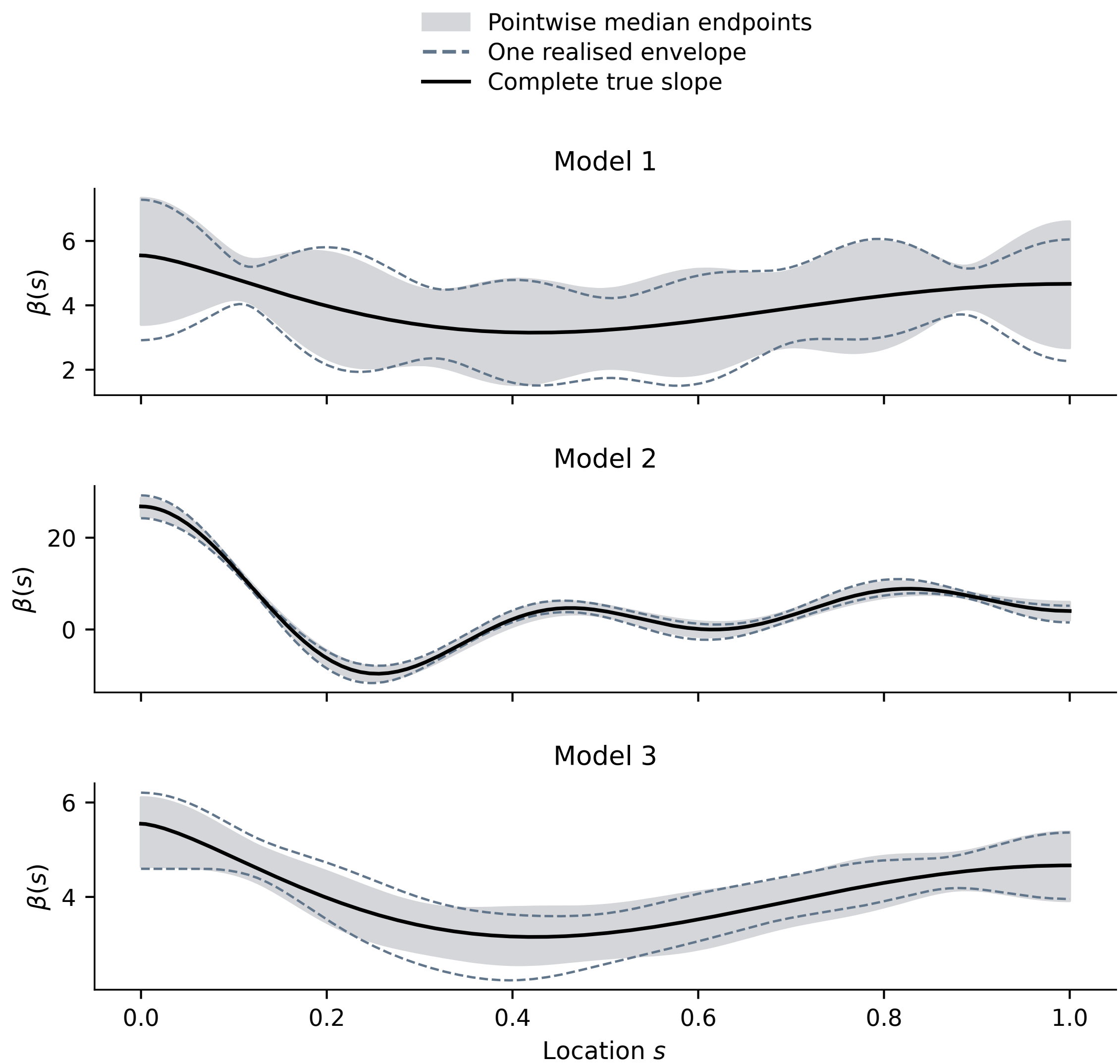


Figure 7.4: Confidence-set illustration at $n = 100$, using $m_C = 10$, a five-dimensional analytic ellipsoid, and the oracle 5% critical value. The black curve is the complete true slope. The grey region joins the pointwise medians of the lower and upper boundaries from 5,000 separately calculated envelopes; it is not a single 95% confidence band. The dashed boundaries come from the sample whose mean envelope width is closest to the median width. All grey and dashed envelopes range only over the five-dimensional basis span. A curve can lie between the pointwise boundaries without belonging to the coefficient ellipsoid.

Table 7.4: Summary of the analytic five-dimensional inverted sets. The first two entries are acceptance frequencies in percent. For each sample, width is averaged over the 200-point functional grid; the table gives the mean and median of these sample-specific widths. The final entry gives empty and unbounded frequencies in percent. Every quantity uses 5,000 simulated samples.

Model	Full-slope non-rejection	Five-term acceptance	Mean width	Median width	Median rank	Empty/ unbounded
Model 1	94.20	94.32	2.810	2.798	5	0/0
Model 2	94.20	94.12	2.824	2.817	5	0/0
Model 3	94.02	93.94	1.118	1.114	5	0/0

All 15,000 sets are non-empty, bounded, and have rank five. The complete true slope is not rejected in 94.20%, 94.20%, and 94.02% of samples; its five-term truncation is accepted in 94.32%, 94.12%, and 93.94%. The latter frequencies concern the displayed five-dimensional candidate space. Data still come from the complete slope, whose non-zero remainder is not modelled as a nuisance component. These are therefore not coverage results for a projection model with an unknown remainder.

The full-slope non-rejection rates are not exact complements of the principal null-size results. The calculations use the same samples and oracle critical values, but one evaluates $\mathcal{T}_{n,10}$ and the other $\mathcal{T}_{n,70}$. This changes only 5, 6, and 6 of the 5,000 decisions in Models 1–3. Non-rejection at $m_C = 10$ is 94.20%, 94.20%, and 94.02%, compared with 94.10%, 94.16%, and 93.90% from subtracting the reported $m = 70$ sizes from one. The agreement supports the stopping diagnostics; shared samples make it a paired comparison, rather than independent coverage evidence.

Models 1 and 2 have almost identical mean widths, about 2.81 and 2.82, consistent with their shared predictor covariance. Model 3's mean width is 1.12, approximately 60% smaller. Stronger leading score variances provide more information about these five coefficient directions, despite Model 3's larger critical value. This advantage concerns the chosen subspace and parameterisation, not uniformly narrower uncertainty for the entire infinite-dimensional slope.

Neither the median envelope nor the representative boundaries should be interpreted as a realised simultaneous confidence band. The interpretation of the $m_C = 10$ inversion also depends on the finite-iteration residual being sufficiently small in these particular designs.

The adaptive estimation variant leaves a large moment residual on the root-n scale and severely distorts the test. Continuing the stabilised Krylov path makes this error negligible in these designs. At $m = 70$ and $n = 100$, rejection is slightly above nominal under continuous-design oracle calibration; the gap includes finite-sample effects, grid approximation, and uncertainty in the simulated critical value. With empirical null calibration, direct-moment power rises sharply with sample size. Inversion gives non-empty, bounded, full-rank finite-dimensional sets in all three designs.

These findings concern dense Gaussian curves, homoskedastic errors, one perturbation direction, and oracle or simulation-based null calibration. They do not establish the performance of feasible plug-in or bootstrap calibration. Chapter 8 examines prediction with the four regularisation procedures on spectroscopy curves. Formal testing there would require feasible calibration whose finite-sample behaviour has yet to be assessed. Appendix D provides the full preprocessing, tuning, stability, and sensitivity details.

Chapter 8

Empirical Application: Pharmaceutical-Tablet Spectroscopy

In Chapters 5 and 7, the known slope and covariance operator made it possible to distinguish the effects of regularisation from numerical error. This chapter examines whether the same patterns appear in observed data. It uses near-infrared (NIR) spectra to predict the active-ingredient assay of pharmaceutical tablets with CG–FPLS, raw FPLS, Arnoldi FPLS, and FPCR. The focus is prediction and numerical stability. Because the true slope is unknown, the application cannot assess slope recovery or use the oracle-assisted inference studied in Chapters 6–7.

The evaluation rules were fixed before the four final models were applied to the supplied test set. However, the test responses and preliminary ordinary PCR/PLS results had already been inspected during dataset selection. The test set is therefore called a *designated benchmark*, rather than a prospectively blind holdout. The results compare the fitted procedures on these tablets; they do not establish how well the methods would perform in future use.

8.1 Data and prediction problem

The data come from the 2002 pharmaceutical-tablet NIR "Shoot-out" described by Hopkins (2003) and distributed in MATLAB format by Eigenvector Research (2026). The supplied file contains measurements from two spectrometers, divided into 155 calibration tablets, 40 validation tablets, and 460 test tablets. The total is 655, although the online catalogue reports 654. All 655 complete records are retained, and this discrepancy is documented. Both instruments measured the same tablets, so their spectra are treated as paired measurements, not additional independent observations.

The main analysis uses instrument 1 and all $T = 650$ wavelengths, from 600 to 1898 nm at 2-nm intervals. Assay, the third response column in the data file, is the sole response. The file does not specify its physical unit, so errors are reported in the supplied assay units. Combining the calibration and validation tablets gives a development sample of 195; the 460 test tablets form the designated benchmark. Figure 8.1 shows common patterns across the spectra, a less regular high-wavelength tail, and an unusually narrow assay range in the validation split.

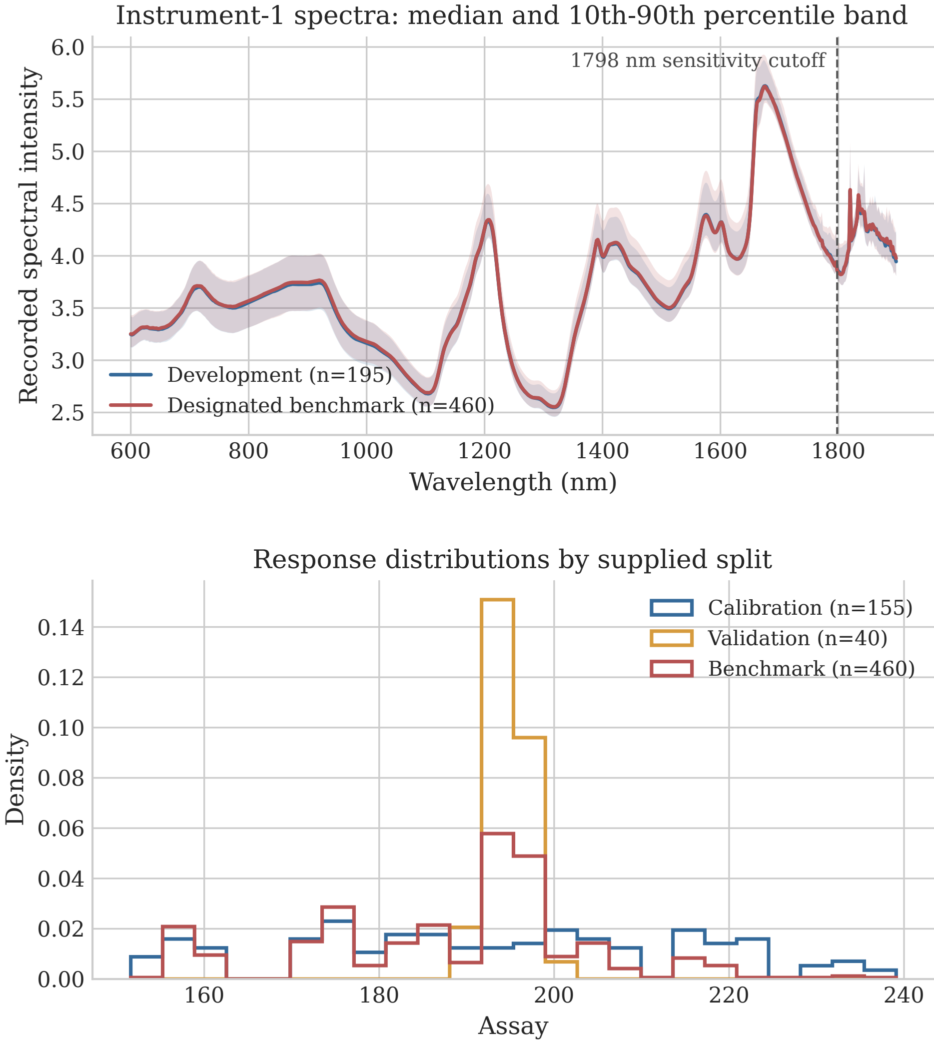


Figure 8.1: Instrument-1 spectra and supplied assay distributions. The spectral panel gives the median and 10th–90th percentile band for the development and designated-benchmark samples; the dashed line marks the prespecified 1798 nm boundary used only in a sensitivity analysis. The response panel separates the calibration, validation, and benchmark splits.

There are 650 spectral coordinates and only 195 development observations. After centring, the sample covariance has rank at most 194, leaving at least 456 directions that the development data cannot identify. This is the finite-sample form of the ill-posed problem discussed earlier.

The equally weighted grid convention from Section 3.1 is retained:

$$\langle f, g\rangle_T = T^{-1} f^\top g, \qquad \widehat{K} = \frac{X^\top X}{nT}, \qquad \widehat{r} = \frac{X^\top y}{n}. \tag{8.1}$$

The resulting prediction for a new spectrum x is

$$\widehat{Y}(x) = \bar{Y}_{\text{train}} + \frac{1}{T}(x - \bar{X}_{\text{train}})^\top \widehat{\beta}. \tag{8.2}$$

The prediction uses the means calculated from the training sample and preserves the scaling used throughout the thesis.

8.2 Evaluation procedure

In the main analysis, each wavelength and the response are centred using only the observations in the training fold. The same training means are then used for the corresponding validation observations. No other preprocessing is applied: there is no variance scaling, smoothing, differentiation, standard-normal-variate transformation, multiplicative-scatter correction, baseline correction, or wavelength selection. This keeps additional regularisation through preprocessing out of the main comparison. The planned analyses of trimmed spectra, smoothed spectra, and instrument 2 are reported in Appendix D.

Each method keeps its own regularisation rule. Raw and Arnoldi FPLS independently select a dimension by minimising prediction mean squared error over at most 70 requested Krylov dimensions, using the same five inner folds. Raw FPLS uses unscaled covariance powers and response-space normal equations. Arnoldi uses twice reorthogonalised Gram–Schmidt and response-space QR. FPCR selects a spectral cut-off by response-space GCV. CG–FPLS uses the conjugate-residual recurrence from the estimation study and the discrepancy rule with $\tau = 1.01$ and $\delta = 0.10$, including its internal moment-GCV variance pilot. A component count therefore does not represent the same effective degrees of freedom across methods. The comparison evaluates the four complete procedures, following the three-axis framework in Section 3.10.

Development performance is assessed using five repeats of five-fold outer cross-validation. The calibration and validation tablets are split separately, with folds balanced over assay. Each outer fold contains 31 calibration and 8 validation tablets. Within every outer training sample, Raw and Arnoldi share a new five-fold inner partition for tuning. Each repeat gives an out-of-fold performance summary. The five summaries describe development performance, while the 25 overlapping fits are not treated as independent experiments.

After the development analysis and implementation checks, each method is fitted once to all 195 development tablets. Raw and Arnoldi use one fixed, shared inner partition for final tuning. The fitted models then predict the 460 benchmark tablets once. RMSE is the primary measure of performance; MAE, signed bias (prediction minus observation), and $R^2 = 1 - \text{SSE}/\text{SST}$ are secondary measures.

Uncertainty is summarised using 10,000 paired iid bootstrap resamples of the benchmark tablets. Every method uses the same resampled tablets, and the fitted models are held fixed. Circular moving-block bootstraps with lengths 5, 10, and 20 assess sensitivity to possible dependence in the ordering of the rows. The exact cross-validation folds and bootstrap indices are saved with the supporting results.

8.3 Benchmark prediction

All four final fits and their benchmark predictions are finite. Table 8.1 reports the measures chosen in advance. Every method explains substantial variation in benchmark assay, with R^2 between 0.887 and 0.914. Raw FPLS has the lowest point estimate of RMSE, 4.607, followed by Arnoldi FPLS at 4.643. FPCR and CG–FPLS have RMSEs of 5.185 and 5.293, respectively.

Table 8.1: Primary designated-benchmark results for full-spectrum instrument 1 ($n = 460$). Brackets give 95% percentile intervals from 10,000 paired iid bootstrap resamples of the benchmark tablets. The intervals beside RMSE describe each method separately. Differences between methods are assessed using paired contrasts in Table 8.2.

Method	m	RMSE [95% interval]	MAE	Bias	R^2
CG–FPLS	3	5.293 [4.367, 6.477]	3.673	0.177	0.887
Raw FPLS	55	4.607 [3.645, 5.785]	3.036	-0.303	0.914
Arnoldi FPLS	5	4.643 [3.693, 5.805]	3.086	-0.301	0.913
FPCR	53	5.185 [4.299, 6.255]	3.547	-0.628	0.892

Figure 8.2 shows that the methods tend to struggle with the same tablets. Raw and Arnoldi give particularly similar predictions: their correlation is 0.999934, their root mean squared difference is 0.174 assay units, and their maximum difference is 0.445. The close agreement allows their RMSE difference to be estimated more precisely than either RMSE alone. Thus the wide individual intervals in Figure 8.3 can overlap almost completely while the paired comparison gives a much narrower interval.

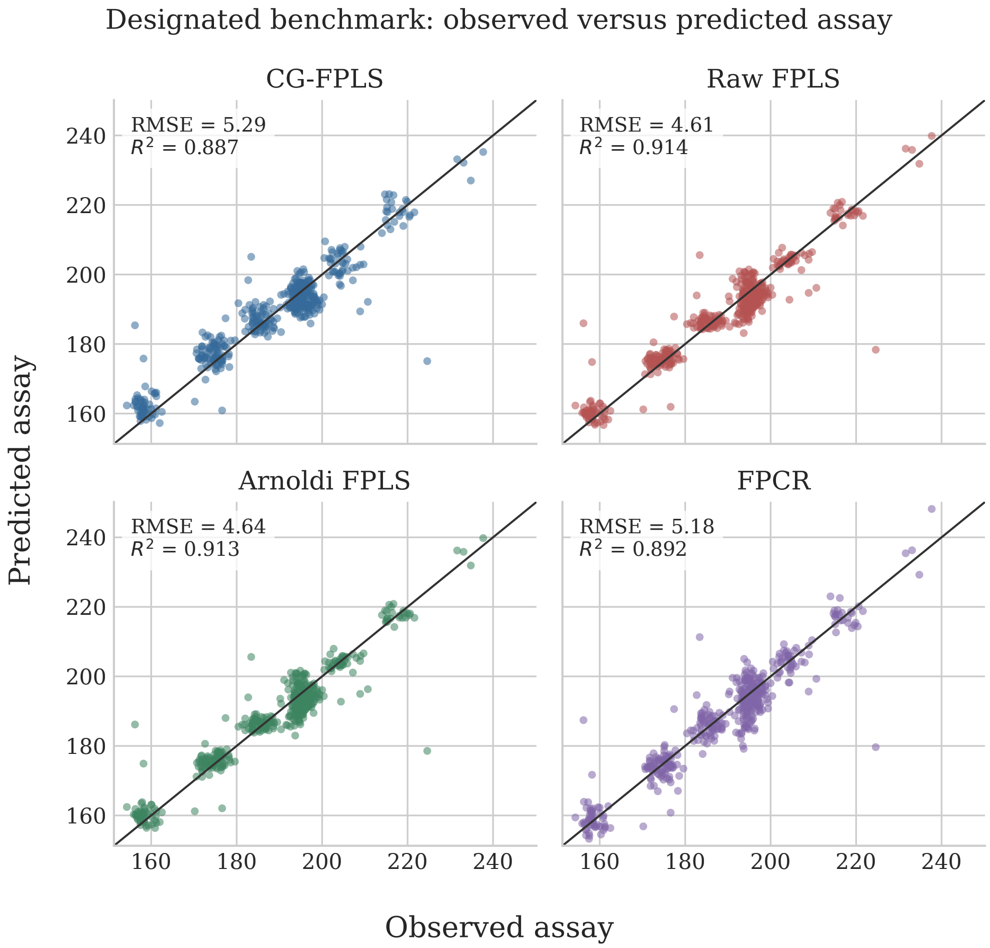


Figure 8.2: Observed and predicted assay on the designated benchmark. The diagonal is the one-to-one line. The four methods capture the main linear signal and share several difficult tablets near the extremes of the assay range.

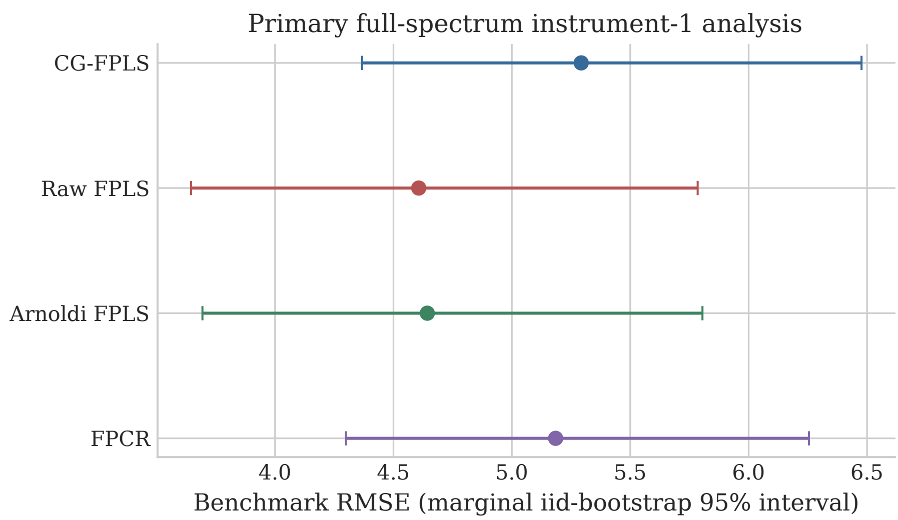


Figure 8.3: Benchmark RMSE estimates and individual 95% percentile intervals from the paired iid bootstrap. Their overlap does not test whether two methods differ. The prespecified paired differences are reported in Table 8.2.

The main paired comparison in Table 8.2 is $\mathrm{RMSE}_{\mathrm{raw}} - \mathrm{RMSE}_{\mathrm{Arnoldi}} = -0.036$ assay units. Its iid percentile interval excludes zero, as do all three moving-block intervals. Yet the difference is only 0.78% of the Arnoldi RMSE and is small relative to the prediction errors themselves. Judging its practical importance would require a known assay scale and an application-specific tolerance. The interval compares these two fitted models on the supplied tablets; it does not show that the raw representation is generally better. Arnoldi's RMSE advantages over CG–FPLS and FPCR are larger, approximately 0.65 and 0.54 units, and remain under the principal block-length-10 bootstrap.

Table 8.2: Declared benchmark RMSE contrasts. Positive values favour Arnoldi FPLS. The iid and circular block-length-10 columns give 95% percentile intervals from 10,000 paired resamples. Models are held fixed throughout the bootstrap.

Contrast	Difference	Iid interval	Block-10 interval
Raw − Arnoldi	-0.036	[-0.057, -0.017]	[-0.062, -0.014]
CG − Arnoldi	0.650	[0.397, 0.921]	[0.257, 1.017]
FPCR − Arnoldi	0.542	[0.286, 0.844]	[0.201, 0.922]

8.4 Selection and numerical stability

The leading pair differs more in numerical stability than in prediction. Across the 25 development outer fits, Arnoldi selects between four and six components, with median five and no numerical flag. Raw FPLS selects between four and 70 requested components, with median 44. It reaches the cap twice and is flagged in 24 of 25 fits. Across the five repeats, RMSE has mean and standard deviation 5.082 ± 0.517 for Raw, compared with 4.917 ± 0.131 for Arnoldi. Much of this difference comes from one poor raw repeat, and the order of the two methods reverses when contiguous folds are used. These CV results share observations: they describe stability, without establishing a separate statistical ranking.

The final raw fit selects nominal dimension 55. Both its normalised power-basis condition number and its response-design condition number reach the reporting ceiling, 4.50×10^{15}, and the fit is flagged. The reported value is capped, so it should not be read as an exact condition number. The fixed $1/\sqrt{\epsilon}$ threshold is already crossed by $m = 6$. Thus, finite predictions alone do not show that the raw coordinates are numerically reliable. Arnoldi selects five components, with basis condition number one, response-design condition number 11.95, and orthogonality defect 7.4×10^{-16}. It therefore achieves essentially the same predictions as Raw using stable five-dimensional coordinates. The complete paths appear in Appendix D.

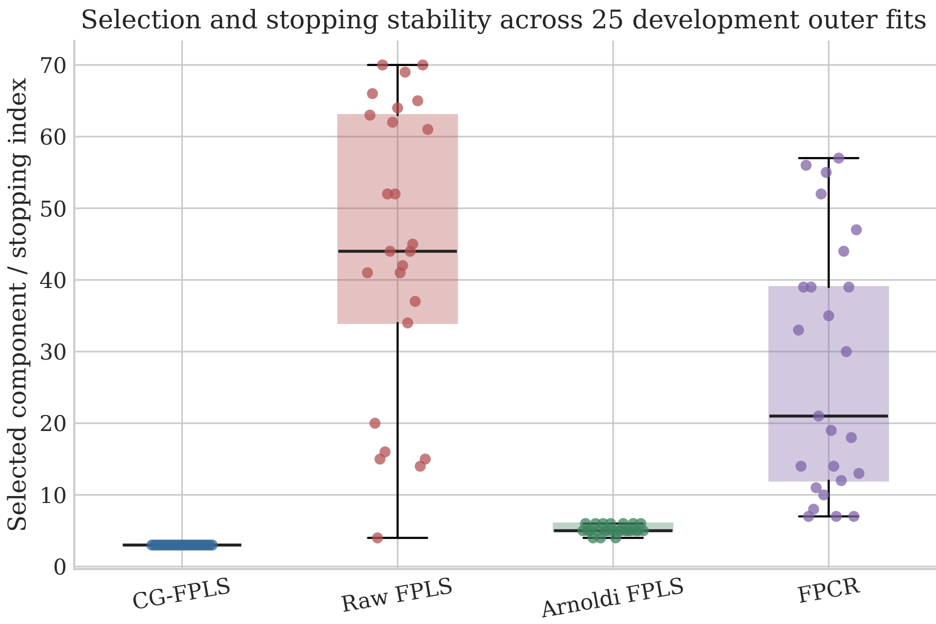


Figure 8.4: Selection and stopping indices across the 25 development outer fits. Points show individual fits and boxes summarise their distributions. Raw and Arnoldi use inner cross-validation, FPCR uses GCV, and CG–FPLS uses discrepancy stopping. These counts therefore do not measure the same effective degrees of freedom.

CG–FPLS stops at three components in every development fit and in the final fit. This consistent early stopping applies strong regularisation and accompanies a larger prediction error. The result concerns the particular discrepancy rule used here, including its variance pilot; it does not show that conjugate-gradient regularisation is generally inferior. FPCR selects between seven and 57 components in development and 53 in the final fit. Component counts therefore reflect different search spaces and tuning rules; they are not directly comparable measures of complexity.

8.5 Robustness, interpretation, and scope

The planned sensitivity analyses show that the detailed ranking depends on the instrument and preprocessing choices. Truncating instrument 1 at 1798 nm lowers the RMSEs of CG–FPLS, Arnoldi, and FPCR, bringing all four results into the range 4.63–4.77. Arnoldi then has the smallest point RMSE. Fixed Savitzky–Golay smoothing favours Raw at 4.66, but does not improve every method. Using the full spectrum from instrument 2 increases RMSE for all four methods, with Raw and Arnoldi again leading. Instrument 2 measures the same tablets, so this comparison assesses sensitivity to the instrument; it provides neither independent external validation nor a calibration-transfer experiment. These changes do not establish that the visible spikes at high wavelengths caused particular prediction errors.

The residual plots in Appendix D show a few large errors shared by the methods, as well as predictions drawn toward the centre of the assay range. All these tablets remain in the analysis. Without batch, acquisition-order, or laboratory information, patterns in row order cannot be given a physical explanation. Nor can the fitted coefficient vectors identify a unique functional slope or wavelengths with a chemically causal role in this problem with 650 coordinates and rank at most 194.

Three limitations are central to interpreting the results. First, the designated benchmark had been inspected during dataset selection, so the reported errors do not come from a prospectively blind validation. Second, the paired-bootstrap intervals resample the supplied benchmark tablets while keeping the fitted models fixed. They do not include uncertainty from the development sample, preprocessing, or tuning. Third, the study examines one response from one tablet dataset, without independent validation across batches, laboratories, or acquisition periods. Within this scope, the application illustrates two findings from the controlled experiments: stopping too early can reduce prediction accuracy, and related Krylov fits can give almost identical predictions despite large differences in their numerical reliability.

Chapter 9

Conclusion and Outlook

The main finding of this thesis is that regularisation must be judged by both its purpose and its numerical reliability. A stopping rule that works well for estimating the slope may be less effective for prediction or unsuitable for inference. Even procedures with the same theoretical Krylov space can behave very differently when calculated in finite precision.

These issues arise because functional linear regression is an inverse problem. The covariance operator provides little information about weak directions, and its finite-rank sample estimate cannot identify every part of the slope. Estimation therefore requires decisions about which information to retain and how to carry out the calculation. The thesis examined these decisions through spectral truncation, raw and orthogonalised Krylov least squares, and direct conjugate-gradient iteration.

The evidence combines mathematical analysis, specified algorithms, controlled simulations, a separate inference study, and a spectroscopic application. Its conclusions depend on the designs and procedures studied. They explain how the aim of the analysis, the stopping rule, and numerical stability affect the results, without identifying a method that is best in every setting.

9.1 Answers to the research questions

9.1.1 Spectral and iterative regularisation

The first question asked how spectral and iterative regularisation compare when the covariance structure and slope change. The answer depends on what is being estimated. Under the discrepancy rule used here, CG–FPLS stops very early. This often reduces variance in the ordinary L^2 slope norm. It gives CG–FPLS the lowest mean ISE in all three models, with slope errors concentrated in a narrow range. However, early stopping can also leave out information useful for predicting the response. CG–FPLS has higher mean prediction error than Arnoldi FPLS in all three models, most clearly in Model 2, where their mean MSPEs are 1.999 and 1.442.

FPCR and FPLS use information in different orders. FPCR follows predictor variance, while FPLS builds its spaces using both $\widehat{K}$ and $\widehat{r}$. The response information can help FPLS find a useful predictive relation with relatively few components. Those components do not have the same meaning as an equal number of principal components. FPCR and Arnoldi have almost identical mean prediction errors in Models 1 and 3. In Model 2, Arnoldi has the lowest mean and median MSPE among the stable methods. These results show a trade-off between estimation aims; neither spectral nor iterative regularisation is consistently preferable.

The tablet results are consistent with a prediction cost from the implemented CG stopping rule. It stops at three steps and gives benchmark RMSE 5.293, compared with 4.607 for Raw and 4.643 for Arnoldi. The true slope is unknown, so these data cannot show whether early stopping improves slope recovery. The comparison also evaluates complete procedures and does not isolate the effect of adding CG iterations.

9.1.2 Numerical reliability of Krylov methods

The second question concerned FPLS formulations that are equivalent in exact arithmetic. At the same valid dimension, Raw and Arnoldi span the same Krylov space and minimise the same response least-squares criterion within it. Their simulation results are also almost identical on the shared set of samples for which the raw fit is numerically stable. In that comparison, each procedure still selects its own dimension. The difficulty is that raw covariance powers can become nearly dependent. Raw FPLS is flagged in 0.40%, 8.18%, and 0.32% of the 5,000 simulations in Models 1–3. The unstable fits produce extremely large ISE and MSPE values that are absent after orthogonalisation. A comparison based only on medians would miss this difference in reliability.

The tablet data make the distinction particularly clear. Twenty-four of 25 raw development fits and the final raw fit are flagged. Selected dimensions range from four to 70. For the final $m = 55$ fit, the basis and design condition numbers are recorded at the reporting ceiling of 4.50×10^{15}. Arnoldi selects between four and six components; its final $m = 5$ basis has condition number one and orthogonality defect 7.4×10^{-16}. Despite this contrast, the two final prediction vectors have correlation 0.999934 and a root mean squared difference of only 0.174 assay units. Their RMSEs differ by 0.036.

Orthogonalisation therefore offers numerical reliability without guaranteeing a lower prediction error in every dataset. It preserves the same exact fixed-m target while allowing that fit to be calculated reliably when raw power coordinates become unstable. Finite prediction errors alone do not establish that a numerical method is trustworthy.

9.1.3 Stopping for estimation and inference

The third question asked why inference may require more CG iterations than estimation. Early stopping helps estimation by limiting the amplification of sampling noise. The theory ensures that the CG-based statistic approximates the direct-moment statistic when the residual $\widehat{r} - \widehat{K}\widehat{\beta}_m$ is negligible on the root-n sampling scale. Meeting an estimation-oriented discrepancy threshold does not ensure that this stronger condition holds.

The stopping study demonstrates this with a variant of the estimation procedure. It retains the earlier discrepancy formula and constants, but limits the variance pilot to 20 components instead of 70 and uses the stabilised moment-residual calculation in place of the short recurrence. It therefore differs from the complete procedure in the estimation study. The variant usually selects one or two components and meets its own threshold, yet rejects a true null at the 5% level in 52.70%, 96.02%, and 71.94% of samples in Models 1–3. The remaining fitting residual is too large on the root-n scale.

Further iterations reduce both the scaled residual and the difference from the direct statistic. At $m = 70$, the two versions give identical decisions in the stopping study, with rejection rates of 6.20%, 6.00%, and 5.26%. In the separate principal size study, the late-stopped test with oracle calibration gives rates of 5.90%, 5.84%, and 6.10%; the direct statistic gives the same decisions. Both studies suggest slightly excessive rejection at $n = 100$. Once the iteration error is negligible, differences from the continuous-model oracle reference reflect finite-sample calibration, grid approximation, and simulation uncertainty in the reference distribution and critical value.

With empirical null calibration, the direct-moment test gains power sharply when n increases from 100 to 200. The separate confidence-set illustration uses $m_C = 10$ and $p = 5$. Every simulated sample gives a non-empty, bounded, full-rank set in this restricted space. This does not establish coverage of the full slope or turn the plotted envelopes into simultaneous confidence bands. Nor does it make $m = 70$ a universal rule. Reliable inference requires checking the scaled moment residual and assessing a calibration method that can be used with observed data.

9.1.4 Regularisation in the tablet data

The fourth question asked whether the same trade-offs appear in observed functional data. All four methods explain substantial assay variation on the designated benchmark, with R^2 from 0.887 to 0.914. Raw and Arnoldi give the strongest predictions. The paired iid interval for Raw-minus- Arnoldi RMSE is $[-0.057, -0.017]$, but the point difference is less than 1% of Arnoldi's RMSE, and the two methods make very similar errors on individual tablets. This difference between fitted models is small relative to the prediction errors. Its practical importance remains unclear without a known assay scale and an application-specific tolerance.

Arnoldi obtains essentially the same predictions with five orthonormal components as Raw does with a severely ill-conditioned nominal 55-component fit. CG's higher prediction error is consistent with the cost of early stopping seen in the simulations. FPCR's larger selected dimension reflects its different ordering of information. Trimming, smoothing, and changing instrument alter prediction errors and selected dimensions, and can reverse the Raw–Arnoldi ordering. The application thus illustrates the effects found in the controlled studies, while showing why they do not support a universal ranking.

9.2 Practical implications

A regularisation rule should reflect its purpose. Slope recovery, prediction, and hypothesis testing place different demands on the fitted model. Early stopping can reduce variation in the slope estimate while discarding useful predictive information. A good prediction rule can, in turn, stop before the moment residual is small enough for inference. One measure of fit cannot show whether all three aims have been met.

The numerical representation also matters. Methods using the same Krylov space in exact arithmetic still need a reliable basis and least-squares solution on a computer. Raw powers may work at a small dimension and fail abruptly later. To follow the APLS path, Arnoldi with reorthogonalisation and QR is a reasonable default, especially when its predictions are effectively tied with the raw fit. This does not make it best for every purpose. Condition numbers, loss of orthogonality, effective dimensions, and invalid candidates can reveal problems hidden by prediction errors alone.

Component counts also need context. A principal component, a Krylov direction, and a CG iteration are not equal units of model complexity. Each procedure therefore keeps its own fitting objective and tuning rule. The results compare complete procedures; equal component counts would not make their effective complexity equal.

Readers should be able to check more than final averages. The saved estimation results contain individual simulation losses, selected dimensions, instability measures, and quantities used for the distributional figures. Inference records contain statistics, confidence-set summaries, diagnostics, and experiment settings. Spectroscopy records also include individual predictions, coefficients, tuning paths, resampling indices, source-file copies, computing-environment details, and checksums. These records support closer examination of the findings; exact reproduction depends on the information retained for each study.

9.3 Limitations

The estimation study covers three designs with densely observed curves, Gaussian scores, no measurement error, homoskedastic errors, $n = 100$, and fixed implementation rules. These choices isolate particular mechanisms but omit irregular or sparse observations, measurement error, heavy-tailed or dependent errors, and many slope and covariance structures. The rankings depend on the chosen tuning procedures and establish neither minimax performance nor uniform superiority across broader model classes.

The inference study does not yet provide a validated procedure for observed data. Its main critical values use continuous-design population quantities known from the simulation, whose covariance weights differ slightly from those on the fixed grid. The power study instead uses critical values estimated from samples generated under the null. It considers only one direction of departure from that null, and the confidence-set display allows only five basis coefficients to vary. The reported results do not validate feasible plug-in, multiplier, or bootstrap calibration. The tablet analysis therefore makes no formal claim about functional inference.

The empirical study uses one response from one public dataset. It provides no independent test across tablet batches, laboratories, sites, or acquisition periods, and instrument 2 measures the same tablets again. The benchmark responses had already influenced dataset selection, so the supplied test split was not prospectively blind. The bootstrap intervals keep the fitted models fixed and omit uncertainty from development sampling, preprocessing, and tuning. Batch identifiers are unavailable to explain possible dependence along the dataset's original ordering, assay units are not explicit in the MAT file, and the true physical slope is unknown. The fitted coefficients allow the analysis to be reproduced and its numerical behaviour compared; they do not establish chemical causality. The public download also contains no explicit open-data licence, which limits redistribution of the raw file.

9.4 Directions for further work

The next methodological step is calibration using observed data. Plug-in, multiplier, and bootstrap approaches need finite-sample assessment that separates critical-value uncertainty from iteration error. A related task is a data-dependent stopping rule that makes the scaled moment residual sufficiently small. A useful prediction index need not meet this inferential requirement.

Numerical reliability could enter the tuning rule through conditioning and rank requirements. Among candidates with practically indistinguishable validation errors, the rule could favour the simplest stable fit. Comparisons with ridge, spline, and other functional regularisers would test whether the findings extend beyond truncation and Krylov methods.

For spectroscopy, the main priority is external validation across independent tablet batches, instruments, laboratories, and acquisition periods. Group information would support resampling that respects batch membership or observation order. A bootstrap that refits preprocessing, tuning, and the models would also capture more sources of uncertainty than the fixed-model intervals reported here. Measurement-error models and methods for irregular grids would bring the analysis closer to practical data collection. The primary response and preprocessing rules should be fixed before the external outcomes are examined.

9.5 Final perspective

The choice of regularisation involves both statistical and numerical judgement. Early stopping can improve slope estimation while still stopping too soon for valid inference. Equivalent Krylov spaces can produce very different results when their bases are computed, and a strong prediction score can coexist with an unstable fit. The practical lesson is to choose regularisation for the question being asked, use a reliable numerical representation, and examine the diagnostics alongside the reported accuracy.

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Appendix A

Definitions and Supporting Proofs

This appendix collects the definitions and mathematical arguments used from Chapter 1 through the estimation and inference chapters. Its purpose is to make the main text self-contained without interrupting the exposition with long proofs. A definition fixes the meaning of a term and therefore is not a statement that can be proved. What can be proved are the properties and consequences that follow from those definitions. Accordingly, Section A.1 records the terminology, and the remaining sections prove the results invoked in the thesis. The presentation follows standard Hilbert-space and inverse-problem arguments; see Lin (2021), Hsing and Eubank (2015), and Engl et al. (1996) for broader treatments.

Throughout, $\mathcal{H}$ is a real separable Hilbert space with inner product $\langle \cdot, \cdot \rangle$ and norm $\| \cdot \|$. In the functional regression model, $\mathcal{H} = L^2(\mathcal{I})$ for a compact interval $\mathcal{I}$. All equalities between elements of $L^2(\mathcal{I})$ are understood up to sets of Lebesgue measure zero. When the covariance operator K is not injective, slope parameters are reported through their minimum-norm representatives in the identified space

$$\mathcal{H}_{\text{id}} = \text{Null}(K)^{\perp} = \overline{\text{Range}(K)}.$$

Statements about a globally identified slope implicitly restrict the parameter space to $\mathcal{H}_{\text{id}}$, unless injectivity of K is assumed explicitly.

A.1 Definitions and notation

Definition A.1 (Inner product, norm, and orthogonality)**.** An *inner product* on a real vector space is a symmetric bilinear map $\langle \cdot, \cdot \rangle$ satisfying $\langle f, f \rangle > 0$ for every $f \neq 0$. Its induced norm is $\|f\| = \langle f, f \rangle^{1/2}$. Elements f and g are *orthogonal*, written $f \perp g$, when

$\langle f, g \rangle = 0$.

Definition A.2 (Hilbert space and orthonormal basis)**.** A *Hilbert space* is an inner-product space that is complete in its induced norm. A sequence $\{e_j\}_{j\geq 1}$ is *orthonormal* if $\langle e_j, e_k \rangle$ equals one for $j = k$ and zero otherwise. It is an *orthonormal basis* if the closure of its linear span is the whole space.

Definition A.3 (ℓ^p spaces, conjugate exponents, and dual balls)**.** For $1 \leq p < \infty$, the sequence space ℓ^p consists of all real sequences $x = (x_j)_{j\geq 1}$ for which

$$\|x\|_p = \left(\sum_{j\geq 1} |x_j|^p \right)^{1/p} < \infty.$$

The space ℓ^∞ consists of bounded sequences and has norm $\|x\|_\infty = \sup_j |x_j|$. Exponents $p, q \in [1, \infty]$ are *Hölder conjugates* when $p^{-1} + q^{-1} = 1$, using the convention $\infty^{-1} = 0$. The continuous dual E^* of a normed space E is the space of its continuous linear functionals, equipped with

$$\|\varphi\|_* = \sup_{\|x\|_E \leq 1} |\varphi(x)|.$$

In $\mathbb{R}^d$, write $B_p^d = \{x : \|x\|_p \leq 1\}$. Its polar under the standard pairing is

$$(B_p^d)^\circ = \left\{ y : \sup_{x \in B_p^d} |y^\mathsf{T} x| \leq 1 \right\}.$$

The norm-induced distance is $d_p(x, z) = \|x - z\|_p$; its radius-r ball centred at z is $z + rB_p^d$.

Definition A.4 (Linear operators)**.** For a linear operator $A : \mathcal{H} \to \mathcal{H}$, its null space and range are

$$\mathrm{Null}(A) = \{f : Af = 0\}, \qquad \mathrm{Range}(A) = \{Af : f \in \mathcal{H}\}.$$

The operator is *injective* when $\mathrm{Null}(A) = \{0\}$ and is *bounded* when its operator norm is finite, where

$$\|A\|_{\mathrm{op}} = \sup_{\|f\| \leq 1} \|Af\|.$$

A bounded operator is *self-adjoint* if $\langle Af, g \rangle = \langle f, Ag \rangle$ for all f, g, and *non-negative* if $\langle Af, f \rangle \geq 0$ for every f. An *eigenpair* (λ, v) satisfies $Av = \lambda v$ with $v \neq 0$.

Definition A.5 (Compact and trace-class operators)**.** A bounded operator A is *compact* if it maps every bounded sequence to a sequence whose image has a convergent subsequence. A non-negative self-adjoint operator is *trace class* if, for one and hence every orthonormal basis $\{e_j\}$, $\mathrm{tr}(A) = \sum_j \langle Ae_j, e_j \rangle < \infty$. Every trace-class operator is compact.

Definition A.6 (Rank-one and covariance operators). For $u, v \in \mathcal{H}$, the rank-one operator $u \otimes v$ is defined by $(u \otimes v)f = \langle v, f\rangle u$. If X is a random element with mean μ_X and $\mathbb{E}\|X\|^2 < \infty$, its covariance operator is

$$K = \mathbb{E}\{(X - \mu_X) \otimes (X - \mu_X)\}.$$

Definition A.7 (Well-posedness, ill-posedness, and regularisation). An inverse problem is *well posed* in the sense relevant here if a solution exists, is unique on the chosen parameter space, and depends continuously on the observed moments. Failure of one of these conditions makes the problem *ill posed*. A *regularisation method* replaces an unstable inverse by a family of stable approximations indexed by a tuning parameter, truncation level, or stopping index.

Definition A.8 (Condition number and Krylov subspace). For a tall matrix $A \in \mathbb{R}^{p\times m}$ with $p \geq m$, its spectral condition number is

$$\kappa_2(A) = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)}$$

when A has full column rank, and $\kappa_2(A) = \infty$ otherwise. For a symmetric positive-definite square matrix this reduces to $\kappa_2(A) = \lambda_{\max}(A)/\lambda_{\min}(A)$. For an operator A, a right-hand side r, and an integer $m \geq 1$, the order-m Krylov subspace is

$$\mathcal{K}_m(A, r) = \operatorname{span}\{r, Ar, \ldots, A^{m-1}r\}.$$

Definition A.9 (Sampling operator, adjoint, and moment residual). For centred functional observations $X_1^c, \ldots, X_n^c$, the sampling operator is $(T_n b)_i = \langle X_i^c, b\rangle$. Relative to the sample inner product $\langle a, c\rangle_n = n^{-1}a^\top c$, its adjoint is $T_n^* a = n^{-1}\sum_i a_i X_i^c$. Hence $\widehat{K} = T_n^* T_n$, $\widehat{r} = T_n^* y$, and the empirical moment residual at b is $e(b) = \widehat{r} - \widehat{K}b = T_n^*(y - T_n b)$.

Definition A.10 (PLS1, NIPALS, and APLS). *PLS1* denotes partial least squares with one scalar response. *NIPALS* is a classical sequential PLS algorithm that constructs a response-covariant score, estimates predictor and response loadings, and deflates both residual objects after each component. In the present functional setting, *APLS* denotes the alternative representation that minimises response least squares over $\mathcal{K}_m(\widehat{K}, \widehat{r})$. Under the full-rank conditions stated below, these are two representations of the same fixed-component standard functional PLS fit.

Definition A.11 (Arnoldi basis and conjugacy). An Arnoldi basis of $\mathcal{K}_m(A, r)$ is an orthonormal sequence generated by repeatedly applying A to the latest basis vector and orthogonalising the result against the existing basis. Non-zero vectors $p_0, \ldots, p_k$ are *A-conjugate* if $\langle Ap_j, p_\ell\rangle = 0$ whenever $j \neq \ell$.

Definition A.12 (Stochastic order and weak convergence). For random quantities Z_n, the notation $Z_n = O_{\mathbb{P}}(1)$ means that $\{Z_n\}$ is bounded in probability, while $Z_n = o_{\mathbb{P}}(1)$ means $Z_n \to 0$ in probability. The notation $Z_n \rightsquigarrow Z$ denotes convergence in distribution. These definitions also apply to norms of random elements in $\mathcal{H}$.

A.2 Hilbert-space geometry

Theorem A.13 (Hölder duality for ℓ^p). *Let $1 \leq p < \infty$ and let q be its Hölder conjugate. If $x \in \ell^p$ and $y \in \ell^q$, then*

$$\left| \sum_{j \geq 1} x_j y_j \right| \leq \|x\|_p \|y\|_q. \tag{A.1}$$

The map $y \mapsto \varphi_y$, where $\varphi_y(x) = \sum_{j\geq 1} x_j y_j$, is an isometric isomorphism from ℓ^q onto $(\ell^p)^*$. In finite dimension this implies

$$(B_p^d)^\circ = B_q^d, \qquad 1 \leq p \leq \infty.$$

Consequently, ℓ^2 is self-dual and $(B_2^d)^\circ = B_2^d$.

Proof. First suppose $1 < p < \infty$. Young's inequality $ab \leq a^p/p + b^q/q$ for $a, b \geq 0$ gives, after normalising non-zero x and y by their respective norms,

$$\sum_{j\geq 1} \frac{|x_j|}{\|x\|_p} \frac{|y_j|}{\|y\|_q} \leq \frac{1}{p} \sum_{j\geq 1} \frac{|x_j|^p}{\|x\|_p^p} + \frac{1}{q} \sum_{j \geq 1} \frac{|y_j|^q}{\|y\|_q^q} = 1.$$

This proves (A.1); if either sequence is zero the claim is immediate. When $p = 1$ and $q = \infty$, the same conclusion follows directly from $\sum_j |x_j y_j| \leq (\sup_j |y_j|) \sum_j |x_j|$.

Hölder's inequality shows that every $y \in \ell^q$ defines a continuous functional with $\|\varphi_y\|_* \leq \|y\|_q$. For $1 < p < \infty$ and $y \neq 0$, take N such that $\sum_{k=1}^N |y_k|^q > 0$; this holds for all sufficiently large N. Equality is obtained as a limit of the finite-support vectors

$$x_j^{(N)} = \begin{cases} \dfrac{\operatorname{sgn}(y_j)|y_j|^{q-1}}{\left(\sum_{k=1}^N |y_k|^q\right)^{(q-1)/q}}, & j \leq N, \\ 0, & j > N, \end{cases}$$

for which $\|x^{(N)}\|_p = 1$ and $\varphi_y(x^{(N)}) = (\sum_{j=1}^N |y_j|^q)^{1/q}$. For $p = 1$, choose signed coordinate vectors whose coordinates approach $\|y\|_\infty$. Hence $\|\varphi_y\|_* = \|y\|_q$ in both cases.

Conversely, let $\varphi \in (\ell^p)^*$ and put $y_j = \varphi(e_j)$ for the standard coordinate vectors. If

$p = 1$, then $|y_j| \le \|\varphi\|_*$, so $y \in \ell^\infty$. If $1 < p < \infty$, apply φ to the preceding norm-one construction based on $(y_1, \dots, y_N)$ whenever its qth-power sum is positive. When that sum is zero, the following inequality is immediate. Thus, in either case,

$$\left(\sum_{j=1}^{N} |y_j|^q\right)^{1/q} \le \|\varphi\|_* \qquad \text{for every } N,$$

and therefore $y \in \ell^q$. On every finitely supported x, $\varphi(x) = \sum_j x_j y_j$. Such sequences are dense in ℓ^p for $p < \infty$, so continuity extends the identity to every $x \in \ell^p$. Thus $y \mapsto \varphi_y$ is onto as well as isometric.

In $\mathbb{R}^d$, the dual-norm identity says $\sup_{x \in B_p^d} |y^\mathsf{T} x| = \|y\|_q$, which is exactly $(B_p^d)^\circ = B_q^d$. The endpoint $p = \infty$, $q = 1$ follows by taking $x_j = \operatorname{sgn}(y_j)$; this is a finite-dimensional polar identity, not an assertion that $(\ell^\infty)^* = \ell^1$. Finally, $p = q = 2$ yields self-duality. The Hilbert-space version of the same identification is the Riesz representation theorem proved below. □

Remark A.14 (The ℓ^∞ endpoint). The finite-dimensional identity $(B_\infty^d)^\circ = B_1^d$ does not extend to an identification $(\ell^\infty)^* = \ell^1$. Every $y \in \ell^1$ does define a continuous functional on ℓ^∞ through coordinate pairing, but the full dual $(\ell^\infty)^*$ is strictly larger. Establishing that strict inclusion requires additional functional-analysis machinery and is not used elsewhere in the thesis; it is stated here only to prevent the finite-dimensional drawing from being overinterpreted (Lin 2021).

Theorem A.15 (Cauchy–Schwarz inequality). *For all $f, g \in \mathcal{H}$,*

$$|\langle f, g\rangle| \le \|f\|\,\|g\|.$$

Proof. If $g = 0$, the assertion is immediate. Otherwise, non-negativity of the squared norm gives, for every $t \in \mathbb{R}$,

$$0 \le \|f - tg\|^2 = \|f\|^2 - 2t\langle f, g\rangle + t^2\|g\|^2.$$

Choose $t = \langle f, g\rangle / \|g\|^2$. Substitution yields $0 \le \|f\|^2 - |\langle f, g\rangle|^2/\|g\|^2$, which is equivalent to the claimed inequality. □

Theorem A.16 (Completeness and separability of L^2). *For a compact interval $\mathcal{I}$, the space $L^2(\mathcal{I})$ with its usual inner product is a separable Hilbert space.*

Proof. It remains to verify completeness and separability. Let $\{f_n\}$ be Cauchy in $L^2(\mathcal{I})$. Choose a subsequence $\{f_{n_k}\}$ such that

$$\|f_{n_{k+1}} - f_{n_k}\|_2 \le 2^{-k}.$$

Put $g_k = f_{n_{k+1}} - f_{n_k}$. Since $\mathcal{I}$ has finite measure, Cauchy–Schwarz gives $\|g_k\|_1 \le |\mathcal{I}|^{1/2}\|g_k\|_2$. Hence $\sum_k \int_{\mathcal{I}} |g_k| < \infty$, so $\sum_k |g_k(t)| < \infty$ for almost every t. Define $f = f_{n_1} + \sum_{k\ge 1} g_k$ at those points. Minkowski's inequality and Fatou's lemma imply

$$\|f - f_{n_m}\|_2 \le \sum_{k=m}^{\infty} \|g_k\|_2 \le \sum_{k=m}^{\infty} 2^{-k} \longrightarrow 0.$$

Thus the subsequence converges in L^2; because the original sequence is Cauchy, the whole sequence converges to the same f. This proves completeness.

For separability, consider step functions taking rational values on finite partitions whose endpoints are rational points of $\mathcal{I}$. This family is countable. Simple functions are dense in L^2, and measurable sets of finite measure can be approximated in measure by finite unions of intervals; their endpoints and the corresponding function values can then be approximated by rationals. The stated countable family is therefore dense in $L^2(\mathcal{I})$. $\square$

Theorem A.17 (Orthogonal projection)**.** *Let M be a closed linear subspace of $\mathcal{H}$. For every $f \in \mathcal{H}$, there exists a unique $P_M f \in M$ such that $f - P_M f \perp M$. Moreover,*

$$\|f - P_M f\| = \inf_{v\in M} \|f - v\|.$$

If $M = \operatorname{span}\{e_1, \dots, e_m\}$ for orthonormal vectors, then $P_M f = \sum_{j=1}^{m} \langle f, e_j\rangle e_j$.

Proof. Put $d = \inf_{v\in M} \|f-v\|$ and choose $v_n \in M$ with $\|f-v_n\| \to d$. The parallelogram identity and $(v_n + v_k)/2 \in M$ imply

$$\|v_n - v_k\|^2 \le 2\|f - v_n\|^2 + 2\|f - v_k\|^2 - 4d^2,$$

so $\{v_n\}$ is Cauchy. Completeness of $\mathcal{H}$ and closedness of M give a limit $v \in M$ with $\|f - v\| = d$. For $h \in M$, the function $t \mapsto \|f - v - th\|^2$ is minimised at zero; its linear term therefore vanishes, giving $\langle f - v, h\rangle = 0$. If v' has the same property, then $v - v' \in M$ and $v - v' \perp M$, so $v = v'$. This proves existence, orthogonality, and uniqueness. Pythagoras gives $\|f - w\|^2 = \|f - v\|^2 + \|v - w\|^2$ for every $w \in M$, which establishes best approximation. In the finite orthonormal case, direct substitution shows that $f - \sum_{j=1}^{m} \langle f, e_j\rangle e_j$ is orthogonal to every basis vector and hence to M. $\square$

Corollary A.18 (Basis expansion and Parseval's identity)**.** *If $\{e_j\}_{j\ge1}$ is an orthonormal basis of $\mathcal{H}$, then, for every $f \in \mathcal{H}$,*

$$f = \sum_{j=1}^{\infty} \langle f, e_j\rangle e_j, \qquad \|f\|^2 = \sum_{j=1}^{\infty} |\langle f, e_j\rangle|^2,$$

where the series converges in norm.

Proof. Let $P_m f = \sum_{j=1}^{m} \langle f, e_j \rangle e_j$. Because the linear span of the basis is dense, the best-approximation property in Theorem A.17 implies $\|f - P_m f\| \to 0$. Orthogonality and Pythagoras give

$$\|f\|^2 = \|f - P_m f\|^2 + \sum_{j=1}^{m} |\langle f, e_j \rangle|^2.$$

Letting $m \to \infty$ proves both assertions. □

Theorem A.19 (Riesz representation)**.** *Every continuous linear functional $\ell : \mathcal{H} \to \mathbb{R}$ has a unique representative $b \in \mathcal{H}$ such that $\ell(f) = \langle f, b \rangle$ for all $f \in \mathcal{H}$.*

Proof. If $\ell = 0$, take $b = 0$. Otherwise, $M = \mathrm{Null}(\ell)$ is a closed subspace. Select $z \notin M$ and put $u = z - P_M z$. Then $u \neq 0$ and $u \perp M$. For arbitrary f, the element

$$f - \frac{\ell(f)}{\ell(u)} u$$

belongs to M. Consequently, $f = m + \{\ell(f)/\ell(u)\}u$ for some $m \in M$. With $b = \ell(u)u/\|u\|^2$, orthogonality gives $\langle f, b \rangle = \ell(f)$. If both b and b' represent ℓ, then $\langle f, b - b' \rangle = 0$ for every f; taking $f = b - b'$ proves $b = b'$. □

A.3 Random functions and covariance operators

Proposition A.20 (Existence of the Hilbert-space mean)**.** *If X is a random element of $\mathcal{H}$ with $\mathbb{E}\|X\|^2 < \infty$, there is a unique $\mu_X \in \mathcal{H}$ satisfying*

$$\langle \mu_X, f \rangle = \mathbb{E}\langle X, f \rangle \quad \textit{for every } f \in \mathcal{H}.$$

Moreover, $\|\mu_X\| \leq (\mathbb{E}\|X\|^2)^{1/2}$.

Proof. The map $\ell(f) = \mathbb{E}\langle X, f \rangle$ is linear. By Theorem A.15 and Cauchy–Schwarz for scalar random variables,

$$|\ell(f)| \leq \mathbb{E}(\|X\|\|f\|) \leq (\mathbb{E}\|X\|^2)^{1/2}\|f\|.$$

It is therefore continuous. Theorem A.19 gives the unique representative μ_X, and the displayed bound gives the norm inequality. □

Theorem A.21 (Properties of a covariance operator)**.** *Let $Z = X - \mu_X$ and suppose $\mathbb{E}\|Z\|^2 < \infty$. The operator*

$$Kf = \mathbb{E}\{\langle Z, f \rangle Z\}$$

is bounded, self-adjoint, non-negative, trace class, and compact. In particular,

$$\|K\|_{\mathrm{op}} \le \mathbb{E}\|Z\|^2, \qquad \mathrm{tr}(K) = \mathbb{E}\|Z\|^2.$$

Proof. For $f, g \in \mathcal{H}$,

$$\langle Kf, g\rangle = \mathbb{E}\{\langle Z, f\rangle\langle Z, g\rangle\}.$$

The expression is symmetric in f and g, proving self-adjointness. With $g = f$ it is non-negative. Moreover,

$$|\langle Kf, g\rangle| \le \mathbb{E}(\|Z\|^2)\|f\|\|g\|,$$

which proves boundedness and the operator-norm bound.

For any orthonormal basis $\{e_j\}$, Tonelli's theorem and Corollary A.18 yield

$$\sum_{j=1}^{\infty}\langle Ke_j, e_j\rangle = \mathbb{E}\sum_{j=1}^{\infty}|\langle Z, e_j\rangle|^2 = \mathbb{E}\|Z\|^2 < \infty.$$

Thus, K is trace class and has the asserted trace.

For completeness, compactness can also be seen directly. Let P_m project onto

$$M_m = \mathrm{span}\{e_1, \ldots, e_m\},$$

and set $K_m = P_m K P_m$, which has finite rank. For unit vectors f, g, Cauchy–Schwarz gives

$$|\langle (I - P_m)Kf, g\rangle| \le \{\mathbb{E}\|Z\|^2\}^{1/2}\{\mathbb{E}\|(I - P_m)Z\|^2\}^{1/2}.$$

The final expectation tends to zero by Parseval and dominated convergence. The same bound applies to $P_m K(I - P_m)$, so $\|K - K_m\|_{\mathrm{op}} \to 0$. An operator-norm limit of finite-rank operators is compact. $\square$

Proposition A.22 (Covariance-kernel representation). *Suppose that $Z = X - \mu_X$ has pointwise second moments and that the integrability conditions for Fubini's theorem hold. If $c(s,t) = \mathbb{E}\{Z(s)Z(t)\}$, then, for almost every s,*

$$(Kf)(s) = \int_{\mathcal{I}} c(s,t)f(t)\,\mathrm{d}t.$$

Proof. Using the definition of K and then Fubini's theorem,

$$\begin{aligned}(Kf)(s) &= \mathbb{E}\left[Z(s)\int_{\mathcal{I}} Z(t)f(t)\,\mathrm{d}t\right]\\ &= \int_{\mathcal{I}} \mathbb{E}\{Z(s)Z(t)\}f(t)\,\mathrm{d}t = \int_{\mathcal{I}} c(s,t)f(t)\,\mathrm{d}t.\end{aligned}$$

The stated integrability assumptions justify interchanging expectation and integration. □

Theorem A.23 (Spectral representation)**.** *Let K be compact, self-adjoint, and non-negative. Its positive eigenvalues can be indexed as $\{\lambda_j : j \in \mathcal{J}_K\}$ in non-increasing order, where $\mathcal{J}_K = \{1, \dots, q\}$ if K has finite rank q and $\mathcal{J}_K = \mathbb{N}$ if it has infinite rank (with $\mathcal{J}_K = \varnothing$ when $K = 0$). There are corresponding orthonormal eigenfunctions $\{v_j : j \in \mathcal{J}_K\}$ spanning $\overline{\mathrm{Range}(K)}$ such that*

$$Kf = \sum_{j \in \mathcal{J}_K} \lambda_j \langle f, v_j \rangle v_j.$$

If K has infinite rank, then $\lambda_j \to 0$.

Proof. If $K = 0$, the result is immediate. Otherwise, define the maximal Rayleigh quotient $\lambda_1 = \sup_{\|f\|=1} \langle Kf, f \rangle > 0$. Take a maximising sequence. The unit ball is weakly sequentially compact, and compactness of K converts a weakly convergent subsequence $f_n \rightharpoonup v_1$ into strong convergence $Kf_n \to Kv_1$. Hence $\langle Kf_n, f_n \rangle \to \langle Kv_1, v_1 \rangle = \lambda_1$. Since $\langle Kv_1, v_1 \rangle \le \lambda_1 \|v_1\|^2$ and $\|v_1\| \le 1$, it follows that $\|v_1\| = 1$. Considering the Rayleigh quotient along $v_1 + th$ for $h \perp v_1$ shows that $\langle Kv_1, h \rangle = 0$; therefore $Kv_1 = \lambda_1 v_1$.

Self-adjointness makes $v_1^\perp$ invariant under K. Repeating the same argument on successive orthogonal complements produces orthonormal eigenvectors with non-increasing eigenvalues until the restriction of K is zero. In the infinite-rank case, if infinitely many eigenvalues were bounded below by some $c > 0$, then the sequence $Kv_j = \lambda_j v_j$ would have pairwise distances at least $\sqrt{2}c$ and hence no convergent subsequence, contradicting compactness. Thus $\lambda_j \to 0$ in that case.

Finally, if a vector orthogonal to every constructed v_j had non-zero image under K, the Rayleigh-quotient argument on that invariant orthogonal complement would produce a unit eigenvector with a positive eigenvalue ρ. In a finite construction this contradicts termination. In an infinite construction that vector belongs to every preceding orthogonal complement, so maximality of each Rayleigh quotient gives $\lambda_j \ge \rho$ for every j, contradicting $\lambda_j \to 0$. Hence K is zero on the remaining complement. This gives the expansion, and the eigenvectors with positive eigenvalues span $\overline{\mathrm{Range}(K)}$. □

Proposition A.24 (Karhunen–Loève expansion and optimality)**.** *Under the assumptions of Theorem A.21, let (λ_j, v_j) be the positive eigenpairs of K and set $\xi_j = \langle X - \mu_X, v_j \rangle$. Then*

$$X = \mu_X + \sum_{j \in \mathcal{J}_K} \xi_j v_j \quad \textit{in mean square,}$$

$\mathbb{E}\xi_j = 0$, *and* $\mathbb{E}(\xi_j\xi_k) = \lambda_j$ *for* $j = k$ *and zero otherwise. For each* $m \leq \operatorname{rank}(K)$, *among all* m*-dimensional subspaces,* $\operatorname{span}\{v_1, \ldots, v_m\}$ *minimises the expected squared reconstruction error. If* K *has finite rank, the full expansion is the case* $m = \operatorname{rank}(K)$.

Proof. Centredness gives $\mathbb{E}\xi_j = 0$, while

$$\mathbb{E}(\xi_j\xi_k) = \langle Kv_j, v_k\rangle = \lambda_j\langle v_j, v_k\rangle.$$

Every component of $Z = X - \mu_X$ in $\operatorname{Null}(K)$ has zero second moment because, for $h \in \operatorname{Null}(K)$,

$$\mathbb{E}\langle Z, h\rangle^2 = \langle Kh, h\rangle = 0.$$

For each $m \leq \operatorname{rank}(K)$, the trace identity therefore implies

$$\mathbb{E}\left\| Z - \sum_{j=1}^{m} \xi_j v_j \right\|^2 = \operatorname{tr}(K) - \sum_{j=1}^{m} \lambda_j.$$

If K has infinite rank, the right-hand side tends to zero as $m \to \infty$; if it has finite rank q, it equals zero at $m = q$. This proves the asserted mean-square expansion.

For any orthonormal $u_1, \ldots, u_m$, the expected reconstruction error is

$$\operatorname{tr}(K) - \sum_{\ell=1}^{m} \langle Ku_\ell, u_\ell\rangle.$$

Writing each u_ℓ in the eigenbasis gives $\sum_\ell \langle Ku_\ell, u_\ell\rangle = \sum_j \lambda_j a_j$, where $0 \leq a_j \leq 1$ and $\sum_j a_j \leq m$. Since the eigenvalues are ordered, this sum cannot exceed $\sum_{j=1}^{m} \lambda_j$. Equality is attained by $u_\ell = v_\ell$, proving optimality. □

Proposition A.25 (Rank of the empirical covariance)**.** *For observations* $X_1, \ldots, X_n$ *and a fixed population mean* μ_X*, the population-centred empirical covariance operator*

$$\widehat{K}_\mu f = \frac{1}{n}\sum_{i=1}^{n} \langle X_i - \mu_X, f\rangle (X_i - \mu_X)$$

has rank at most n*. The sample-centred empirical covariance operator*

$$\widehat{K}_{\overline{X}} f = \frac{1}{n}\sum_{i=1}^{n} \langle X_i - \overline{X}, f\rangle (X_i - \overline{X})$$

has rank at most $n - 1$.

Proof. The range of $\widehat{K}_\mu$ is contained in the span of the n curves $X_i - \mu_X$, so its rank is at most n. The range of $\widehat{K}_{\overline{X}}$ is contained in the span of the n curves $X_i - \overline{X}$. These curves sum to zero, so their span has dimension at most $n-1$. The rank of the corresponding operator cannot exceed that dimension. $\square$

A.4 The functional normal equation and inverse problem

Proposition A.26 (Centring and the normal equation). *Suppose*

$$Y = \alpha + \langle X, \beta \rangle + \varepsilon, \qquad \mathbb{E}(\varepsilon \mid X) = 0,$$

with finite second moments. Then $\alpha = \mu_Y - \langle \mu_X, \beta \rangle$, and the centred variables satisfy $Y^c = \langle X^c, \beta \rangle + \varepsilon$. If $r = \mathbb{E}(Y^c X^c)$ and K is the covariance operator of X, then $K\beta = r$.

Proof. Taking expectations and using $\mathbb{E}\varepsilon = \mathbb{E}\{\mathbb{E}(\varepsilon \mid X)\} = 0$ gives the formula for α and therefore the centred model. For arbitrary $f \in \mathcal{H}$,

$$\begin{aligned}\langle r, f \rangle &= \mathbb{E}\{Y^c \langle X^c, f \rangle\} \\ &= \mathbb{E}\{\langle X^c, \beta \rangle \langle X^c, f \rangle\} + \mathbb{E}\{\varepsilon \langle X^c, f \rangle\}.\end{aligned}$$

The last expectation is zero by iterated expectations. The first equals $\langle K\beta, f \rangle$. Since the equality holds for every f, $r = K\beta$. $\square$

Proposition A.27 (Excess prediction risk). *For $R(b) = \mathbb{E}\{Y^c - \langle X^c, b \rangle\}^2$ under the correctly specified model,*

$$R(b) - R(\beta) = \langle K(b-\beta), b-\beta \rangle = \|K^{1/2}(b-\beta)\|^2.$$

Proof. Let $h = b - \beta$. The prediction residual at b is $\varepsilon - \langle X^c, h \rangle$. Expanding the square gives

$$R(b) = \mathbb{E}\varepsilon^2 - 2\mathbb{E}\{\varepsilon \langle X^c, h \rangle\} + \mathbb{E}\langle X^c, h \rangle^2.$$

The middle term is zero by the conditional mean restriction, and the last term is $\langle Kh, h \rangle$. Since $R(\beta) = \mathbb{E}\varepsilon^2$, the first equality follows. The spectral representation defines $K^{1/2} f = \sum_j \lambda_j^{1/2} \langle f, v_j \rangle v_j$ and gives $\langle Kh, h \rangle = \|K^{1/2} h\|^2$. $\square$

Proposition A.28 (Variance identities for the discrete simulation). *Let*

$$X = \sum_{j=1}^{J} \sqrt{\lambda_j} Z_j \phi_j, \qquad Y = \langle X, \beta \rangle_T + \varepsilon,$$

where the Z_j are independent with mean zero and variance one, ε is independent of them with mean zero and variance σ^2, and $\langle\cdot,\cdot\rangle_T$ is the discrete inner product. Then

$$\begin{aligned}\mathbb{E}\|X\|_T^2 &= \sum_{j=1}^{J} \lambda_j \|\phi_j\|_T^2,\\ \mathrm{Var}\{\langle X,\beta\rangle_T\} &= \sum_{j=1}^{J} \lambda_j \langle \phi_j,\beta\rangle_T^2,\\ \mathrm{Var}(Y) &= \sum_{j=1}^{J} \lambda_j \langle \phi_j,\beta\rangle_T^2 + \sigma^2.\end{aligned}$$

Proof. Write $c_j = \langle \phi_j,\beta\rangle_T$. Bilinearity gives

$$\langle X,\beta\rangle_T = \sum_{j=1}^{J} \sqrt{\lambda_j} Z_j c_j.$$

All cross-products have expectation zero because the scores are independent and centred, while $\mathbb{E}Z_j^2 = 1$. Therefore the variance of this sum is $\sum_j \lambda_j c_j^2$. Similarly,

$$\mathbb{E}\|X\|_T^2 = \sum_{j,k} \sqrt{\lambda_j\lambda_k}\, \mathbb{E}(Z_jZ_k)\langle \phi_j,\phi_k\rangle_T = \sum_j \lambda_j \|\phi_j\|_T^2.$$

Finally, independence of ε and the scores makes the covariance between the noiseless response and the error zero, so their variances add. No discrete orthogonality assumption on the ϕ_j is required. □

Proposition A.29 (Empirical bias–variance identity)**.** *For arbitrary $b_1,\dots,b_R,\beta$ in a real Hilbert space, let $\bar b = R^{-1}\sum_{r=1}^R b_r$. Then*

$$\frac{1}{R}\sum_{r=1}^{R} \|b_r-\beta\|^2 = \|\bar b-\beta\|^2 + \frac{1}{R}\sum_{r=1}^{R} \|b_r-\bar b\|^2.$$

Proof. For every r,

$$b_r-\beta = (b_r-\bar b) + (\bar b-\beta).$$

Expanding the squared norm and averaging yields the two displayed squared terms plus

$$\frac{2}{R}\sum_{r=1}^{R} \langle b_r-\bar b, \bar b-\beta\rangle.$$

The cross-term is zero because $\sum_r (b_r-\bar b)=0$. The argument applies in particular to the grid inner product used in the simulation and explains why the empirical variance uses divisor R when an exact decomposition of the Monte Carlo mean squared error is desired. $\square$

Proposition A.30 (Identification and the minimum-norm target)**.** *Two slopes β and $\widetilde{\beta}$ generate the same centred conditional mean if and only if $\widetilde{\beta} - \beta \in \mathrm{Null}(K)$. In the uncentred model, if $h = \widetilde{\beta} - \beta \in \mathrm{Null}(K)$, the parameter pairs (α, β) and $(\alpha - \langle \mu_X, h\rangle, \widetilde{\beta})$ generate the same conditional mean. Hence the slope is fully identified exactly when K is injective. Otherwise, every identified equivalence class has a unique minimum-norm representative in*

$$\mathrm{Null}(K)^{\perp} = \overline{\mathrm{Range}(K)}.$$

Proof. For $h = \widetilde{\beta} - \beta$,

$$\mathbb{E}\langle X^c, h\rangle^2 = \langle Kh, h\rangle.$$

If $Kh = 0$, the non-negative random variable on the left has expectation zero, so $\langle X^c, h\rangle = 0$ almost surely. Conversely, this almost sure equality implies

$$Kh = \mathbb{E}\{\langle X^c, h\rangle X^c\} = 0.$$

This proves the centred equivalence and the injectivity statement. If $h \in \mathrm{Null}(K)$, then $\langle X, h\rangle = \langle \mu_X, h\rangle$ almost surely. Consequently,

$$\alpha - \langle \mu_X, h\rangle + \langle X, \beta + h\rangle = \alpha + \langle X, \beta\rangle \quad \text{almost surely,}$$

which proves the uncentred assertion.

Every $b \in \mathcal{H}$ has the orthogonal decomposition $b = b_0 + h$ with $b_0 \in \mathrm{Null}(K)^{\perp}$ and $h \in \mathrm{Null}(K)$. All elements $b_0 + h$ give the same conditional mean, while $\|b_0 + h\|^2 = \|b_0\|^2 + \|h\|^2$ is uniquely minimised by $h = 0$. Finally, for every bounded operator, $\mathrm{Range}(K)^{\perp} = \mathrm{Null}(K^*)$; self-adjointness and taking orthogonal complements give $\overline{\mathrm{Range}(K)} = \mathrm{Null}(K)^{\perp}$. $\square$

Theorem A.31 (Picard condition and instability of inversion)**.** *Let K be compact, self-adjoint, and non-negative, with positive eigenpairs (λ_j, v_j). The equation $K\beta = r$ has a solution if and only if $r \perp \mathrm{Null}(K)$ and*

$$\sum_{j:\lambda_j>0} \frac{|\langle r, v_j\rangle|^2}{\lambda_j^2} < \infty.$$

When these conditions hold, the unique minimum-norm solution is

$$\beta^{\dagger} = \sum_{j:\lambda_j>0} \frac{\langle r, v_j\rangle}{\lambda_j} v_j.$$

If K has infinite rank, the restriction of the minimum-norm inverse to $\mathrm{Range}(K)$, $K^\dagger|_{\mathrm{Range}(K)} : \mathrm{Range}(K) \to \mathrm{Null}(K)^\perp$, *is unbounded in the $\mathcal{H}$ norm.*

Proof. If $K\beta = r$, then r lies in $\overline{\mathrm{Range}(K)} = \mathrm{Null}(K)^\perp$ and $\langle r, v_j \rangle = \lambda_j \langle \beta, v_j \rangle$. Parseval therefore implies the summability condition. Conversely, the displayed condition makes the proposed series for $\beta^\dagger$ square summable. Applying K term by term yields the component of r in $\mathrm{Null}(K)^\perp$, which equals r by the first condition. All other solutions differ from $\beta^\dagger$ by an element of $\mathrm{Null}(K)$, so Proposition A.30 gives the minimum-norm claim.

For the final statement, $v_j \in \mathrm{Range}(K)$ because $v_j = K(v_j/\lambda_j)$. Yet

$$\frac{\|K^\dagger v_j\|}{\|v_j\|} = \frac{1}{\lambda_j} \longrightarrow \infty$$

as $\lambda_j \to 0$. Thus no finite operator-norm bound for $K^\dagger$ exists on $\mathrm{Range}(K)$. $\square$

A.5 Regularised and Krylov solutions

Proposition A.32 (Tikhonov solution)**.** *For $\rho > 0$, the criterion $R(b) + \rho\|b\|^2$ has the unique minimiser*

$$\beta_\rho = (K + \rho I)^{-1} r.$$

If $r \perp \mathrm{Null}(K)$, its spectral form is

$$\beta_\rho = \sum_{j:\lambda_j > 0} \frac{\langle r, v_j \rangle}{\lambda_j + \rho} v_j.$$

Proof. Apart from a constant independent of b, the population risk equals $\langle Kb, b \rangle - 2\langle r, b \rangle$. The spectral representation shows directly that $K + \rho I$ is invertible and that $\|(K + \rho I)^{-1}\|_{\mathrm{op}} \le 1/\rho$. The directional derivative of the penalised criterion at b in direction h is

$$2\langle (K + \rho I)b - r, h \rangle.$$

It vanishes for every h exactly when $(K + \rho I)b = r$. Since $\langle (K + \rho I)h, h \rangle \ge \rho\|h\|^2$, the criterion is strictly convex and the solution is unique. Expansion in the eigenbasis produces the stated filter. $\square$

Proposition A.33 (FPCR restriction)**.** *Let $V_m = \mathrm{span}\{v_1, \ldots, v_m\}$ with $\lambda_1, \ldots, \lambda_m > 0$.*

The minimiser of $R(b)$ over V_m is

$$\beta_m^{\mathrm{FPCR}} = \sum_{j=1}^{m} \frac{\langle r, v_j \rangle}{\lambda_j} v_j.$$

The empirical formula follows after replacing (K, r) by $(\widehat{K}, \widehat{r})$.

Proof. Write $b = \sum_{j=1}^{m} b_j v_j$. Up to an additive constant,

$$R(b) = \sum_{j=1}^{m} \{\lambda_j b_j^2 - 2b_j \langle r, v_j \rangle\}.$$

Each term is a strictly convex quadratic whose minimiser is $b_j = \langle r, v_j \rangle / \lambda_j$. Combining the coefficients proves the result. □

Proposition A.34 (Polynomial form of Krylov estimators)**.** *Every $\beta_m \in \mathcal{K}_m(K, r)$ can be written as $\beta_m = p_{m-1}(K)r$ for a polynomial of degree at most $m - 1$. Its moment residual satisfies*

$$r - K\beta_m = \pi_m(K)r, \qquad \pi_m(\lambda) = 1 - \lambda p_{m-1}(\lambda).$$

Proof. Membership in $\mathcal{K}_m(K, r)$ means that there are coefficients $a_0, \ldots, a_{m-1}$ with $\beta_m = \sum_{j=0}^{m-1} a_j K^j r$. Taking $p_{m-1}(\lambda) = \sum_{j=0}^{m-1} a_j \lambda^j$ gives the first assertion. Then $r - K\beta_m = \{I - Kp_{m-1}(K)\}r = \pi_m(K)r$ by direct algebra. □

Proposition A.35 (Raw and Arnoldi Krylov spaces)**.** *In exact arithmetic, if Arnoldi orthogonalisation does not break down before step m, its vectors $q_1, \ldots, q_m$ form an orthonormal basis of $\mathcal{K}_m(K, r)$. Consequently, a uniquely defined restricted least-squares solution is invariant to whether the raw power basis or the Arnoldi basis is used.*

Proof. The first Arnoldi vector is $q_1 = r/\|r\|$ and hence spans $\mathcal{K}_1(K, r)$. Suppose inductively that $\mathrm{span}\{q_1, \ldots, q_j\} = \mathcal{K}_j(K, r)$. Arnoldi forms Kq_j and removes its projections onto the existing vectors. The remainder therefore belongs to $\mathcal{K}_{j+1}(K, r)$ and is orthogonal to $\mathcal{K}_j(K, r)$. Absence of breakdown means that this remainder is non-zero, so its normalisation adds one new direction and spans $\mathcal{K}_{j+1}(K, r)$. Induction proves the first claim. The optimisation problem depends on the subspace, not on its coordinates; uniqueness therefore makes its solution basis-invariant. Finite-precision discrepancies arise because computed raw powers may lose linear independence, not because the exact subspaces differ. □

Proposition A.36 (Response-space Krylov characterisation)**.** *Let $T : \mathcal{H} \to \mathbb{R}^n$ be bounded, let $K = T^*T$ and $r = T^*y$, and put $M_m = \mathcal{K}_m(K, r)$. If T is injective on M_m,* *the restricted response least-squares problem*

$$\min_{b \in M_m} \|y - Tb\|_n^2$$

has a unique solution β_m^Y. Its moment residual $e_m^Y = r - K\beta_m^Y$ satisfies

$$e_m^Y \perp M_m.$$

Equivalently, β_m^Y uniquely minimises $\Phi(b) = \frac{1}{2}\langle Kb, b\rangle - \langle r, b\rangle$ over M_m.

Proof. Expanding the response criterion and using $K = T^*T$ and $r = T^*y$ gives

$$\frac{1}{2}\|y - Tb\|_n^2 = \frac{1}{2}\|y\|_n^2 + \frac{1}{2}\langle Kb, b\rangle - \langle r, b\rangle.$$

The first term is constant in b. For $h \in M_m$, the directional derivative of the remaining expression is

$$\frac{\mathrm{d}}{\mathrm{d}t}\Phi(\beta_m^Y + th)\bigg|_{t=0} = \langle K\beta_m^Y - r, h\rangle.$$

It vanishes for every $h \in M_m$ precisely when $e_m^Y \perp M_m$. Injectivity of T on the finite-dimensional space gives $\langle Kh, h\rangle = \|Th\|_n^2 > 0$ for every non-zero $h \in M_m$. The criterion is therefore strictly convex on M_m, so this stationary point exists uniquely and is the claimed minimiser. □

Proposition A.37 (NIPALS–APLS equivalence). *Let $T : \mathcal{H} \to \mathbb{R}^n$ be bounded, set $K = T^*T$ and $r = T^*y$, and consider scalar-response PLS with common centring and inner products. Suppose that the first m score directions generated by the NIPALS-type score-and-deflation construction are non-zero and that T is injective on $M_m = \mathcal{K}_m(K, r)$. Then the NIPALS score space is TM_m, its fitted response is the orthogonal projection of y onto TM_m, and its reconstructed slope equals the unique APLS solution in Proposition A.36.*

Proof sketch. At the first step, the NIPALS weight is proportional to $T^*y = r$, so its score lies in TM_1. Predictor deflation at each subsequent step replaces the current sampling operator T_{h-1} by $(I - P_{t_h})T_{h-1}$, where P_{t_h} is the response-space orthogonal projection onto the current score. Response deflation applies the same projection to the current response. Expanding these projections and repeatedly using $K = T^*T$ gives the standard PLS1 polynomial induction: the unnormalised hth weight has the form

$$c_h = \pi_{h-1}(K)r,$$

where π_{h-1} has degree $h - 1$. Its leading coefficient is non-zero whenever the new score is non-zero. Consequently,

$$\operatorname{span}\{c_1, \dots, c_h\} = \operatorname{span}\{r, Kr, \dots, K^{h-1}r\} = M_h,$$

because the change-of-basis matrix between the two displayed sequences is triangular with non-zero diagonal. Applying T gives equality of the score space and TM_h.

The predictor deflation makes successive scores mutually orthogonal. At each step, the NIPALS response loading is the least-squares coefficient of the current residual response on the new score. Hence the sum of the first m fitted score contributions is the orthogonal projection of y onto their span, namely TM_m. Minimising $\|y - Tb\|_n^2$ over $b \in M_m$ produces the same projection. Injectivity on M_m makes the restricted slope unique, so the reconstructed NIPALS slope and the APLS slope coincide. This operator argument is the standard fixed-component equivalence established for functional PLS by Delaigle and Hall (2012). □

Proposition A.38 (Response and moment Krylov objectives)**.** *Let K be self-adjoint and non-negative, $r \in \mathcal{H}$, and $M_m = \mathcal{K}_m(K, r)$. Assume that $\dim(M_m) = m$ and that K is injective on M_m. Then both restricted quadratic criteria below are strictly convex. Define*

$$\beta_m^Y = \arg\min_{b \in M_m} \left\{ \frac{1}{2}\langle Kb, b\rangle - \langle r, b\rangle \right\}, \qquad \beta_m^M = \arg\min_{b \in M_m} \frac{1}{2}\|r - Kb\|^2.$$

These response-space and moment-residual solutions are characterised by

$$\begin{aligned} r - K\beta_m^Y &\perp M_m, \\ r - K\beta_m^M &\perp KM_m. \end{aligned}$$

With $g_j = K^{j-1}r$ and $\beta_m = \sum_{j=1}^m a_j g_j$, their coefficient systems are respectively

$$\begin{aligned} \sum_{k=1}^m \langle r, K^{j+k-1}r\rangle a_k &= \langle r, K^{j-1}r\rangle, \\ \sum_{k=1}^m \langle r, K^{j+k}r\rangle a_k &= \langle r, K^j r\rangle, \qquad j = 1, \dots, m. \end{aligned}$$

Thus, the two methods use the same Krylov space but do not generally have the same finite-m solution.

Proof. For the response criterion

$$\Phi(b) = \frac{1}{2}\langle Kb, b\rangle - \langle r, b\rangle,$$

the directional derivative at b in direction $h \in M_m$ is

$$\frac{\mathrm{d}}{\mathrm{d}t}\Phi(b+th)\bigg|_{t=0} = -\langle r - Kb, h\rangle.$$

It vanishes for every $h \in M_m$ if and only if $r - Kb \perp M_m$. For the moment criterion

$$\Psi(b) = \frac{1}{2}\|r - Kb\|^2,$$

the directional derivative at b in direction $h \in M_m$ is

$$\frac{\mathrm{d}}{\mathrm{d}t}\Psi(b+th)\bigg|_{t=0} = -\langle r - Kb, Kh\rangle.$$

It vanishes for every $h \in M_m$ if and only if $r - Kb \perp KM_m$. For non-zero $h \in M_m$, injectivity and non-negativity of K imply $\langle Kh, h\rangle > 0$ and $\|Kh\|^2 > 0$. These are the quadratic forms governing the response and moment criteria, respectively, so both restrictions are strictly convex and the two stationarity conditions are sufficient and uniquely determine their minimisers.

For the response solution, take the inner product of $r - K\sum_k a_k g_k$ with g_j. Self-adjointness and $g_j = K^{j-1}r$ give

$$\langle Kg_k, g_j\rangle = \langle K^k r, K^{j-1}r\rangle = \langle r, K^{j+k-1}r\rangle,$$

which yields the first coefficient system. For the moment solution, take the inner product of the same residual with $Kg_j = K^j r$. Then

$$\langle Kg_k, Kg_j\rangle = \langle K^k r, K^j r\rangle = \langle r, K^{j+k}r\rangle,$$

which yields the second system. The one-power shift proves that equality is not implied merely by using the same M_m. □

Proposition A.39 (Finite termination with conjugate directions)**.** *Let $A \in \mathbb{R}^{d\times d}$ be symmetric positive definite and let $p_0, \ldots, p_{d-1}$ be non-zero, pairwise A-conjugate directions. Starting from any x_0, successively minimise $f(x) = \frac{1}{2}x^\top Ax - b^\top x$ along these directions without changing the coefficients already selected. After at most d directions, the iterate is the unique minimiser $A^{-1}b$. Standard conjugate gradient constructs such directions in exact arithmetic.*

Proof. If $\sum_j c_j p_j = 0$, multiplication by $p_k^\top A$ gives $c_k p_k^\top A p_k = 0$. Positive definiteness implies $c_k = 0$ for every k, so the d directions are linearly independent and form a basis of $\mathbb{R}^d$. Write an iterate as $x_0 + \sum_j \eta_j p_j$. Then

$$f\left(x_0 + \sum_j \eta_j p_j\right) = f(x_0) + \sum_j \eta_j p_j^\top (Ax_0 - b) + \frac{1}{2} \sum_j \eta_j^2 p_j^\top A p_j,$$

because all cross terms vanish by conjugacy. The minimising coefficient for one direction is therefore independent of every other coefficient. After all basis directions have been minimised, every directional derivative is zero; hence $Ax - b = 0$ and $x = A^{-1}b$. The standard CG recurrence produces pairwise A-conjugate search directions by induction from the residual orthogonality relations, so the conclusion applies to its exact-arithmetic iterates. This result concerns the standard finite-dimensional CG geometry used in Figure 3.2; the released functional recurrence is defined separately in (3.34)– (3.38). □

A.6 Late stopping and inference

The asymptotic arguments in this section follow the inference construction of Babii et al. (2026), with the identification qualification made explicit.

The sampling operator retains the notation T_n used throughout the thesis. To avoid confusing that operator with the scalar test statistic, the latter is denoted by $\mathcal{T}_n$. If K is not injective, the statistically identifiable null is $H_0 : K\beta = Kb$; this becomes the literal slope null $H_0 : \beta = b$ when β and b lie in $\mathcal{H}_{\mathrm{id}}$. Chapter 6 writes $\mathcal{T}_{n,m}$ when the finite iteration must be visible and abbreviates $\mathcal{T}_{n,m_n}$ to $\mathcal{T}_n$ for an inference-specific index sequence.

Proposition A.40 (Finite-iteration decomposition). *For any candidate slope b and any iterate $\widehat{\beta}_m$, define*

$$u_n(b) = \widehat{r} - \widehat{K}b, \qquad e_m = \widehat{r} - \widehat{K}\widehat{\beta}_m.$$

Then

$$\widehat{K}(\widehat{\beta}_m - b) = u_n(b) - e_m.$$

Proof. Add and subtract $\widehat{r}$:

$$\widehat{K}(\widehat{\beta}_m - b) = \widehat{K}\widehat{\beta}_m - \widehat{K}b = (\widehat{r} - \widehat{K}b) - (\widehat{r} - \widehat{K}\widehat{\beta}_m).$$

□

Theorem A.41 (Late-stopping equivalence and null limit). *Suppose that $(X_i, Y_i)_{i=1}^n$*

are independent and identically distributed and satisfy

$$Y_i^c = \langle X_i^c, \beta \rangle + \varepsilon_i, \qquad \mathbb{E}(\varepsilon_i \mid X_i) = 0.$$

Assume $\mathbb{E}\|X^c\|^4 < \infty$ *and* $\mathbb{E}(\varepsilon^2 \mid X) \le \sigma_{\max}^2 < \infty$ *almost surely, and let the identifiable null* $H_0 : K\beta = Kb$ *hold. Equivalently, if* $\beta, b \in \mathcal{H}_{\mathrm{id}}$*, assume* $H_0 : \beta = b$*. If, for an index* $m = m_n$ *that may depend on* n*,*

$$\sqrt{n}\, \|e_m\| = o_{\mathbb{P}}(1),$$

then

$$\mathcal{T}_n(b) = n\|\widehat{K}(\widehat{\beta}_m - b)\|^2 = n\|u_n(b)\|^2 + o_{\mathbb{P}}(1).$$

Let

$$V = \mathbb{E}\{\varepsilon^2 (X^c \otimes X^c)\}$$

be the covariance operator of εX^c*. If its positive eigenvalues are* $\{\omega_j : j \in \mathcal{J}_V\}$*, where* $\mathcal{J}_V$ *is finite or countably infinite, then*

$$\mathcal{T}_n(b) \rightsquigarrow \sum_{j \in \mathcal{J}_V} \omega_j Z_j^2, \qquad Z_j \overset{\text{iid}}{\sim} \mathcal{N}(0,1).$$

The sum is finite when V *has finite rank and converges almost surely and in* L^1 *otherwise.*

Proof. Under $H_0 : K\beta = Kb$, Proposition A.30 implies $\langle X^c, \beta - b \rangle = 0$ almost surely. With population centring,

$$u_n(b) = \frac{1}{n} \sum_{i=1}^{n} \varepsilon_i X_i^c.$$

If sample means are used instead, the exact expression is

$$u_n(b) = \frac{1}{n} \sum_{i=1}^{n} \varepsilon_i X_i^c - \overline{\varepsilon}\, \overline{X}^c.$$

The two sample means are $O_{\mathbb{P}}(n^{-1/2})$ under the stated second moments, so their product is $O_{\mathbb{P}}(n^{-1})$ and is negligible after multiplication by $\sqrt{n}$.

The random element εX^c is mean zero because $\mathbb{E}(\varepsilon \mid X) = 0$, and it has a finite second moment by the conditional variance bound and $\mathbb{E}\|X^c\|^2 < \infty$. The central limit theorem for independent, identically distributed random elements in a separable Hilbert space therefore gives $\sqrt{n}\, u_n(b) \rightsquigarrow G$, where G is a mean-zero Gaussian element with covariance operator V. In particular, $\sqrt{n}\|u_n(b)\| = O_{\mathbb{P}}(1)$.

By Proposition A.40,

$$\begin{aligned}\left|\mathcal{T}_n(b) - n\|u_n(b)\|^2\right| &\le 2\sqrt{n}\|u_n(b)\|\sqrt{n}\|e_m\| + n\|e_m\|^2 \\ &= o_{\mathbb{P}}(1).\end{aligned}$$

The continuous mapping theorem now gives $n\|u_n(b)\|^2 \rightsquigarrow \|G\|^2$. Applying the covariance spectral representation to G yields $G = \sum_j \sqrt{\omega_j} Z_j w_j$ in mean square for orthonormal eigenfunctions w_j. Parseval's identity therefore gives $\|G\|^2 = \sum_j \omega_j Z_j^2$. Since $\sum_j \omega_j = \operatorname{tr}(V) = \mathbb{E}\|\varepsilon X^c\|^2 < \infty$, the non-negative series converges almost surely and in L^1, completing the proof. □

Proposition A.42 (Homoskedastic form of the variance operator)**.** *If $\mathbb{E}(\varepsilon^2 \mid X) = \sigma^2$ almost surely, then the variance operator in Theorem A.41 satisfies*

$$V = \sigma^2 K.$$

Consequently, if $Kv_j = \lambda_j v_j$, then $Vv_j = \sigma^2 \lambda_j v_j$ and the null limit is $\sum_j \sigma^2 \lambda_j Z_j^2$ over the positive spectral indices.

Proof. For arbitrary $g, h \in \mathcal{H}$, iterated expectation gives

$$\begin{aligned}\langle Vh, g\rangle &= \mathbb{E}\big[\varepsilon^2 \langle X^c, h\rangle\langle X^c, g\rangle\big] \\ &= \mathbb{E}\big[\mathbb{E}(\varepsilon^2 \mid X)\langle X^c, h\rangle\langle X^c, g\rangle\big] \\ &= \sigma^2\langle Kh, g\rangle.\end{aligned}$$

Equality of these bilinear forms for every g and h proves $V = \sigma^2 K$. The spectral conclusion follows immediately. □

Corollary A.43 (Asymptotic size)**.** *Suppose the conditions of Theorem A.41 hold and V has at least one positive eigenvalue. Let $q_{1-\alpha}$ be a continuity-point $(1-\alpha)$ quantile of $Q = \sum_j \omega_j Z_j^2$. Then the rejection rule*

$$\varphi_n(b) = \mathbf{1}\{\mathcal{T}_n(b) > q_{1-\alpha}\}$$

has limiting null rejection probability α.

Proof. Theorem A.41 gives $\mathcal{T}_n(b) \rightsquigarrow Q$. Since at least one weight is positive, Q has a continuous distribution: condition on all terms except one positive weighted chi-squared term and integrate its continuous distribution function. The portmanteau theorem at $q_{1-\alpha}$ therefore yields

$$\mathbb{P}\{\mathcal{T}_n(b) > q_{1-\alpha}\} \longrightarrow \mathbb{P}\{Q > q_{1-\alpha}\} = \alpha.$$

□

Proposition A.44 (Inference under fixed and local alternatives)**.** *Assume the conditions needed for the Hilbert-space central limit theorem in Theorem A.41,* $\|\widehat{K}-K\|_{\mathrm{op}} = o_{\mathbb{P}}(1)$*, and* $\sqrt{n}\|e_{m_n}\| = o_{\mathbb{P}}(1)$*.*

For a fixed alternative satisfying $K(\beta-b)\neq 0$*,*

$$n^{-1}\mathcal{T}_n(b) \xrightarrow{\mathbb{P}} \|K(\beta-b)\|^2,$$

so $\mathcal{T}_n(b)\to\infty$ *in probability. For the triangular sequence* $\beta_n = b + n^{-1/2}\Delta$ *with fixed* $\Delta\in\mathcal{H}$*,*

$$\mathcal{T}_n(b) \rightsquigarrow \|G+K\Delta\|^2.$$

If $V\phi_j=\omega_j\phi_j$*, the latter variable equals*

$$\sum_j \omega_j Z_j^2 + 2\sum_j \sqrt{\omega_j} Z_j\langle\phi_j, K\Delta\rangle + \|K\Delta\|^2.$$

Proof. Under a fixed alternative, the empirical null moment decomposes as

$$u_n(b) = \frac{1}{n}\sum_{i=1}^{n} \varepsilon_i X_i^c + \widehat{K}(\beta-b),$$

apart from the lower-order sample-centring term treated in Theorem A.41. The first term is $O_{\mathbb{P}}(n^{-1/2})$, while operator convergence gives $\widehat{K}(\beta-b)\to_{\mathbb{P}} K(\beta-b)$. Since $e_{m_n}=o_{\mathbb{P}}(n^{-1/2})$, Proposition A.40 implies

$$\widehat{K}(\widehat{\beta}_{m_n}-b) \xrightarrow{\mathbb{P}} K(\beta-b).$$

Squaring the norm proves the first claim.

Under $\beta_n = b+n^{-1/2}\Delta$, multiply the analogous decomposition by $\sqrt{n}$ to obtain

$$\sqrt{n}\,\widehat{K}(\widehat{\beta}_{m_n}-b) = \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\varepsilon_i X_i^c + \widehat{K}\Delta - \sqrt{n}e_{m_n} + o_{\mathbb{P}}(1).$$

The three displayed terms converge jointly to G, $K\Delta$, and zero, respectively. Slutsky's theorem and the continuous mapping theorem give the local limit. Finally write $G=\sum_j\sqrt{\omega_j}Z_j\phi_j$ and expand $\|G+K\Delta\|^2$; Parseval's identity justifies the displayed series. □

Corollary A.45 (Confidence sets by test inversion)**.** *Let* $q_{1-\alpha}$ *be a critical value for which the test that rejects when* $\mathcal{T}_n(b) > q_{1-\alpha}$ *has asymptotic size* α *on an identified* *parameter space $\Theta \subseteq \mathcal{H}_{\mathrm{id}}$. Then*

$$\mathcal{C}_{1-\alpha} = \{b \in \Theta : \mathcal{T}_n(b) \le q_{1-\alpha}\}$$

is an asymptotic $(1-\alpha)$ confidence set for every $\beta \in \Theta$.

Proof. By construction,

$$\mathbb{P}_\beta\{\beta \in \mathcal{C}_{1-\alpha}\} = \mathbb{P}_\beta\{\mathcal{T}_n(\beta) \le q_{1-\alpha}\} = 1 - \mathbb{P}_\beta\{\mathcal{T}_n(\beta) > q_{1-\alpha}\}.$$

The final rejection probability converges to α, so the coverage probability converges to $1-\alpha$. □

Proposition A.46 (Finite-dimensional confidence ellipsoid and envelope)**.** *Let $H \in \mathbb{R}^{T\times p}$ contain the values of p candidate basis functions on an equally weighted grid, set $A = \widehat{K}H$, and let $f = \widehat{K}\widehat{\beta}_m$. For a critical value $q > 0$, define*

$$C_q = \left\{c \in \mathbb{R}^p : \frac{n}{T}\|f - Ac\|_2^2 \le q\right\}.$$

If A has full column rank, put

$$\widehat{c} = (A^\top A)^{-1}A^\top f, \qquad \rho^2 = \frac{Tq}{n} - \|f - A\widehat{c}\|_2^2.$$

Then C_q is empty when $\rho^2 < 0$ and, when $\rho^2 \ge 0$,

$$C_q = \{c : (c - \widehat{c})^\top A^\top A(c - \widehat{c}) \le \rho^2\}.$$

For an evaluation vector $h(s) = (h_1(s), \dots, h_p(s))^\top$, the extrema of $h(s)^\top c$ over a non-empty C_q are

$$h(s)^\top\widehat{c} \pm \sqrt{\rho^2 h(s)^\top(A^\top A)^{-1}h(s)}.$$

If A is rank deficient and C_q is non-empty, its coefficient set is unbounded along $\mathrm{Null}(A)$*; a pointwise functional envelope is unbounded wherever evaluation does not annihilate those directions.*

Proof. The least-squares residual $u = f - A\widehat{c}$ is orthogonal to $\mathrm{Range}(A)$. Therefore, for every c,

$$\begin{aligned}\|f - Ac\|_2^2 &= \|u + A(\widehat{c} - c)\|_2^2 \\ &= \|u\|_2^2 + (c - \widehat{c})^\top A^\top A(c - \widehat{c}).\end{aligned}$$

Substitution into the defining inequality proves the empty-set and ellipsoid claims.

When A has full column rank, write $d = (A^\top A)^{1/2}(c - \widehat{c})$. The constraint becomes $\|d\|_2 \le \rho$, and

$$h(s)^\top(c - \widehat{c}) = \left\{(A^\top A)^{-1/2} h(s)\right\}^\top d.$$

Cauchy–Schwarz gives the displayed upper and lower values, attained when d is parallel or antiparallel to $(A^\top A)^{-1/2}h(s)$.

If A is rank deficient, take non-zero $v \in \mathrm{Null}(A)$. For any accepted c, every $c + tv$ has the same residual for all $t \in \mathbb{R}$. Thus the coefficient set is unbounded, and so is the value $h(s)^\top(c + tv)$ whenever $h(s)^\top v \ne 0$. $\square$

Late stopping removes iteration bias. Identifying a unique slope still requires restriction to $\mathrm{Null}(K)^\perp$.

Appendix B

Supplementary Monte Carlo Diagnostics

This appendix collects the basis-coefficient and pointwise distribution plots supporting Section 5.7. Each page uses the same frozen $R = 5{,}000$ archive as Chapter 5. The horizontal ranges of the histograms are based on the central 99% of the pooled distribution so that the centre remains readable; nonzero omitted tail counts are printed inside the panels as "out: N". Each histogram is normalised over its displayed observations. All replay-based raw-FPLS panels—the basis-coefficient histograms, pointwise slope histograms, and quantile plots—use the archived fits not flagged by the raw stability diagnostic, while all replications are retained for CG–FPLS, Arnoldi FPLS, and FPCR.

In these plots, "CG", "Raw", and "Arnoldi" denote the released implementation of CG–FPLS, raw FPLS, and Arnoldi FPLS. Model identity is given in the caption; method columns retain the same order throughout. The full numerical distribution summaries remain in the frozen archive.

B.1 Leading basis coefficients

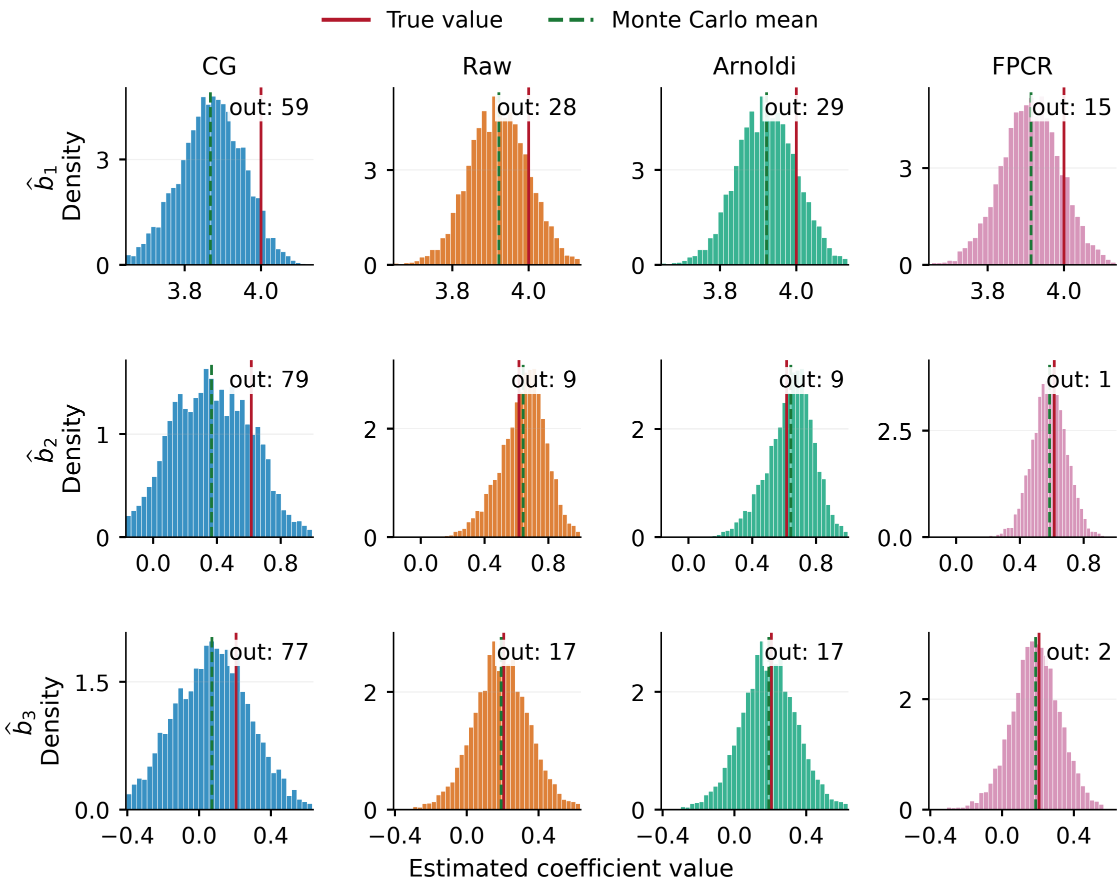

Figure B.1: Empirical distributions of the first three basis coefficients in Model 1. Solid vertical lines mark the true coefficients and dashed lines mark Monte Carlo means.

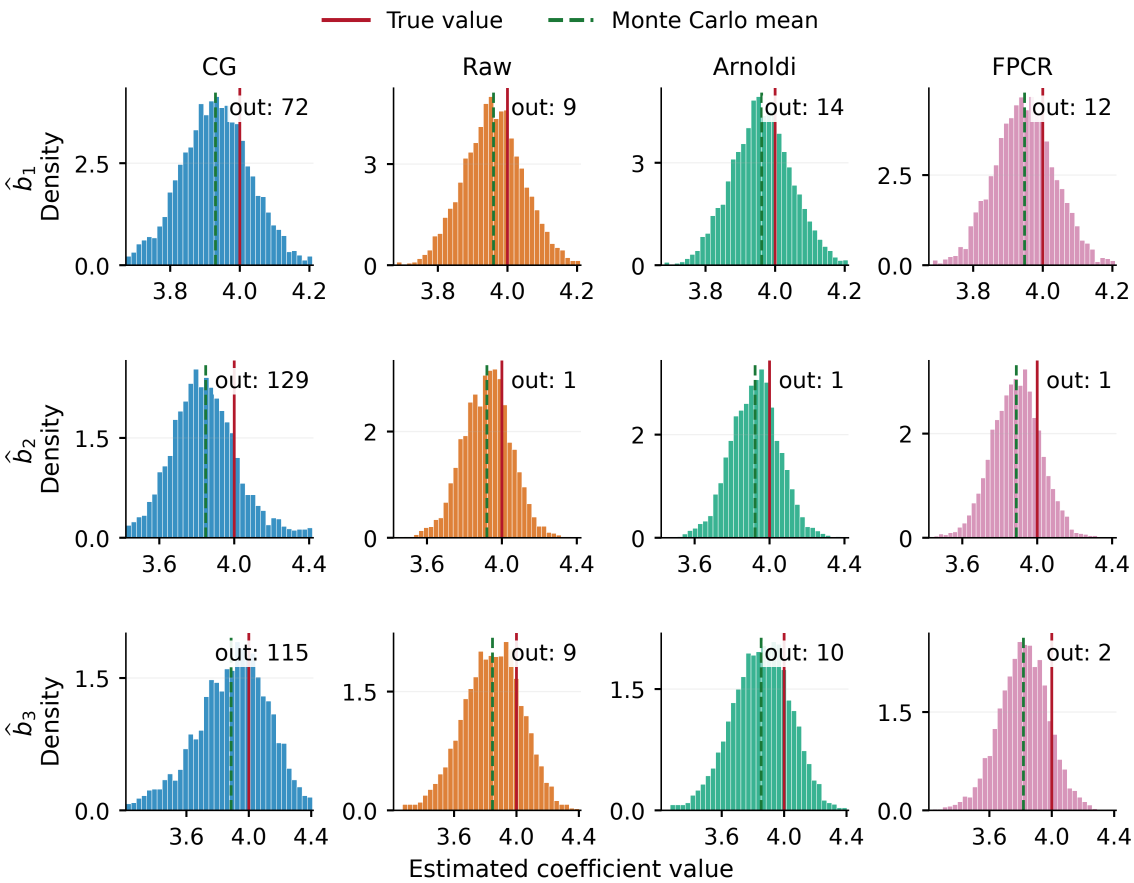

Figure B.2: Empirical distributions of the first three basis coefficients in Model 2.

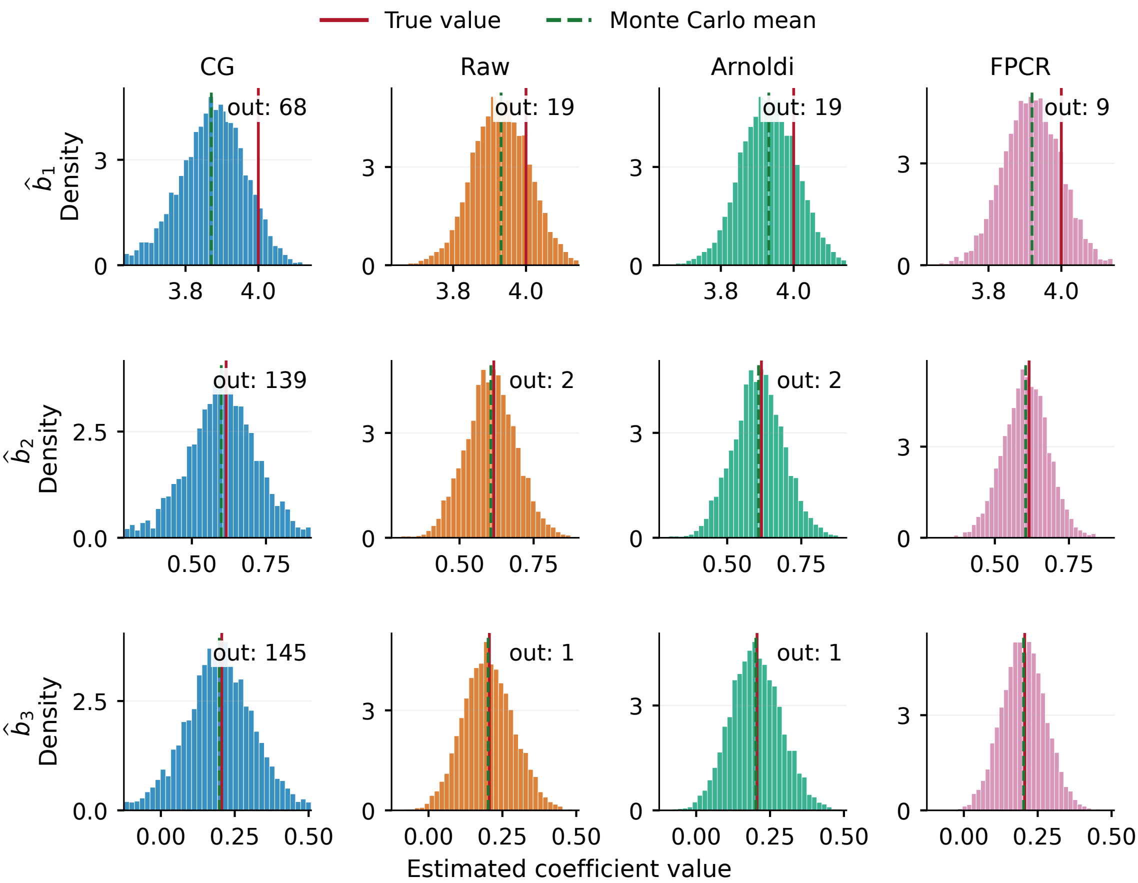

Figure B.3: Empirical distributions of the first three basis coefficients in Model 3.

B.2 Pointwise slope distributions

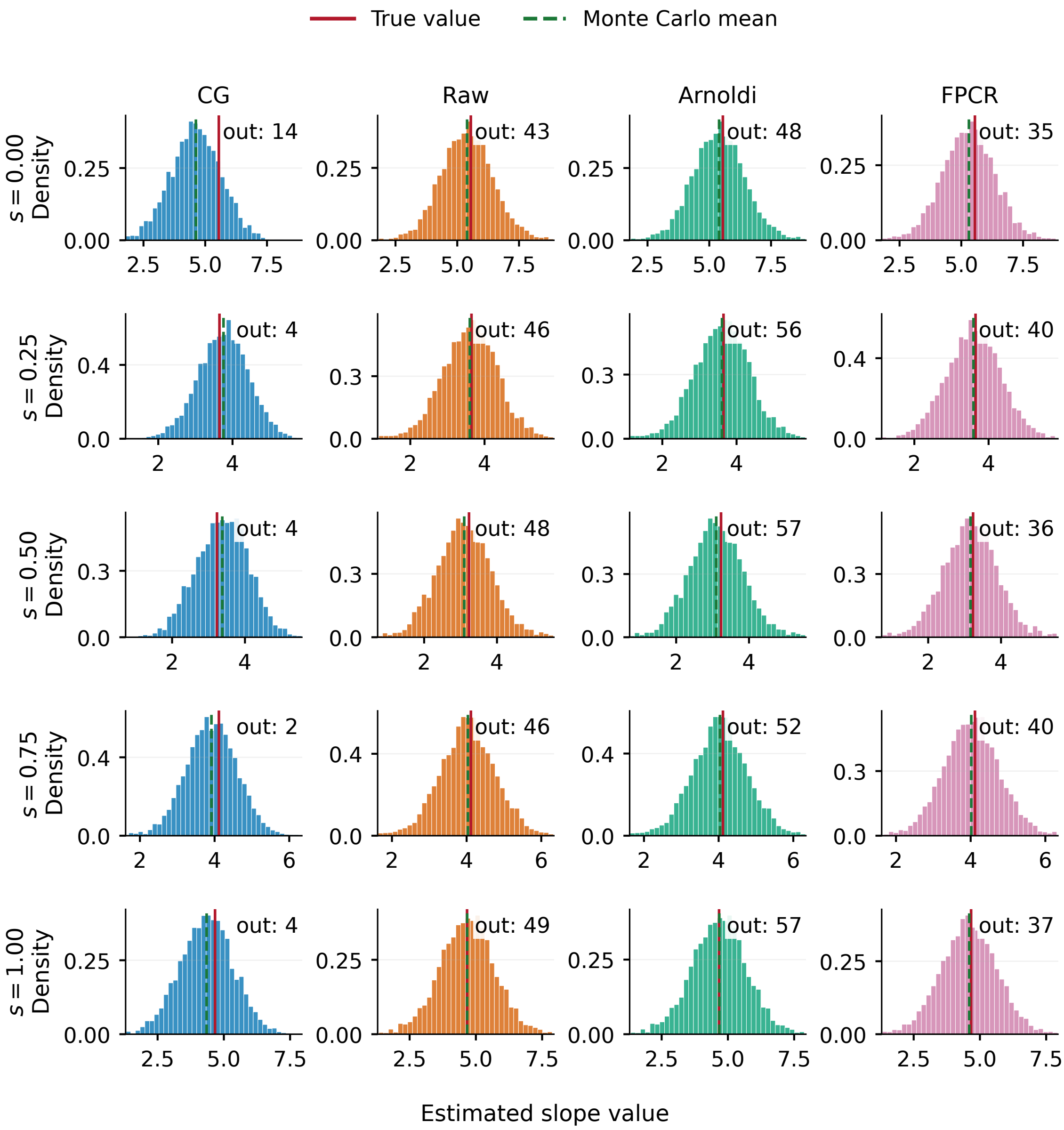

Figure B.4: Pointwise distributions of the selected slope estimator at five locations in Model 1.

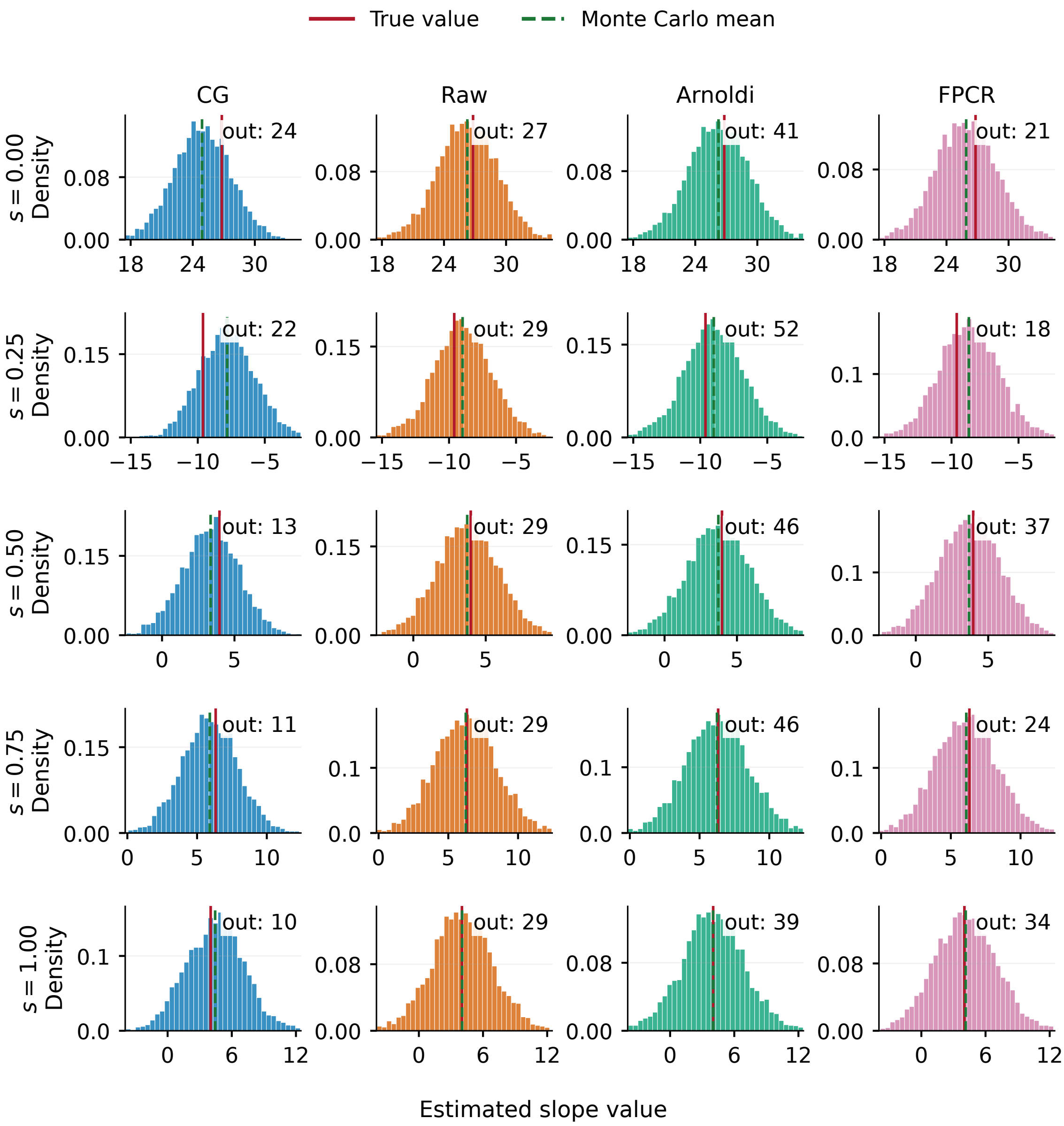

Figure B.5: Pointwise distributions of the selected slope estimator at five locations in Model 2.

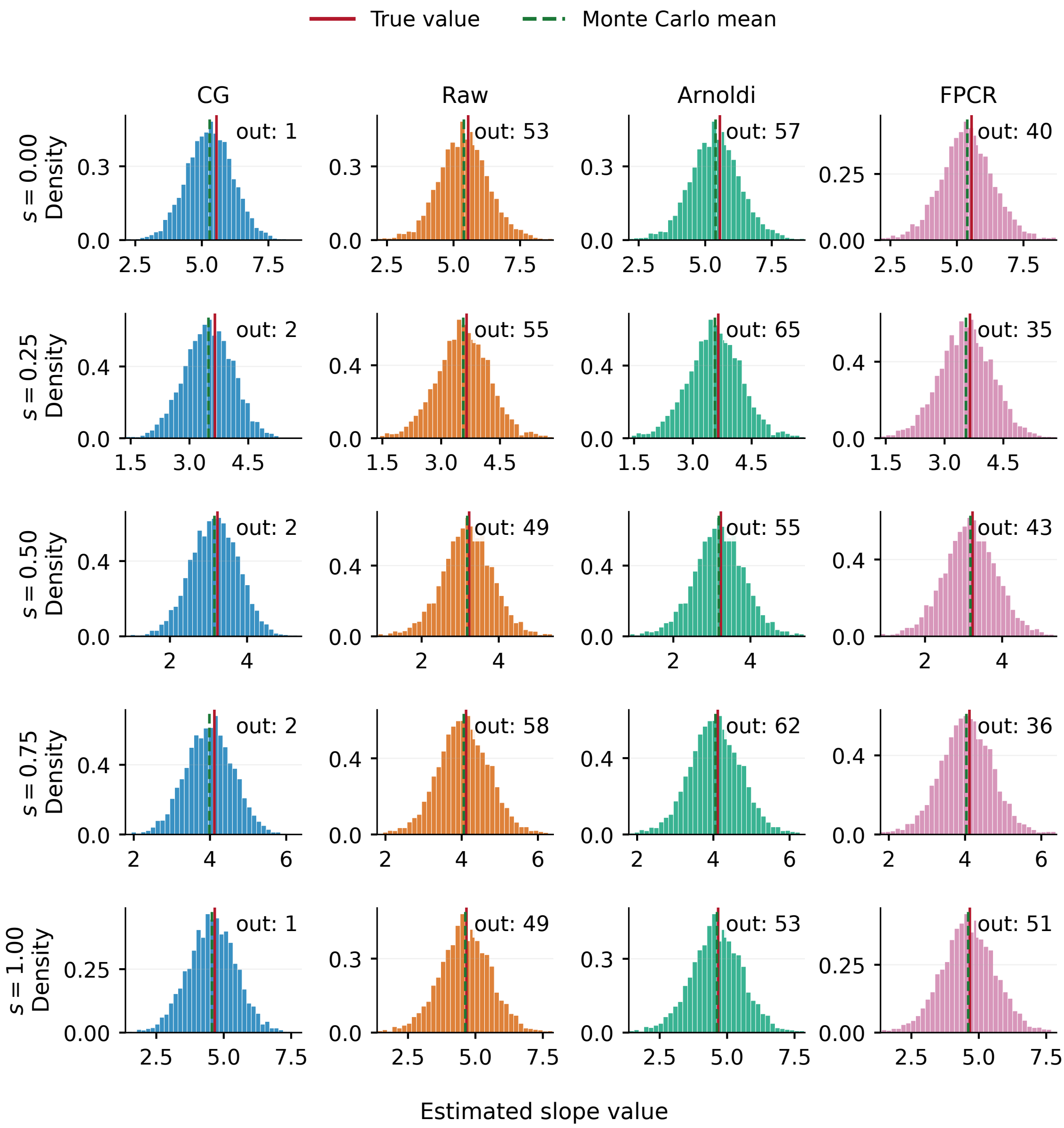

Figure B.6: Pointwise distributions of the selected slope estimator at five locations in Model 3.

B.3 Normal quantile–quantile diagnostics

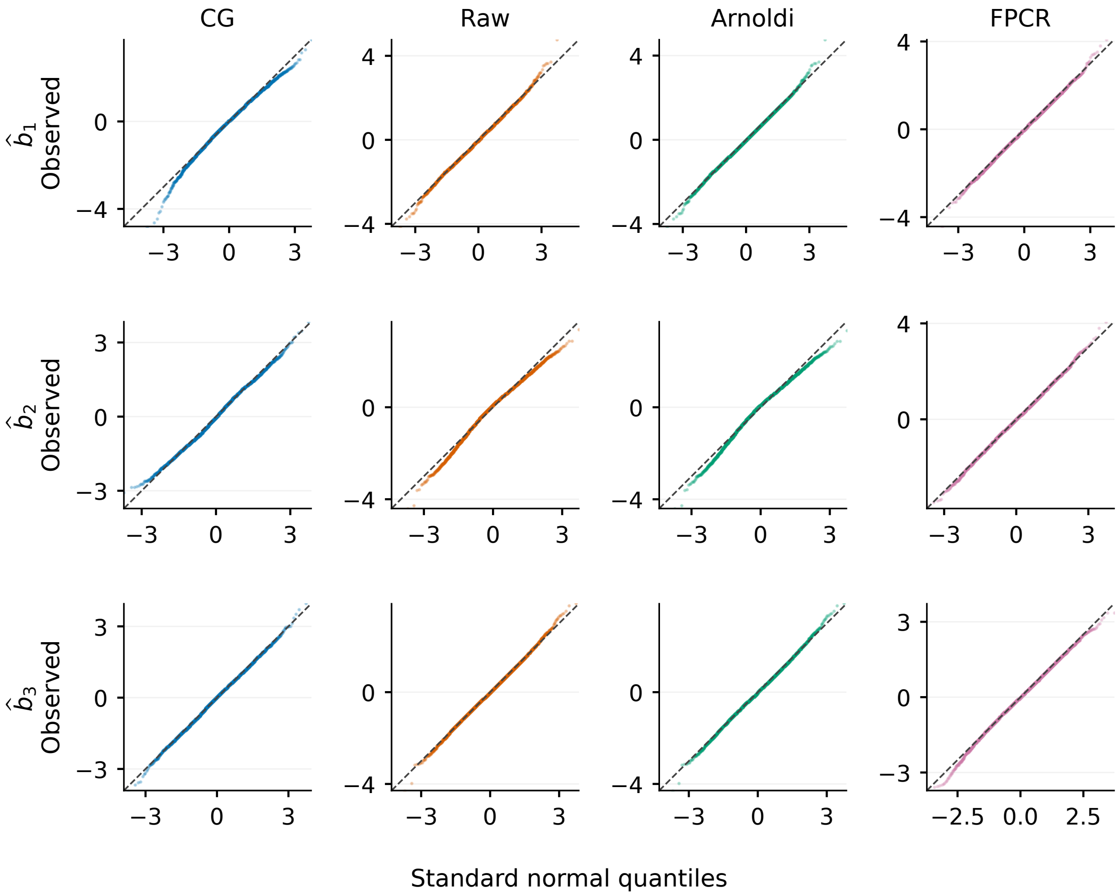

Figure B.7: Normal quantile–quantile plots for the first three estimated basis coefficients in Model 1. Each coefficient sample is standardised by its empirical mean and standard deviation; the dashed line is the 45-degree identity line.

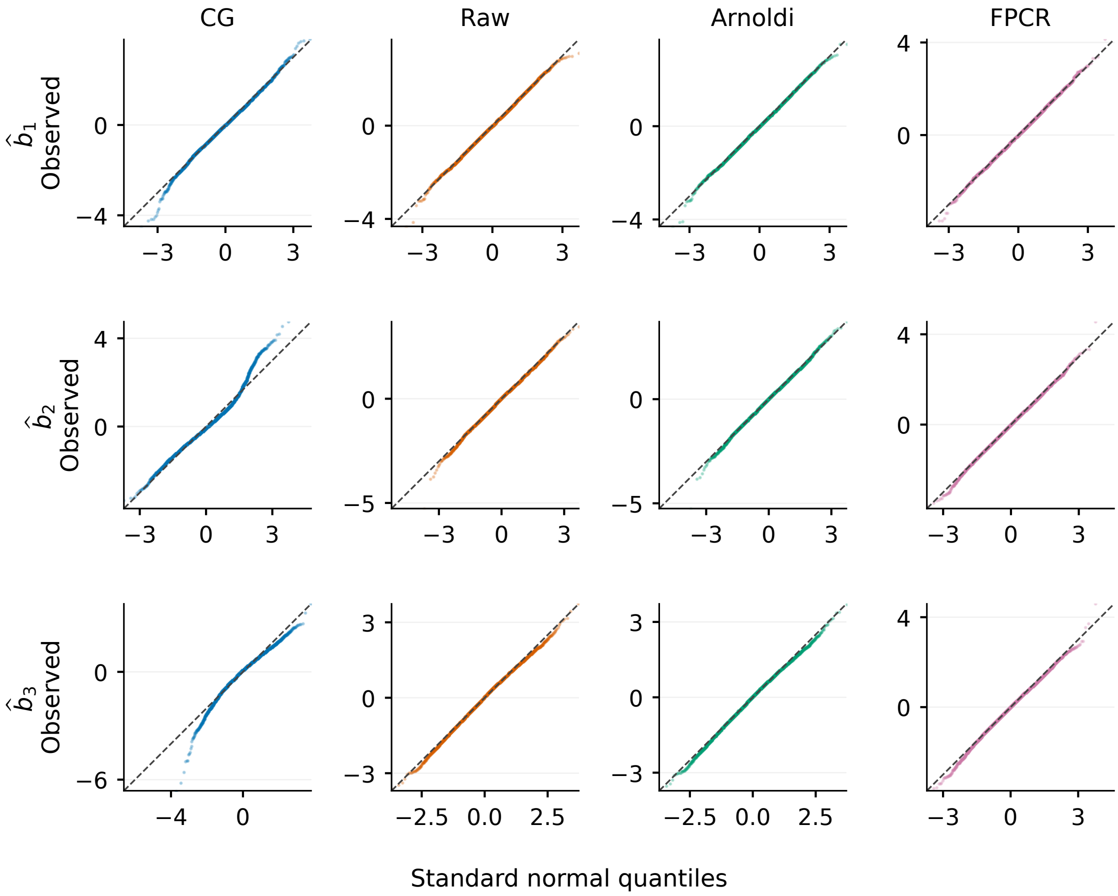

Figure B.8: Normal quantile–quantile plots for the first three estimated basis coefficients in Model 2, using the same empirical standardisation and identity reference line as in Model 1.

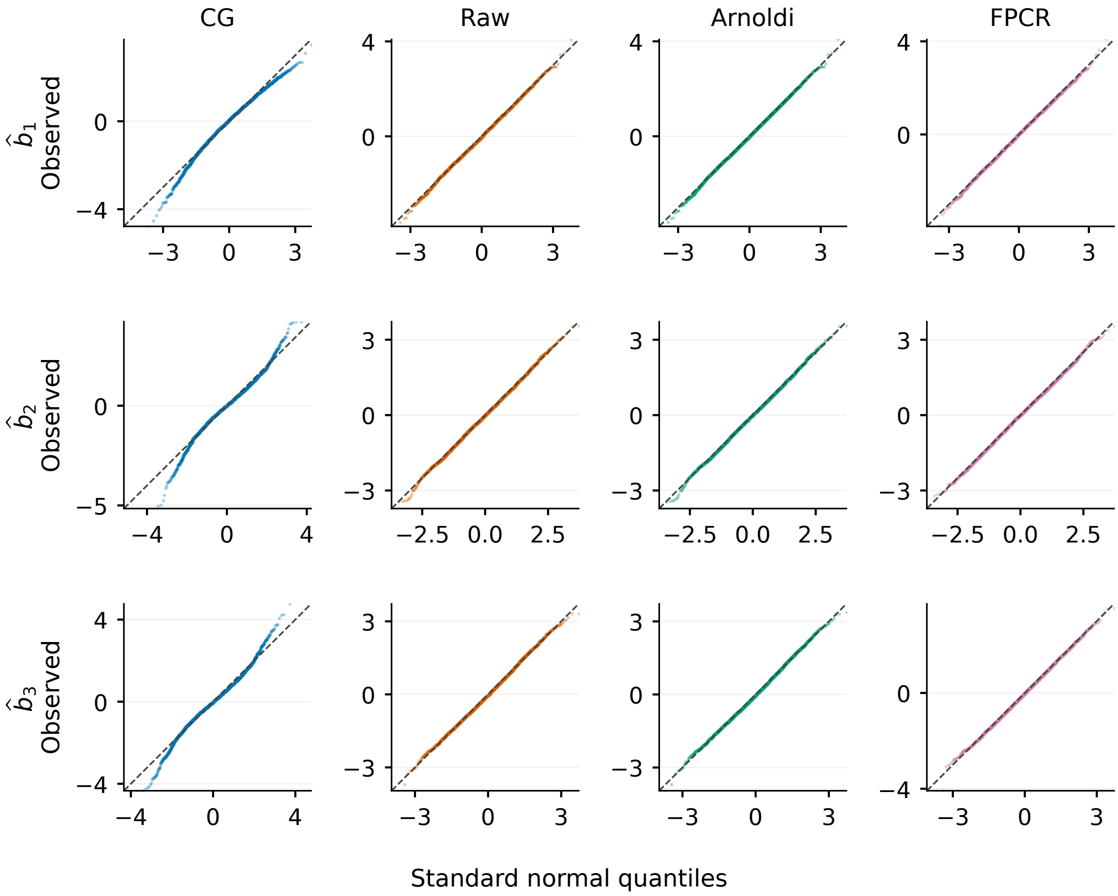

Figure B.9: Normal quantile–quantile plots for the first three estimated basis coefficients in Model 3, using the same empirical standardisation and identity reference line as in Model 1.

Appendix C

Supplementary Inference Diagnostics

Chapter 7 reports the exact rejection frequencies at the 1%, 5%, and 10% levels in Table 7.2. The additional figure below isolates the principal 5% comparison and displays its conditional Monte Carlo uncertainty. It is retained for visual completeness but is placed here because the numerical table communicates the same evidence more compactly.

C.1 Oracle-calibrated size at the 5% level

All three point estimates lie modestly above 5%, matching the heavier finite-sample upper tails documented in Figure 7.2. The figure should therefore be read as a finite-sample size assessment of the oracle reference, not as evidence that oracle calibration is exact or available in an empirical application.

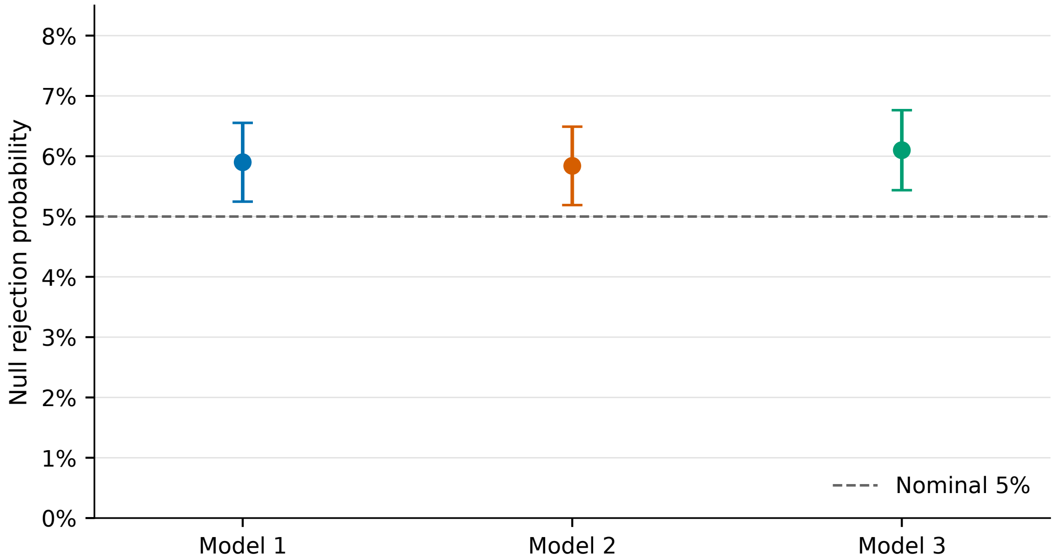

Figure C.1: Empirical null rejection probability of $\mathcal{T}_{n,70}$ at $n = 100$, using the oracle weighted chi-square 5% critical value. Each point is based on 5,000 finite-sample replications; vertical bars are normal-approximation intervals equal to the estimate plus or minus 1.96 binomial Monte Carlo standard errors. They condition on the realised critical values and do not include uncertainty from the 50,000 oracle draws. Since only oracle calibration was selected, the figure compares empirical size with the nominal line rather than comparing alternative feasible calibration procedures.

Appendix D

Supplementary Spectroscopy Analysis and Diagnostics

This appendix is the audit trail for Chapter 8. It records the dataset identity, every preprocessing and validation rule, the complete development and sensitivity summaries, and the numerical diagnostics needed to distinguish finite prediction from reliable Krylov coordinates. The chapter and appendix are derived from one frozen archive; the material below does not redefine the primary analysis after observing the benchmark results.

D.1 Data provenance and structural audit

The source is the 2002 pharmaceutical-tablet NIR Shoot-out (Hopkins 2003; Eigenvector Research 2026). The archive was downloaded from the Eigenvector Research data page on 8 September 2026. Its identifiers are:

- ZIP SHA-256:

  `04c03a4314ab8e8f63e0ecc788b657b5d1b5450791ce465ad41ad90cb5d52125`;

- contained MAT-file SHA-256:

  `129a32ec9e194e568cbd96a200c88aabeb347156081b5dfbb0120372fdd6c22a`.

The distributed file contains 155 calibration, 40 validation, and 460 test rows, giving 655 rather than the catalogue's stated 654 tablets. All rows and all spectral entries are finite, and there are no exact duplicate spectra within either instrument. Every complete row is retained. Because the file does not supply global tablet identifiers, immutable analysis identifiers join the source split to one-based row order, for example `calibrate:001`. Corresponding instrument-1 and instrument-2 rows are treated as paired observations.

Table D.1 quantifies the response imbalance that is visible in Figure 8.1. The validation response has sample standard deviation 2.393 and range 188.2–199.5, whereas the calibration and benchmark distributions are much wider. In fact, 267 of the 460 benchmark assay values fall outside the validation range. Validation-set R^2 therefore has a small denominator and is not comparable with benchmark R^2.

Table D.1: Assay response by supplied archive split. Standard deviations use denominator $n - 1$.

Split	n	Mean	Standard deviation	Minimum	Maximum
Calibration	155	192.885	21.980	151.6	239.1
Validation	40	194.795	2.393	188.2	199.5
Benchmark	460	188.400	15.763	154.3	237.7

The wavelength units and domain are supplied by the external description; they are not embedded as ordinary numeric metadata in the MAT object. The archive and catalogue provide no explicit open-data licence. The frozen empirical archive therefore contains checksums and a reproducible preparation workflow, but does not redistribute the source MAT file.

D.2 Functional representation and preprocessing

Each spectrum is represented by its $T = 650$ evaluations on the uniform 600–1898 nm grid. The discrete geometry is

$$\langle f, g \rangle_T = \frac{1}{T} \sum_{j=1}^{T} f_j g_j, \qquad T = 650. \tag{D.1}$$

For a training sample $\mathcal{A}$, wavelength and response means are computed using only rows in $\mathcal{A}$. The centred estimator has intercept

$$\widehat{\alpha}_{\mathcal{A}} = \overline{Y}_{\mathcal{A}} - \frac{1}{T} \overline{X}_{\mathcal{A}}^{\top} \widehat{\beta}_{\mathcal{A}}, \tag{D.2}$$

which is algebraically equivalent to the prediction form in (8.2). The training means are applied unchanged to held-out observations. The source code also archives the resulting original-input coefficient vector and intercept for every fitted model, so its stored prediction can be reconstructed directly.

No primary operation other than this fold-local centring is applied. In particular, there is no variance scaling, smoothing, derivative construction, wavelength selection, standard-normal-variate transformation, multiplicative- scatter correction, or baseline correction. These choices matter because every such operation can add regularisation or alter the covariance geometry.

The primary final fit has $650 > 195$ and numerical covariance rank 194 after centring. At least 456 grid-coordinate null directions therefore remain, so a unique unrestricted physical slope is not identified. Coefficient vectors are archived for replay and numerical comparison, not interpreted as recovered chemical effects.

Three alternatives were fixed before benchmark fitting:

1. instrument 1 restricted to 600–1798 nm inclusive, retaining 600 wavelengths;
2. instrument 1 smoothed row by row with a 15-point cubic Savitzky–Golay filter, derivative order zero and interpolation boundary mode; and
3. the complete 650-wavelength instrument-2 spectrum for the same tablets.

Each transformed dataset is centred within the relevant training sample. The fixed smoothing operator itself does not estimate parameters from held-out rows; all sample-dependent centring remains fold local.

D.3 Evaluation design and leakage safeguards

Table D.2 summarises the complete evaluation design. Random folds are formed separately within the supplied calibration and validation sources. Within each source, observations are ordered by assay in groups of five and assigned one of the five fold labels at random. Every development outer fold thus contains exactly 31 calibration and 8 validation tablets. This preserves source composition while avoiding folds dominated by one part of the response range.

Across five repeats, $195 \times 5 = 975$ out-of-fold predictions are stored for each method. These are repeated predictions of the same 195 tablets, not 975 independent observations. Correspondingly, repeat summaries and fold distributions are descriptive diagnostics rather than a basis for standard independent-sample tests. The development completion record additionally states `benchmark_variables_loaded=false`: the development stage did not load the test variables. Only after its checks passed was the full stage allowed to fit the 195-row models and evaluate the benchmark.

The benchmark bootstrap resamples tablet indices in common across methods. This preserves pairing and estimates conditional variation over the supplied benchmark

Table D.2: Frozen spectroscopy evaluation design. Exact assignments and resampling matrices, rather than only seeds, are included in the archive.

Stage	Frozen rule
Repeated development	Five repeats of five source-specific, assay-balanced outer folds; one complete out-of-fold metric per repeat.
Inner selection	New five-fold balanced partitions inside each outer training set; Raw and Arnoldi share indices but choose independently.
Order sensitivity	Five source-specific contiguous outer folds, each holding out 31 calibration and 8 validation rows.
Original split	Fit and tune on 155 calibration tablets; evaluate once on the 40 validation tablets.
Final development fit	Fit on all 195 development tablets; Raw and Arnoldi use one frozen common balanced inner partition.
Designated benchmark	Predict the 460 supplied test tablets once; no benchmark outcome enters centring, tuning, repair, or variant selection.
Conditional bootstrap	10,000 paired iid draws and 10,000 circular moving-block draws at each block length 5, 10, and 20; models are held fixed.
Randomness	Master seed 20260908 with separate deterministic streams for outer folds, inner folds, final tuning, iid bootstrap, and block bootstrap.

empirical distribution. It does not refit centring, tuning, or the models and therefore excludes development-sample and selection uncertainty. The circular block calculations preserve local row-order sequences, but no batch or acquisition-order metadata are available to determine whether archive order has a physical interpretation.

D.4 Complete procedures and tuning rules

The estimator formulas are given in Chapter 3; Table D.3 records the application-specific tuning and diagnostics. The requested path maximum is 70, further limited by grid size, centred training-sample size, numerical rank, and recurrence breakdown.

Table D.3: Estimator-specific regularisation rules in the frozen application. The procedures retain different finite-m objectives and stopping rules.

Method	Regularisation choice	Principal diagnostics
CG–FPLS	Released-code conjugate-residual recurrence and first discrepancy crossing; $\tau = 1.01$, $\delta = 0.10$, with an internal moment-GCV FPCR variance pilot.	Threshold, crossing status, moment-residual path, pilot dimension, and variance estimate.
Raw FPLS	Minimum five-fold inner prediction MSE over the unscaled raw Krylov power sequence, fitted by response-space normal equations.	Candidate validity, raw-basis and response-design condition numbers, and Krylov column norms.
Arnoldi FPLS	Independent minimum on the same inner folds, using twice reorthogonalised Gram–Schmidt and response-space QR.	Effective dimension, rank breakdown, response-design condition, and basis orthogonality defect.
FPCR	Response-space GCV over covariance-eigenvalue cut-offs.	Numerical rank, retained eigenvalue, and retained spectral condition number.

Invalid or non-finite raw candidates are excluded from selection. They are not silently repaired by a pseudoinverse, ridge penalty, or substitute method. A complete method failure would remain in the diagnostics and create missing predictions rather than a dropped fold. In the realised study, all 140 stored fits and all predictions are finite. These safeguards make clear why selected component counts are method-specific indices rather than a common complexity scale.

D.5 Development evidence

Table D.4 gives the repeat-level development results. Arnoldi has the smallest mean repeated-CV RMSE, while Raw has the second-smallest mean but a much larger standard deviation. Raw's five repeat RMSEs are 4.850, 4.960, 5.997, 4.844, and 4.760; the single 5.997 repeat creates its adverse tail. The contiguous-fold check reverses the top pair, with Raw at 4.913 and Arnoldi at 5.120. Hence development prediction supports Raw and Arnoldi as the leading pair but does not establish a stable ordering between them.

Table D.4: Development RMSE. The first three columns summarise five complete repeated nested-CV out-of-fold estimates. The final two columns are single prespecified sensitivities.

Method	Mean (SD)	Median	Range	Contiguous folds	Original validation
CG–FPLS	5.924 (0.083)	5.940	5.784–5.990	6.149	5.998
Raw FPLS	5.082 (0.517)	4.850	4.760–5.997	4.913	4.132
Arnoldi FPLS	4.917 (0.131)	4.954	4.741–5.084	5.120	4.150
FPCR	5.364 (0.219)	5.396	5.074–5.624	5.349	3.978

The original 155-to-40 split gives the metrics in Table D.5. All biases are positive, but the small validation set and its restricted assay support make this a weak basis for general calibration claims. In particular, every R^2 is negative because its SSE is compared with the unusually small validation SST. This is compatible with RMSE around four for three methods and does not contradict the broader benchmark signal.

Table D.5: Original supplied-split sensitivity: fitting on the 155 calibration tablets and evaluating on the 40 validation tablets. Bias is prediction minus observation.

Method	RMSE	MAE	Bias	R^2
CG–FPLS	5.998	4.820	2.109	-5.442
Raw FPLS	4.132	3.566	1.460	-2.057
Arnoldi FPLS	4.150	3.558	1.406	-2.085
FPCR	3.978	3.431	1.512	-1.834

Selection is more revealing than the small differences among the leading development prediction errors. Table D.6 records every outer-fit range. CG stops at three throughout.

Arnoldi remains tightly concentrated between four and six. Raw spans almost the entire requested path, reaches the cap in two fits, and triggers the fixed instability rule in 24 of 25. FPCR also varies widely, although its spectral truncation index is not directly comparable with a Krylov iteration.

Table D.6: Selection and numerical stability over 25 repeated-development outer fits. Quartiles use the stored selected indices.

Method	Minimum	Q1	Median	Q3	Maximum	Cap / flags
CG–FPLS	3	3	3	3	3	0 / 0
Raw FPLS	4	34	44	63	70	2 / 24
Arnoldi FPLS	4	5	5	6	6	0 / 0
FPCR	7	12	21	39	57	0 / n/a

D.6 Benchmark uncertainty and residuals

Chapter 8 gives all four primary point estimates and marginal iid intervals. Table D.7 supplies the complete prespecified RMSE sensitivity for the three declared paired contrasts. The focal Raw-minus-Arnoldi interval remains below zero for every block length, as do the larger CG-minus-Arnoldi and FPCR-minus-Arnoldi differences. This order robustness applies to conditional RMSE contrasts. Bias conclusions are less stable: the longer-block bias intervals for the secondary contrasts cross zero.

Table D.7: Declared benchmark RMSE contrasts and paired 95% percentile intervals. Circular blocks follow the supplied archive order.

Contrast	Estimate	Iid	Block 5	Block 10	Block 20
Raw − Arnoldi	-0.0363	[-0.0565, -0.0169]	[-0.0595, -0.0158]	[-0.0622, -0.0141]	[-0.0649, -0.0117]
CG − Arnoldi	0.6500	[0.3973, 0.9208]	[0.3286, 0.9783]	[0.2565, 1.0169]	[0.1664, 1.0687]
FPCR − Arnoldi	0.5420	[0.2855, 0.8442]	[0.2403, 0.8839]	[0.2015, 0.9218]	[0.1886, 0.9384]

Figure D.1 shows prediction-minus-observation residuals against observed assay. The methods share a few large errors, including one high-assay tablet predicted near the centre of the range and a low-assay tablet overpredicted by all procedures. These rows remain in every reported metric and in the bootstrap resampling population. The common extremes widen marginal RMSE intervals while largely cancelling in paired method differences. Their presence does not establish data error or a batch effect in the absence of supporting metadata.

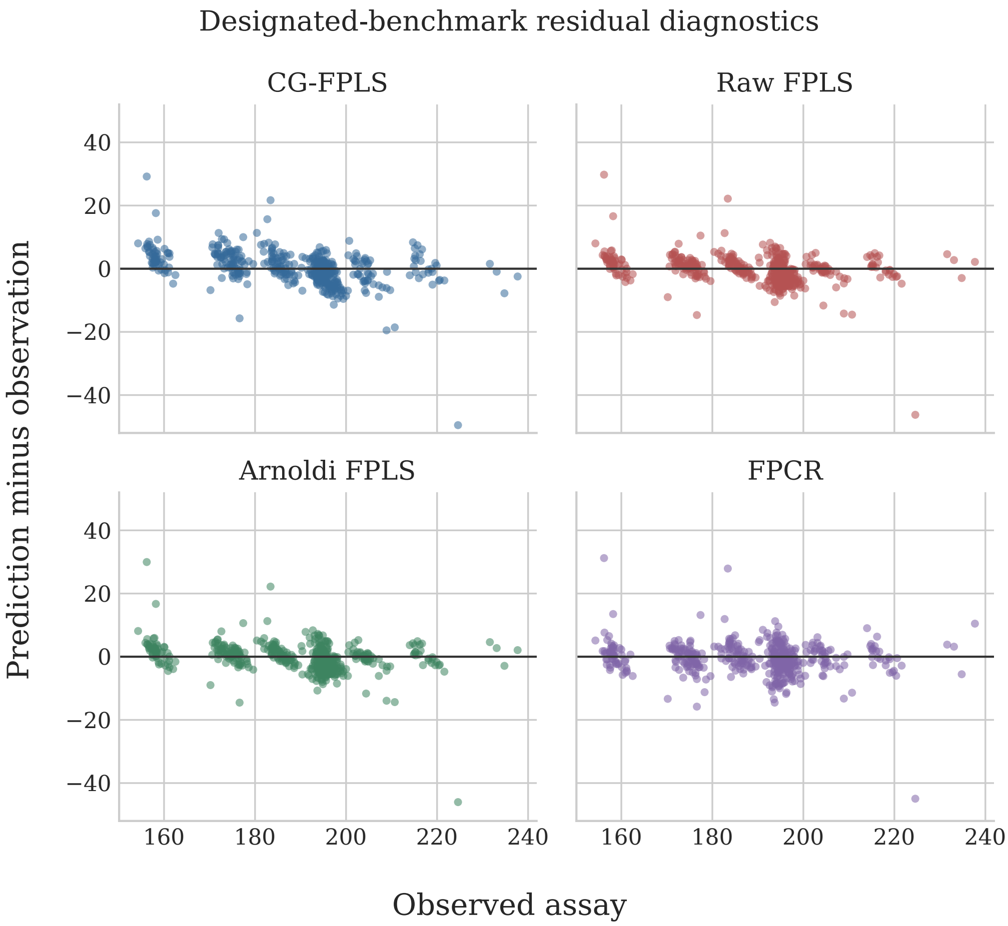

Figure D.1: Designated-benchmark residuals, defined as prediction minus observation, plotted against observed assay. All 460 tablets are retained. Shared large residuals are descriptive; the archive contains no batch or acquisition-order metadata with which to assign a physical cause.

D.7 Final-fit numerical diagnostics

Table D.8 contrasts final training fit with numerical representation. Raw and Arnoldi achieve nearly the same training RMSE and coefficient norm, but through sharply different coordinate systems. Raw's selected condition numbers are at the reporting ceiling, whereas Arnoldi retains an orthonormal basis to machine precision. FPCR has the smallest training RMSE and largest coefficient norm; these in-sample quantities do not override the benchmark ordering.

Table D.8: Final fit on all 195 development tablets. Coefficient norms use the archived original-input vector under the T^{-1} prediction convention.

Method	m	Training RMSE	$\|\widehat{\beta}\|_2$	Principal diagnostic
CG–FPLS	3	5.544	18,997	Pilot $m = 70$, $\widehat{\sigma}^2 = 10.467$; discrepancy threshold 0.191 was crossed.
Raw FPLS	55	4.399	21,494	Basis and response-design conditions both recorded at 4.50×10^{15}; instability flag set.
Arnoldi FPLS	5	4.395	21,573	Basis condition 1; response-design condition 11.95; orthogonality defect 7.40×10^{-16}.
FPCR	53	3.400	45,334	Retained eigenvalue 7.02×10^{-7}; retained spectral condition 35,919.

Figure D.2 gives the complete relevant paths. For Raw, the basis condition is already 5.74×10^8 at $m = 6$, above the fixed $1/\sqrt{\epsilon_{\text{mach}}} = 6.71 \times 10^7$ threshold; the response design is still worse. The raw Krylov column norm falls by more than 80 orders of magnitude by the selected $m = 55$, and the aggregate inner-CV criterion contains invalid candidate dimensions. Its finite selected prediction must therefore not be equated with stable continuation of the intended power basis.

Arnoldi's basis condition remains one along the requested path, and its orthogonality defect stays near machine precision. The defect is exactly zero at $m = 1$, so that point is absent on the logarithmic vertical scale. At the selected $m = 5$, the response-space least-squares problem is also well conditioned. Raw and Arnoldi nevertheless make almost identical benchmark predictions: correlation 0.999934, RMS paired difference 0.174, and maximum absolute difference 0.445 assay units. The empirical role of orthogonalisation is thus reliable representation of essentially the same predictive relation, not guaranteed improvement of every realised loss.

D.8 Preprocessing and instrument sensitivities

Table D.9 and Figure D.3 report every prespecified benchmark variant. The full instrument-1 analysis remains the sole primary result. Trimming the high-wavelength tail improves CG–FPLS, Arnoldi, and FPCR and compresses the four-method RMSE range, but slightly worsens Raw. Smoothing changes both errors and selected indices without

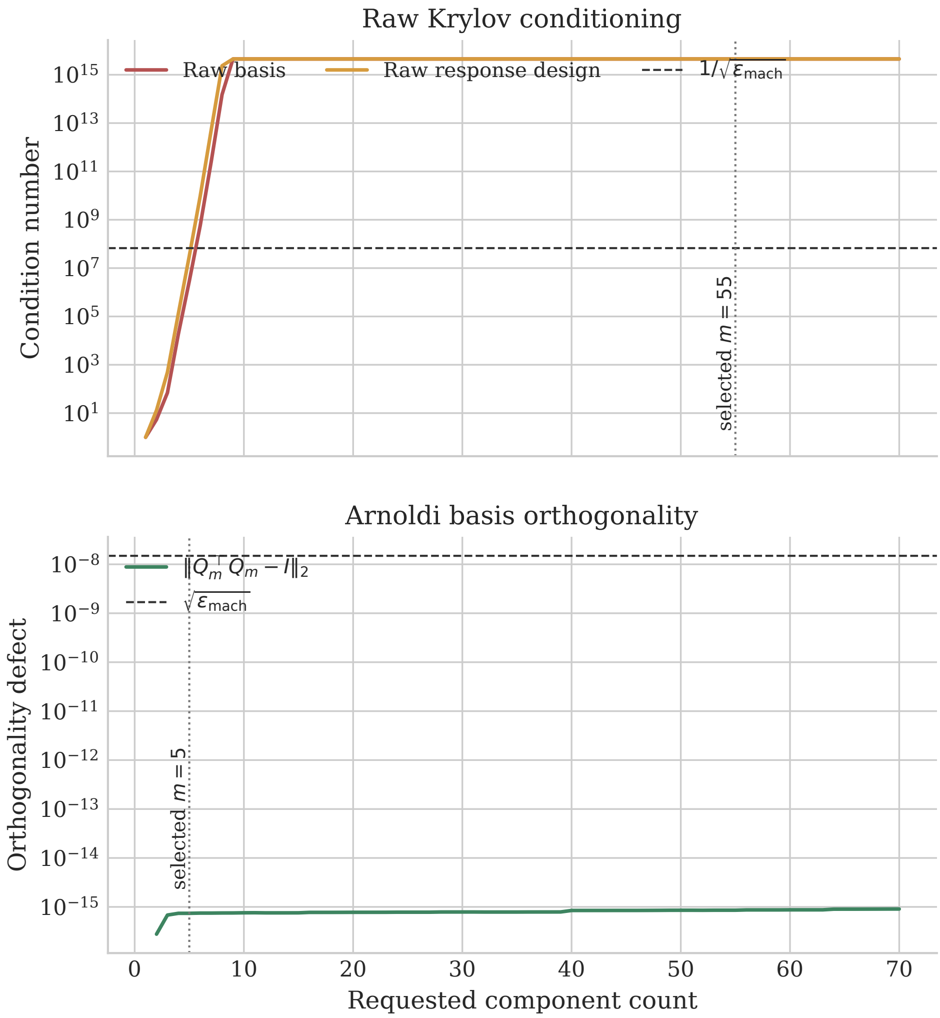

Figure D.2: Final raw and Arnoldi Krylov diagnostics. The upper panel shows raw-basis and response-design condition numbers against the fixed $1/\sqrt{\epsilon_{\mathrm{mach}}}$ flag threshold. The lower panel shows Arnoldi's spectral-norm orthogonality defect; its exactly zero $m = 1$ value is not displayed on the logarithmic scale. Vertical markers identify the selected dimensions 55 and 5.

uniformly improving them. Instrument 2 raises every RMSE; because it measures the same tablets, this is measurement-system sensitivity rather than independent replication.

Table D.9: Complete prespecified benchmark sensitivities. Parenthesised values in the RMSE column are selected component or stopping indices. Sensitivity rows are descriptive point estimates without variant-specific uncertainty intervals.

Variant	Method	RMSE (m)	MAE	Bias	R^2
Instrument 1, full	CG–FPLS	5.293 (3)	3.673	0.177	0.887
	Raw FPLS	4.607 (55)	3.036	-0.303	0.914
	Arnoldi FPLS	4.643 (5)	3.086	-0.301	0.913
	FPCR	5.185 (53)	3.547	-0.628	0.892
Instrument 1, 600–1798 nm	CG–FPLS	4.669 (3)	3.007	-0.456	0.912
	Raw FPLS	4.767 (10)	3.105	-0.476	0.908
	Arnoldi FPLS	4.632 (8)	2.846	-0.513	0.913
	FPCR	4.719 (14)	2.877	-0.575	0.910
Instrument 1, SG smooth	CG–FPLS	5.341 (3)	3.717	0.013	0.885
	Raw FPLS	4.663 (18)	3.056	-0.497	0.912
	Arnoldi FPLS	4.953 (9)	3.042	-0.491	0.901
	FPCR	4.823 (27)	3.053	-0.444	0.906
Instrument 2, full	CG–FPLS	6.764 (3)	5.100	1.022	0.815
	Raw FPLS	5.042 (53)	3.242	-0.601	0.897
	Arnoldi FPLS	5.095 (5)	3.279	-0.562	0.895
	FPCR	6.395 (37)	4.150	-0.921	0.835

The Raw–Arnoldi point ordering changes across variants, reinforcing that the primary 0.036-unit difference is not a universal ranking. Trimming improves all methods except Raw, but this association does not prove that visible high-wavelength spikes caused the primary residuals. Similarly, poorer instrument-2 results do not identify a transfer mechanism without a dedicated calibration-transfer design.

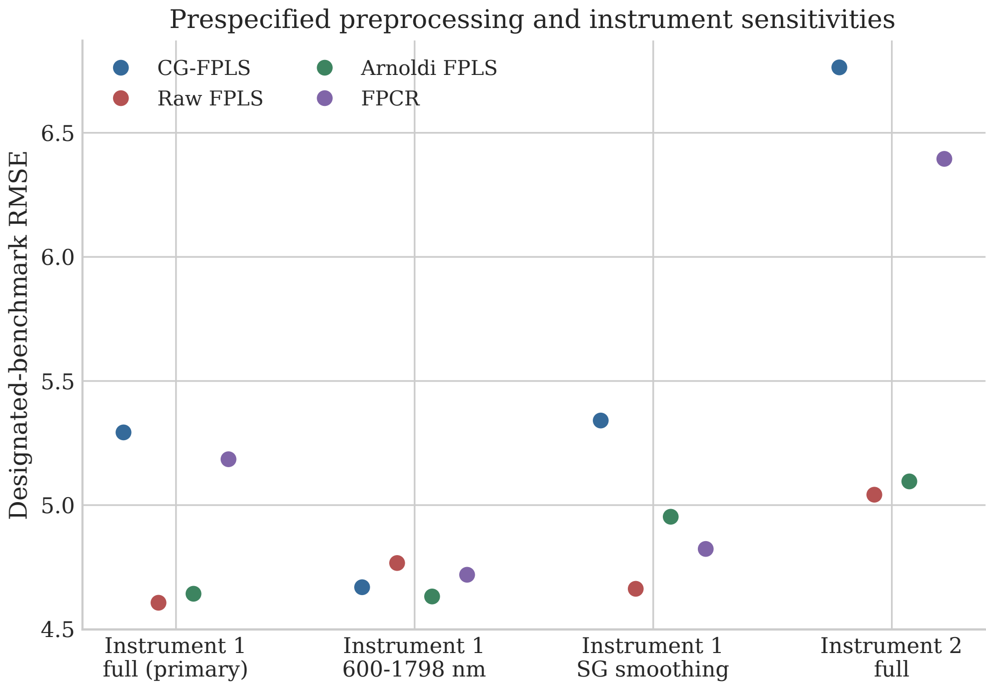

Figure D.3: RMSE across the primary analysis and three prespecified sensitivities. Points are descriptive benchmark estimates without uncertainty intervals. Horizontal comparisons change the spectral representation or instrument and do not define a post hoc replacement for the primary full-spectrum instrument-1 result.